

The University of the South Pacific

CS241: Software Design and Implementation

Assignment 1

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Software Design Report

Case Study: Student Administration System

Proposed by: Group 4

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1.0 Project Title & Team Details

Student Administration System

Software Design Report

For

Bula High School

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2.0 Introduction & Purpose

The software design document is a piece of document that provides a detailed outline or a detailed overview in terms of developing a software system. This document outlines all the system components and processes that are needed to create the final system (Atlassian, n.d.).

The software design document consists of highly detailed information regarding the architecture of the system being built. This includes various visual illustrations through different diagrams such as the ERD, different levels of DFD's, structure chart, use case diagram, domain class diagram, design class diagram, sequence diagram and user interfaces, together with the system risk analysis and the project implementation plan which guides the development team in achieving the technical aspects of the system which will ensure that the final product is well aligned with the client's needs (Understanding PDDs and SDDs: Essential Documentation for Automation Projects, n.d.).

2.1 Social Significance

This software design document holds a great social significance as it identifies the current ongoing issues occurring due to the lack of a Student Administration System in the high schools today. Some of these issues include loss of student data due to manual record keeping techniques such as file systems, delayed and incorrect decisions being made and difficulties and recording class attendance. This document also identifies the solutions to the major issues identified through the design and development of a Student Administration System which will have a favourable return on the day-to-day operations of the high schools. In order to achieve such outcome, this design document will ensure that the relevant stakeholders are well communicated, and the goals of the project and the approach needed to achieve those goals are well understood by all the stakeholders (Atlassian, n.d.).

2.2 Problem Domain

Lack of a proper administrative system is a common issue faced by most secondary schools in Fiji. Such an issue makes it difficult for the academic staff to track students' academic performance, record attendance of students that are present in class, participating in extracurricular activities or those involved in disciplinary incidents as manually kept records are often lost. Along with these, various departments in the schools are faced with lack of communication which results in errors and untimely decisions, the students, teachers and parents also face lack of communication, and the schools are overall inefficient in the enrolment of students and managing schedules.

The Student Administration System (SAS) aims to resolve the above-mentioned issues through providing a secured mean of storing all student related information, class attendance records, extracurricular activity attendance records and disciplinary case records in a centralised database, enabling easy access of these records when need arises. This allows the academic staff to be more efficient as the system minimizes the current academic workload. The system is user friendly as it consists of a communication module that will allow for direct communication between students, teachers and parents.

The system aims to improve the student enrolment process by allowing students and parents to directly enrol via the application instead of waiting in long queues on the enrolment day. In addition, the system will also allow for the creation of schedules such as class and teacher timetables.

2.3 Purpose of the Design Document

The purpose of this design document is to outline a structured approach towards the development of the Student Administration System (SAS) through highlighting all the processes and the components needed to produce the final working system. The design document also consists of diagrams that portray the various key aspects of the proposed system. For instance, the use case diagram is drawn to show the interaction between the users or the actors involved and the system, creating a visual representation on the way in which the various actors of the system interact with the functionalities of the system. The class diagrams will show the classes, attributes and methods of the system and its relationships. This shows how the way in which the components of the system interact. The data flow diagrams will be drawn to show the flow of data as data coming into the system and data going out of the system. The entity relationship diagram is drawn to outline the structure of the database of the system, clearly outlining the entities, attributes and the relationships. The document consists of a sequence diagram as well which shows the interaction of objects in sequential manner and finally the illustration of structure charts portrays the entire system in a hierarchical manner, describing the functions and the sub functions of the system as lowest modules (GeeksforGeeks, n.d.).

Apart from the mentioned illustrated diagrams, the design document provides the development team with the risk analysis of the system along with the implementation plan which will provide guidance to the development team in creating and delivering a system that is fully functional and has met the needs of the client.

2.4 Intended Users

The intended users for this design document are the system architect, database developer, ui/ux designer, integration specialist, QA engineer and the project manager. The system architect will use this document to analyse the stated requirements, design the database structure, the system architecture and the overall architecture of the system. The database developer will use the design document to create the overall schema of the database, establish the relationships and implement the data structure to effectively handle student related data. The UI/UX designer will use this design document to create user friendly interfaces of the system dashboard, attendance tracking and grade recording modules together with the parent portals. The integration specialist will use this document to connect all the modules and create a system that is cohesive together with a consolidated data flow. The QA Engineer will conduct multiple testing which includes unit testing, integration testing, system testing and user acceptance testing through the use of this document and finally the project manager will use the design document to assess the implementation of the system and the go-live activities.

2.5 Organizational Context

The client identified for the Student Administration System is Bula High School, which is a secondary level educational institute that has been encountering increasing number of student enrolments for quite some time now. The school has seen a 10% yearly increase of the total number of students being enrolled. The school currently has 20 academic staff members which includes teachers, and the administrative staff. The school also has 2 librarians who maintain the school library and the records present in the library through a manual file system. The yearly increase in the number of students getting enrolled in the school has necessitated the need for the school to invest in a Student Administration System. The system will allow for efficient record keeping within the various departments of the school, ensuring easy access to information for timely decision making. The proposed system will be easy to use as it will require minimal training for all groups of people that are part Bula High School and is able to handle multiple concurrent users during peak times such as enrolment of students when a new academic term begins.

2.6 Design Standards

The design standards that have been adopted for the Student Administration System include firstly, security. Security is a major concern in any aspect of software development. The proposed system will comprise of multiple layers of security to reduce vulnerabilities relating to confidential student data. Secondly, the system will be developed using the agile methodology which will allow the development team to gain frequent feedback as this methodology provides short feedback loops. This will ensure for continuous improvement of the system that is being developed (10 Software Development Best Practices (2025 Checklist), 2025). Thirdly, the system design will follow the Keep It Simple, Stupid (KISS) principle which focuses on writing easy to understand code, simple architectural design and implementation. Using the KISS principle will ensure that future team members that might work on the system are able to understand and modify the program code (Gupta, 2024) and lastly, the system adheres to the Model View Controller (MVC) framework. This framework is composed of three components which are firstly, the model. This component handles all the data logic and forms an interaction with the database. Secondly, the view component handles the presentation of information to the end users ensuring that it is dynamically rendered and lastly the controller component handles the flow of data as it enables the connection between the model component and the view component. The use of the MVC framework separates the business logic, user interface logic and the input logic (GeeksforGeeks, 2025).

3.0 System Capabilities

3.1 Stakeholders

3.1.1 Internal Stakeholders

Stakeholder	Description
School Administration	Consists of the school's administrative branch including principal, vice principal and board of directors. Responsible for deciding the direction of the school in terms of integrating programs and curriculum
Academic Staff	Includes head of departments and teaching staff. Responsible for creating and assessing assignments, taking attendance and taking classes.
Students	Are the students currently enrolled at the high school. Responsible for selecting and attending class and doing assessments with optional extracurricular participation.

3.1.2 External Stakeholders

Stakeholder	Description
Parents and Guardians	Are the legal guardian for a student that provides financial and emotional aid and attends parent-teacher meetings.
Educational Authorities (Ministry of Education)	Sets policies and mandates that the school must follow. Defines important dates in the academic year such as holidays and examination dates.
Higher Education Institutions (Universities)	Consists of universities. They will require academic documents from students which will be issued by the high school.

3.1.3 System End Users

End User	Role
Students	Are the primary users of the system whereby a student repository is created to contain student information, grades and activities.
School Administrators	Oversees system management such as access role and audit logs.
Faculty Members	Does daily system operations including attendance taking and entering assessment marks.
Legal Guardian	Access to student information and communication with school

3.2 Functional Requirements

3.2.1 User Management

FR1 – Create New User Profile. The system shall allow administrators to create new user accounts with username, email, password, and role assignment.

FR2 – Update User Profile. The system shall allow users to update their personal information including contact details and preferences.

FR3 – Delete User Profile. The system shall allow administrators to delete user accounts while maintaining data integrity.

FR4 – Perform Data Backup. The system shall automatically backup all data daily and allow manual backup operations.

FR5 – Manage User Roles. The system shall allow administrators to assign and modify user roles (Student, Teacher, Administrator, Parent).

FR6 – User Login. The system shall authenticate users with username and password credentials.

FR7 – View User Profiles. The system shall allow authorized users to view user profiles based on their permission level.

FR8 – Edit User Profile. The system shall allow users to edit their own profile information.

FR9 – User Logout. The system shall provide secure logout functionality that terminates user sessions.

3.2.2 Student Information Management

FR10 – Record Student Medical Information. The system shall allow authorized staff to record and maintain confidential student medical information.

FR11 – Import Student Data. The system shall allow importing student data from external files and systems.

FR12 – Export Student Data. The system shall allow exporting student data in various formats (CSV, Excel, PDF).

FR13 – Generate Student Data Report. The system shall generate comprehensive reports on student enrolment and demographics.

3.2.3 Academic Performance Management

FR14 – Enter Student Marks. The system shall allow teachers to enter student marks for assignments, tests, and examinations.

FR15 – View Student Grades. The system shall allow students, parents, and teachers to view grades based on their access level.

FR16 – Update Student Marks. The system shall allow authorized users to update previously entered marks with audit trail.

FR17 – Delete Student Marks. The system shall allow administrators to delete marks while maintaining deletion records.

FR18 – Generate Academic Report. The system shall generate report cards, progress reports, and academic transcripts.

FR19 – MyAPA Assessment. The system shall provide assessment and progress analytics for tracking student learning outcomes.

3.2.4 Attendance and Discipline Management

FR20 – Manage Teacher Leave and Attendance. The system shall allow administrators to manage teacher leave requests and attendance tracking.

FR21 – Record Class Attendance. The system shall allow teachers to record daily class attendance for all students.

FR22 – Record Disciplinary Incidents. The system shall allow staff to record disciplinary incidents with relevant details and actions taken.

FR23 – Notify Attendance Issues. The system shall automatically notify parents and staff about attendance problems.

3.2.5 Timetable and Scheduling Management

FR24 – Create New Class Timetable. The system shall allow administrators to create class timetables with subjects, times, and room assignments.

FR25 – Update Class Timetable. The system shall allow updating existing timetables while checking for conflicts.

FR26 – Delete Class Timetable. The system shall allow authorized users to delete timetables with proper notifications.

FR27 – Detect Scheduling Conflicts. The system shall automatically detect and alert users to scheduling conflicts.

FR28 – View Class Timetable. The system shall allow users to view timetables in daily, weekly, and term formats.

3.2.6 Extracurricular Activities Management

FR29 – Record Extracurricular Activity Attendance. The system shall allow coordinators to record student attendance for extracurricular activities.

FR30 – Enrol in Extracurricular Activities. The system shall allow students to browse and enrol in available extracurricular activities.

3.2.7 Teacher Schedule Management

FR31 – Create Teacher Timetables. The system shall allow administrators to create individual teacher schedules and duty assignments.

FR32 – Update Teacher Timetables. The system shall allow updating teacher schedules with automatic conflict checking.

FR33 – Delete Teacher Timetables. The system shall allow administrators to delete teacher timetables with proper documentation.

FR34 – Create Course Shells for Subjects. The system shall allow administrators to create course templates for each subject.

3.2.8 Communication Management

FR35 – Create Announcement. The system shall allow authorized users to create announcements for specific user groups.

FR36 – Update Announcements. The system shall allow users to modify existing announcements and publication schedules.

FR37 – Delete Announcement. The system shall allow authorized users to delete announcements with confirmation.

FR38 – Share Documents. The system shall provide secure document sharing with appropriate permission controls.

FR39 – Direct Messaging. The system shall provide secure messaging between teachers, students, parents, and administrators.

3.2.9 Library Management

FR40 – Reserve Books. The system shall allow users to search and reserve library books online.

FR41 – View Due Date. The system shall display due dates for borrowed materials and send reminder notifications.

FR42 – Issue Fines. The system shall automatically calculate and issue fines for overdue or damaged library items.

FR43 – Search Resources. The system shall provide search functionality for library books and digital resources.

FR44 – Download Resources. The system shall allow authorized users to download digital educational materials.

3.2.10 AI Integration

FR45 – Chat With AI. The system shall provide an AI chat assistant to help users navigate the system and answer common questions.

3.3 Non-Functional Requirements

3.3.1 Product Requirements

3.3.1.1 Performance Requirements

- NFR1: The system must support at least 1,000 concurrent students, 10 staff members, and 2 administrators at any given time.
- NFR2: The system must record attendance for a class of 40 students with a transaction submission time of 4 seconds.
- NFR3: The system must be hosted on an AWS EC2 instance which shall be configured to have 16 virtual CPUs, 32 GB RAM, and 400 GB storage.
- NFR4: The system must be able to generation academic reports using student grades within 7 seconds of generation prompt.
- NFR5: The system must implement asset compression and lazy loading to ensure all page's load within 3 seconds, supporting users with low bandwidth (3G).
- NFR6: The system must process user registration and login within 5 seconds. Upon which, users must be allocated to their appropriate account.
- NFR7: The system must load the student or teacher dashboard within 4 seconds with subject shells being made available at point of registration.
- NFR8: Database queries for complex reports and academic analytics must return results within 10 seconds for datasets up to 10,000 student records.
- NFR9: The system must maintain performance standards during peak usage periods, including enrolment, exam results release, and report card generation.
- NFR10: Mobile device performance must maintain response times within 7 seconds on 3G networks and 4 seconds on 4G Wi-Fi networks.

3.3.1.2 Dependability Requirements

- NFR11: The system must backup all critical data automatically on a daily basis.
- NFR12: The system must be available to all registered users with 99.9% uptime.
- NFR13: The system must be extensively stress-tested in a user-controlled environment (Docker) prior to deployment.
- NFR14: The system must return to its most stable previous backup in case of critical system failure.

- NFR15: The system must fully be recovered within 30 minutes of critical failure.

3.3.1.3 Reliability and Availability Requirements

- NFR16: The system must have a Recovery Time Objective (RTO) of a maximum of 2 hours and a Recovery Point Objective (RPO) of a maximum of 24 hours.
- NFR17: Automatic failover mechanisms must activate within 30 seconds when primary servers become unavailable.
- NFR18: The system must maintain data consistency across all modules during system failures and recovery.
- NFR19: Scheduled maintenance windows must be limited to 4 hours monthly and occur during off-peak hours with 72 hours of advance notice.

3.3.1.4 Security Requirements

- NFR20: The system must be able to implement multi-factor authentication (MFA) and Role-Based Access Control to allow only registered users to be able to access the system.
- NFR21: The system must hide all sensitive student data from unauthorized users. Only users with viewing privileges must be able to view it.
- NFR22: The system must manage all GET and POST requests with HTTPS protocols.
- NFR23: The system must have mandatory password rotations for school staff and system administrators every 90 days.

3.3.1.5 Usability Requirements

- NFR24: The system must be built with KISS principles for the user interface, ensuring all core features can be accessed within three clicks.
- NFR25: The system must be able to be accessed through all modern browsers built on different render engines – Chromium, WebKit and Firefox.
- NFR26: The system must be fully responsive on all mobile and desktop resolution sizes.
- NFR27: The system shall use graphical elements to reduce cognitive load and guide the user.
- NFR28: New users must be able to complete basic tasks within 30 minutes of their first login with minimal training.
- NFR29: The system must provide contextual help and tooltips for all major functions and comprehensive online documentation.
- NFR30: Error messages must be user-friendly, provide clear guidance for resolution, and include technical support contact information.
- NFR31: Interface localization must support seamless switching between English (UK), Fijian, and Hindi.

3.3.1.6 Technical Requirements

- NFR32: The system must be able to use Jenkins to administer automated deployment pipelines (CI/CD)
- NFR33: The system must be built with modular, component-based frameworks – React.
- NFR34: The system must automatically update all time sensitive events to adjust to UTC-12 (Fiji Time) including during daylight saving transitions.

3.3.1.7 Accessibility Requirements

- NFR35: The system must include accessibility functionality such as hover and focus states on interactive elements for users with disabilities.
- NFR36: All system text shall use a sans-serif font for readability and proper scalability.
- NFR37: The system will use ARIA labels for screen reader users.
- NFR38: The system must include alt text and titles for all elements for visual disabilities and screen readers.

3.3.1.8 Compatibility Requirements

- NFR39: The system must support the latest three versions of major browsers.
- NFR40: Mobile compatibility must include iOS 13+ and Android 9+.
- NFR41: The system must integrate seamlessly with existing school management systems.

3.3.2 Organisational Requirements

3.3.2.1 Environmental Requirements

- NFR42: The system must be hosted on a EC2 instance to avoid need for on-premises servers
- NFR43: The school must integrate WI-FI and workstation upgrades to ensure all students and staff may access them.

3.3.2.2 Operational Requirements

- NFR44: The system shall be at all times monitored and managed by IT administrator.
- NFR45: The school must publish mandatory system training material to all teachers and staff.
- NFR46: The system must only undergo maintenance outside of school hours with a 3-day advance notice to all users.

3.3.2.3 Training and Support Requirements

- NFR47: Technical support must respond to critical issues within 2 hours and general inquiries within 24 hours during business days.
- NFR48: User manuals and technical documentation must be updated within 30 days of any system changes.
- NFR49: System updates must include user notification with detailed change logs and impact assessments 48 hours before implementation.

3.3.2.4 Development Requirements

- NFR50: The system must be built with a hybrid database, using PostgreSQL (relational database) and AWS Dynamo (non-relational database) as well as React.js for the frontend and Django/Node.js for the backend.
- NFR51: The system must be rigorously tested through Unit Testing and User Acceptance Testing prior to deployment.

- NFR52: The system must have all documentation clearly organized and fully traceable before system rollout.
- NFR53: The system must be built with clear and concise naming conventions for all methods and variables for future reference.
- NFR54: The system will use version control (Git) to manage source code.

3.3.2.5 Data Requirements

- NFR55: Student academic records must be retained for a minimum of 7 years, and attendance records for 3 years.
- NFR56: The system must implement automated data archiving for records older than 5 years.
- NFR57: Data migration from legacy systems must maintain 100% accuracy with comprehensive validation and rollback capabilities.
- NFR58: All data input must undergo validation checks before being stored.

3.3.2.6 Audit and Compliance Requirements

- NFR59: The system must log all user actions with timestamps and user identification.
- NFR60: Compliance reports must be automatically generated monthly.
- NFR61: The system must provide data export capabilities in standard formats for regulatory audits.
- NFR62: Data subject rights must be supported, including access to personal data, data portability, and secure data deletion.

3.3.3 External Requirements

3.3.3.1 Regulatory Requirements

- NFR63: The system must comply with all mandatory policies issues by the Ministry of Education, Fiji as well as international data protection treaties.
- NFR64: The system must request for parental confirmation to gain access to a student's personal information to unauthorized users.

3.3.3.2 Accounting Requirements

- NFR65: A full cost-benefit analysis has been conducted, showing fixed costs of \$118,000, ongoing costs of \$80,000/year, and annual benefits of \$90,000.
- NFR66: The projected return on investment (ROI) is \$10,000 annually.

3.3.3.3 Monitoring and Analytics Requirements

- NFR67: The system must implement real-time performance monitoring with automated alerts when resources reach 80% capacity.
- NFR68: User behaviour analytics must track feature usage and common pathways to inform future improvements.
- NFR69: Automated health checks must run every 15 minutes, monitoring database connectivity and server resources.
- NFR70: Capacity planning reports must be generated monthly, showing resource utilization and projected scaling needs.

3.3.3.4 Safety/Security Requirements

- NFR71: The system must have offline data storage capabilities to sync data once an internet connection is available.
- NFR72: The system must under scheduled maintenance only during school term breaks.
- NFR73: The system must have all data be logged into their appropriate databases in real-time if a internet connection is available.
- NFR74: The system must undergo regular penetration tests bi-annually by authorized assessment teams.

3.3.3.5 Environmental Impact Requirements

- NFR75: The system must use energy-efficient cloud hosting with carbon footprint reporting.
- NFR76: Resource optimization must include automatic scaling down during off-peak hours to minimize energy consumption.
- NFR77: Environmental impact reporting must be generated annually.

3.3.3.6 Ethical Requirements

- NFR78: Only authorized personnel may access student performance data.
- NFR79: The system must disclose all data collection, usage and sharing policies to users.

3.3.3.7 Legislative Requirements

- NFR80: The system must adhere to all data protection laws imposed by international laws.
- NFR81: The system must ensure all use of personal and academic data can only be accessed by the authorized parties.
- NFR82: The system must log all activity done by users on the system.

3.3.3.8 Quality Assurance Requirements

- NFR83: The system must maintain a minimum of 90% code coverage through automated testing.
- NFR84: Performance testing must be conducted quarterly, including load and stress testing.
- NFR85: User Acceptance Testing must achieve a 95% satisfaction rating from representative user groups.

3.3.3.9 Localization and Language Requirements

- NFR86: The system must initially launch with English (UK) as the primary preference with the ability to change to different languages.

3.3.3.10 Communication and Integration Requirements

- NFR87: The system must enforce secure encryption and authentication of APIs to integrate third party services into the system.

4.0 Risk analysis

4.1 Database Design Risks

Risk Category	Risk Description	Impact Level	Likelihood	Mitigation Strategies	Implementation Priority
Data Integrity	Due to numerous foreign relationships existing between entities, data corruption and loss of historical records can occur	High	Medium	Implementing explicit cascading rules to automatically manage data relationships such as updates and deletion. Enforce referential integrity by use of constraints and foreign keys in the database.	High
Performance & Scalability	Performance bottlenecks due to complex joins across multiple tables while also enduring thousands of concurrent users	High	High	Implementing caching layer to reduce load on the system. Index all frequently queried fields to ensure quick retrieval during peakhours.	High
Data Security & Privacy	Sensitive student data accessible through multiple pathways	Critical	Medium	Maintaining an extensive and detailed audit log to track and monitor all user data accesses	Critical

4.2 System Architecture Risks

Single Point of Failure	Use of centralized database can result in critical failure as it takes down the entire system if it fails.	High	Low	Database replication to move to a new instance in the case of failure. Implement load balancing to entire traffic is distributed evenly across all database instances.	High
Data Consistency	Real-time updates made by users can cause conflicting information to enter into the system.	Medium	Medium	Implementing a concurrency locking mechanism to prevent two users from editing or updating the same piece of data at the same time.	Medium
Complex Relationships	Many-to-many relationships causing data anomalies	Medium	Medium	Use of bridging entities to act as an intermediary between many-to-many relationships. Define concise business rules to validate the type of data that can be entered into each data entry.	Medium

4.3 Operational Risks

Data Migration	Migrating paper-based and spreadsheet data to new system	High	High	Have rollback procedures to revert the system to a previous state in case of system failure. Validate all data before it is moved into the system.	High
User Adoption	Resistance to complex system with multiple portals	Medium	High	System design to be done in a concise, intuitive manner that is easy to navigate. Actively collect user feedback while system is being constructed to make improvements to ease user adoption during full release.	High
System Integration	Integration challenges with existing and third-party systems	Medium	Medium	Design of system with clear intention to integrate APIs later on to ensure it is not tightly coupled. Set up of a separate integration environment which will act as a testing ground for integrations before it goes live.	Medium

4.4 ERD Design Risk

Missing Critical Entities	ERD may lack entities for complete functionality	Low	Medium	Addition of tables for data recovery and backup.	Low
Attribute Definition	Insufficient constraints or validation rules for attributes	Medium	Medium	Use of proper data types, with attributed having a "NOT NULL" constraint to ensure appropriate data is entered. Including time-based attributes into tables to track changes over time.	Medium

Risk Assessment Legend

Impact Levels:

Critical	System failure, data breach, legal compliance issues
High	Significant operational disruption, user dissatisfaction
Medium	Moderate operational issues, workaround available
Low	Minor inconvenience, minimal business impact

Likelihood:

High	Very likely to occur (>70%)
Medium	Moderately likely (30-70%)
Low	Unlikely to occur (<30%)

Implementation Priority:

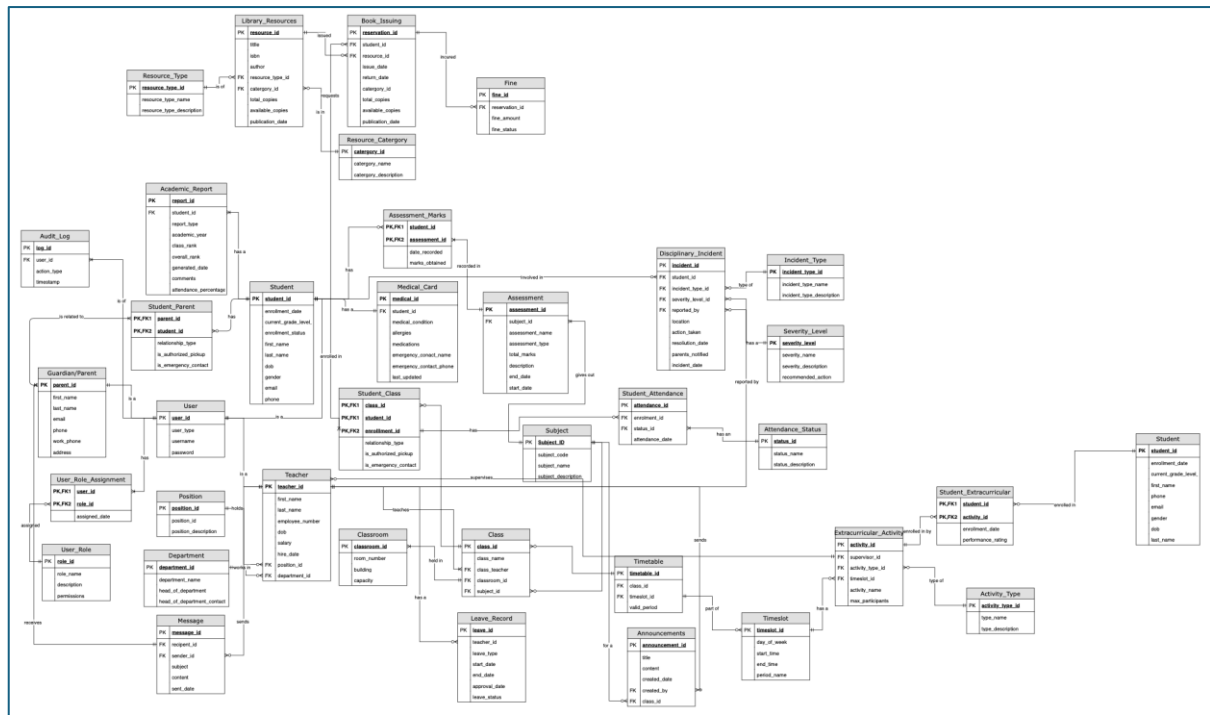
Critical	Must be implemented before system deployment
High	Should be implemented during initial development phase
Medium	Can be implemented in subsequent releases
Low	Can be addressed in future maintenance cycles

4.5 Budget Allocation for Risk Mitigation

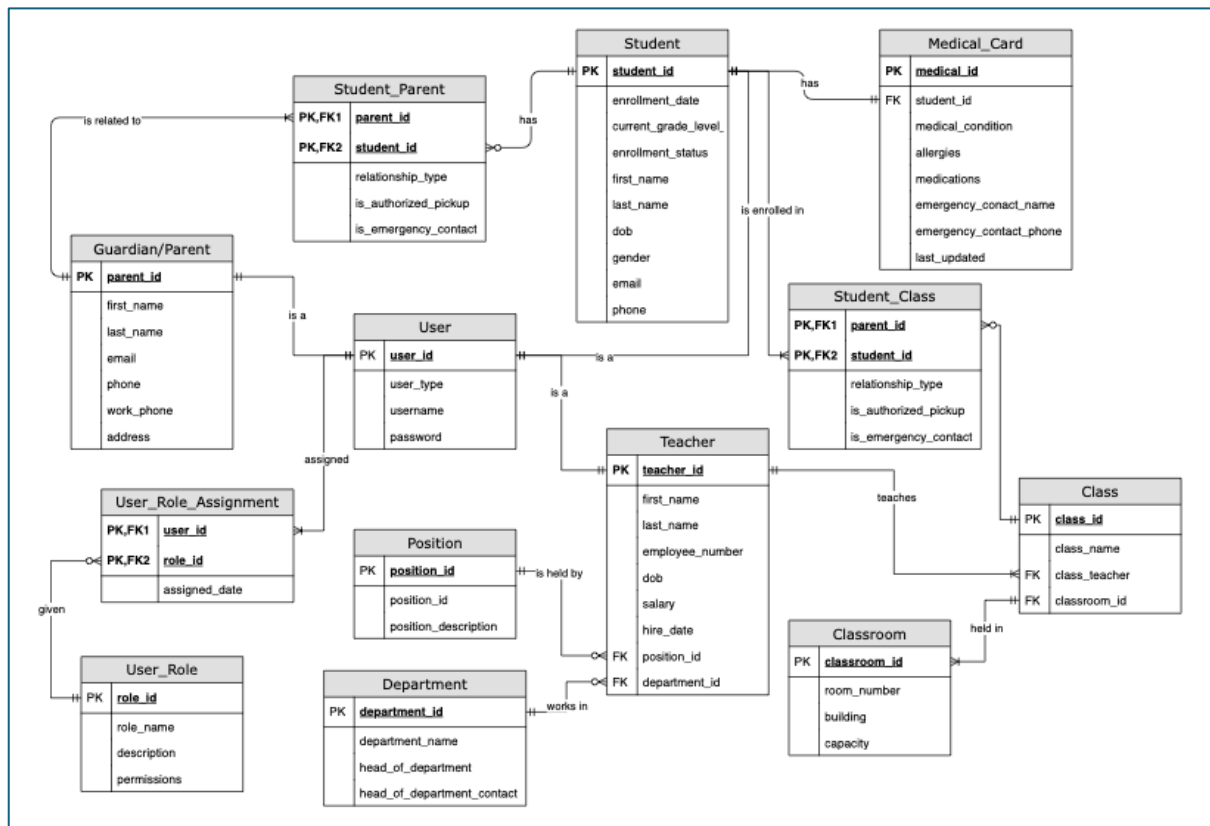
Risk Category	Estimated Cost	Percentage of Total Budget (\$198,000)
Data Security & Privacy	\$25,000	12.6%
Performance & Scalability	\$20,000	10.1%
Data Migration	\$15,000	7.6%
User Training & Adoption	\$8,000	4.0%
System Integration	\$10,000	5.1%
Total Risk Mitigation	\$78,000	39.4%

5.0 Entity Relationship Diagram

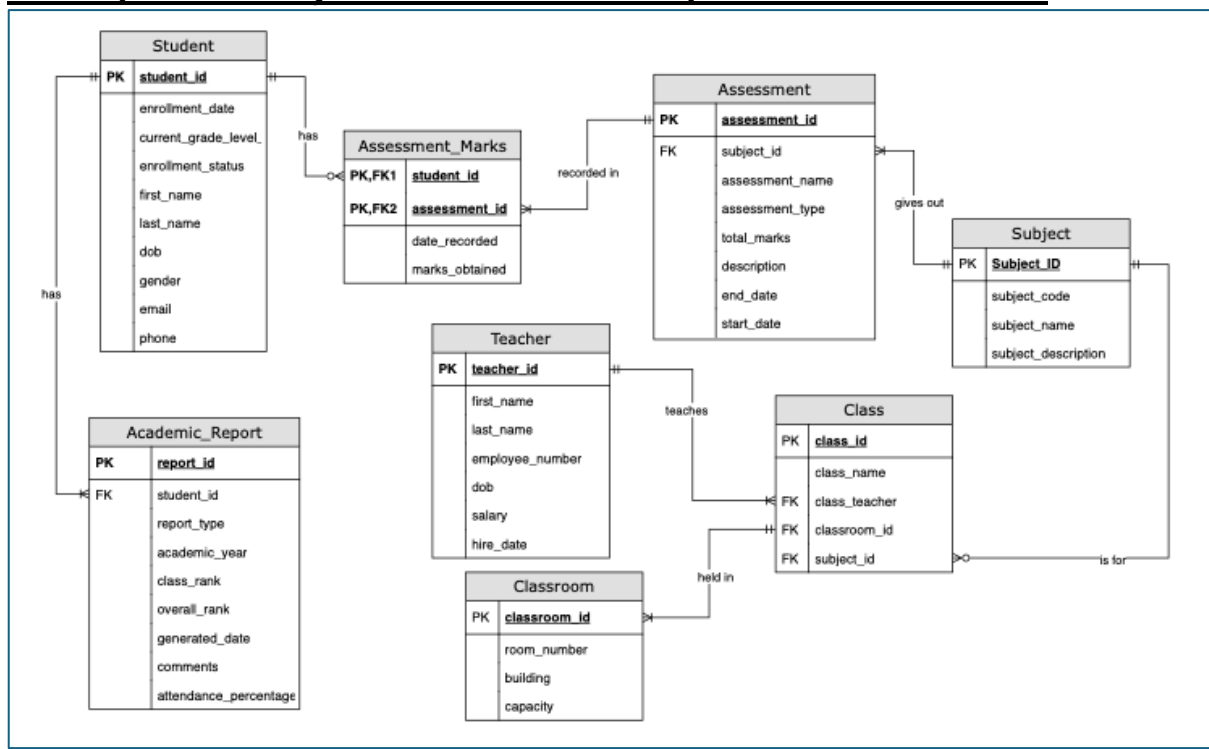
5.1 Complete Diagram



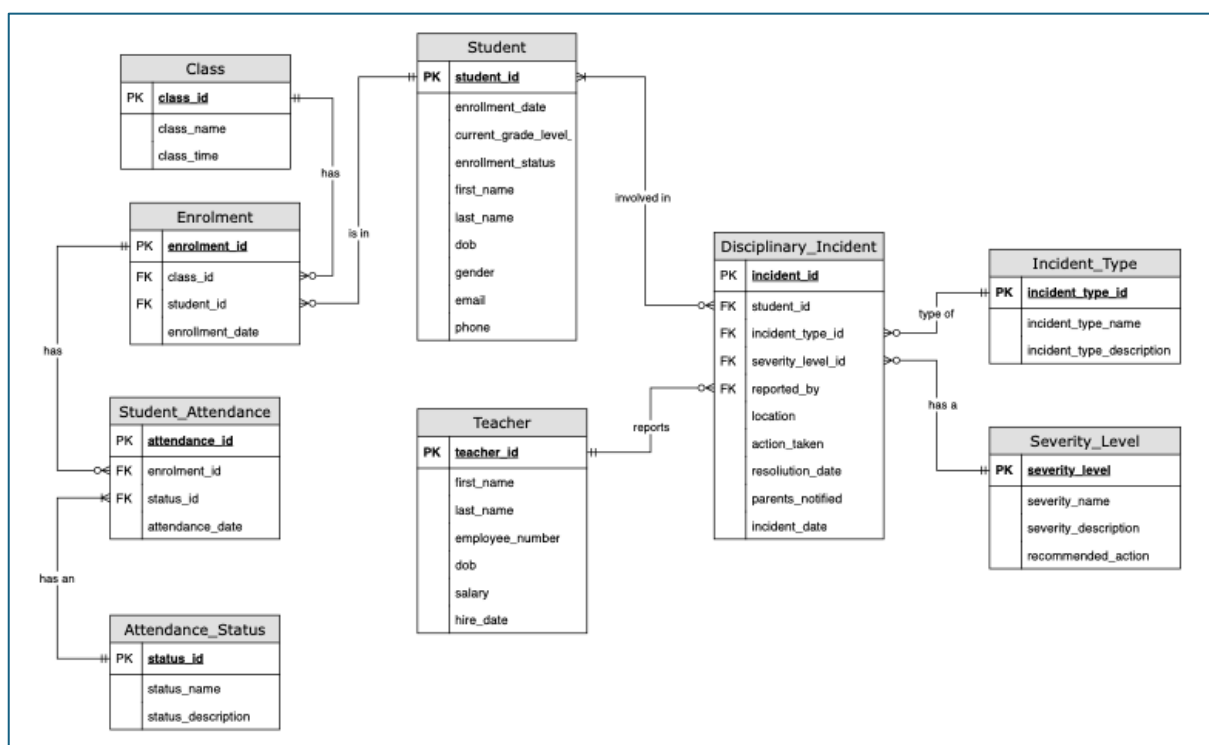
5.2 Repository for student information



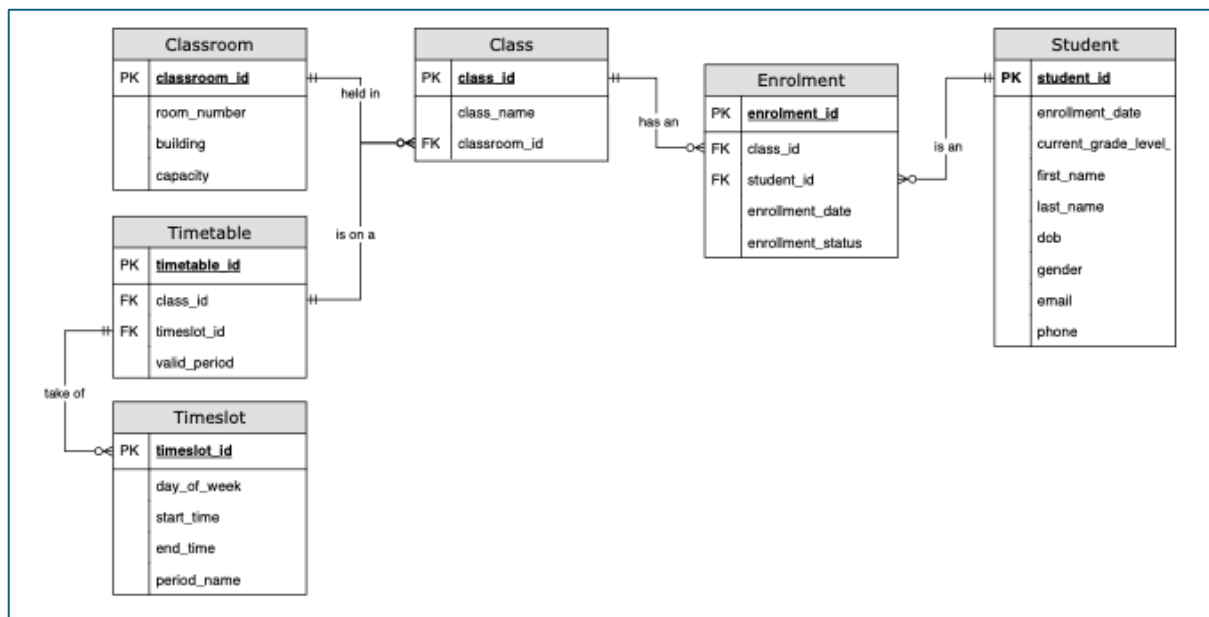
5.3 Report card system and academic performance tracker



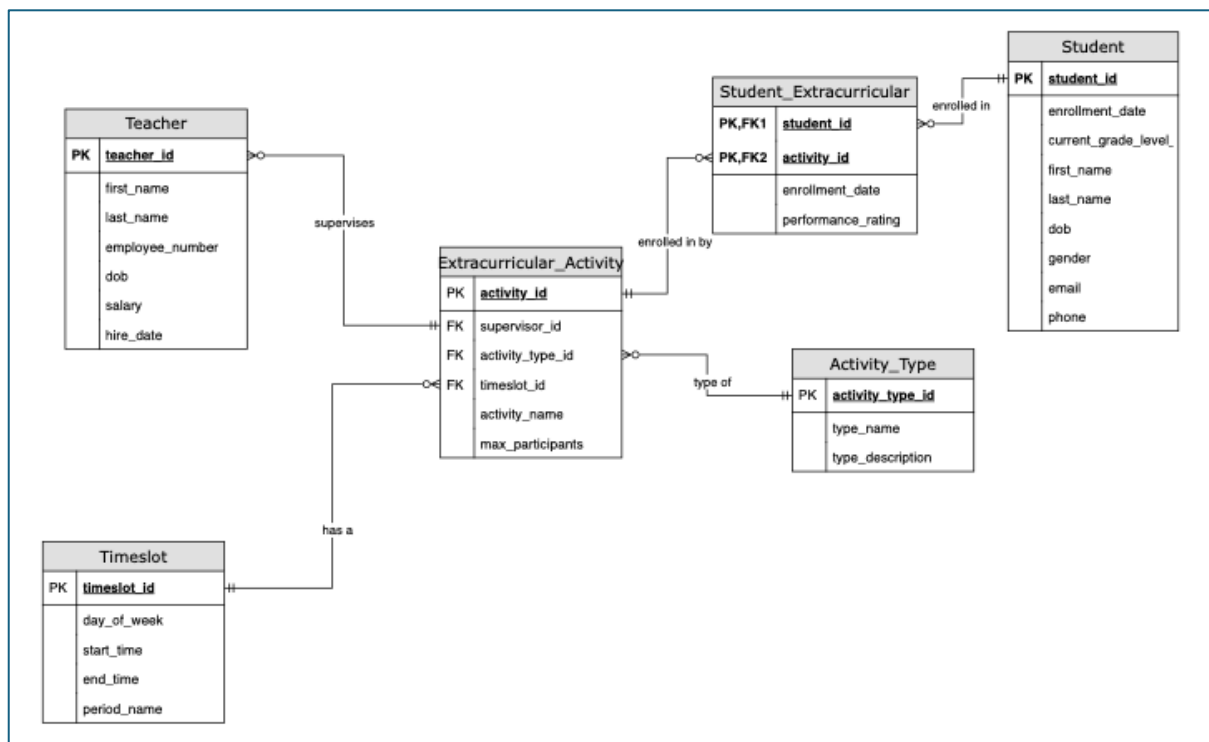
5.4 Discipline and attendance monitoring and management



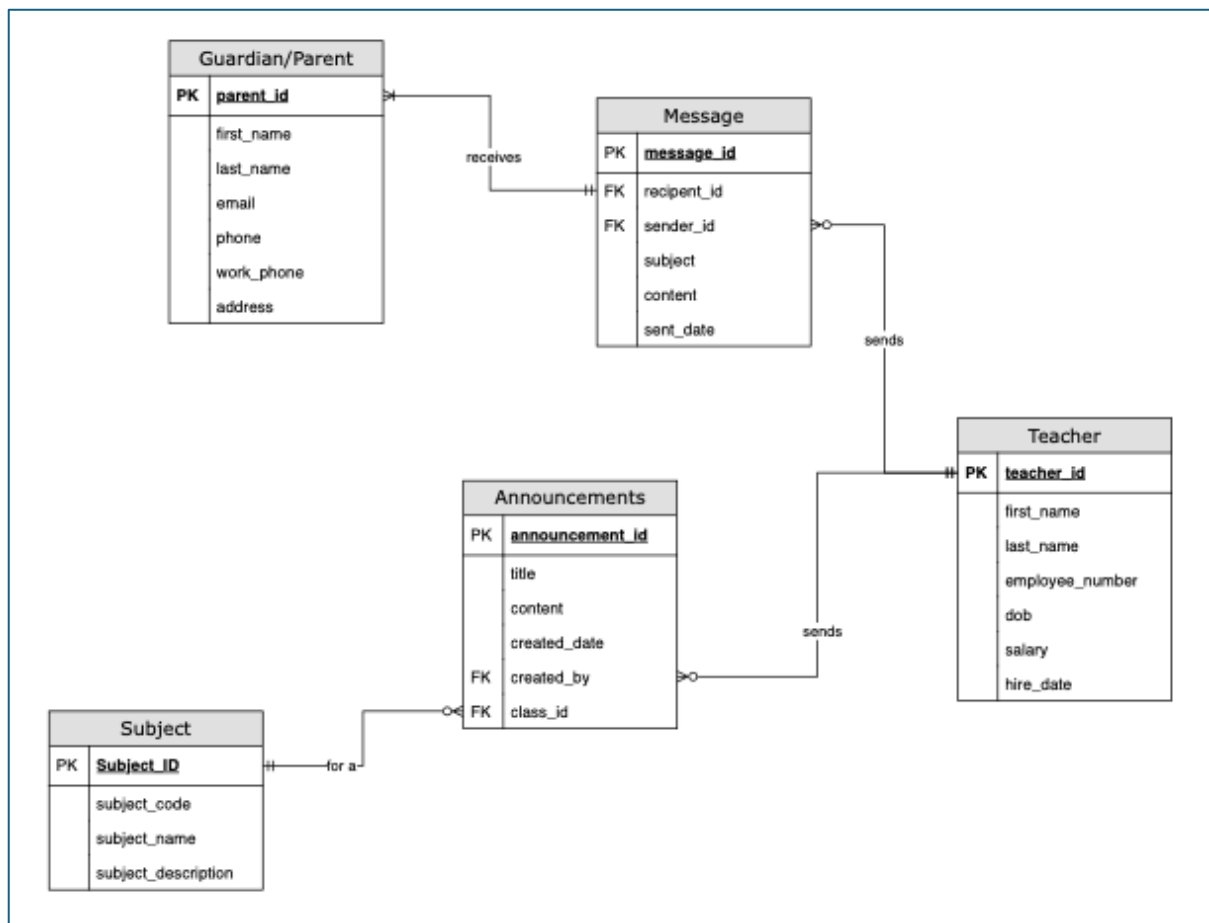
5.5 Class timetables and Scheduling System



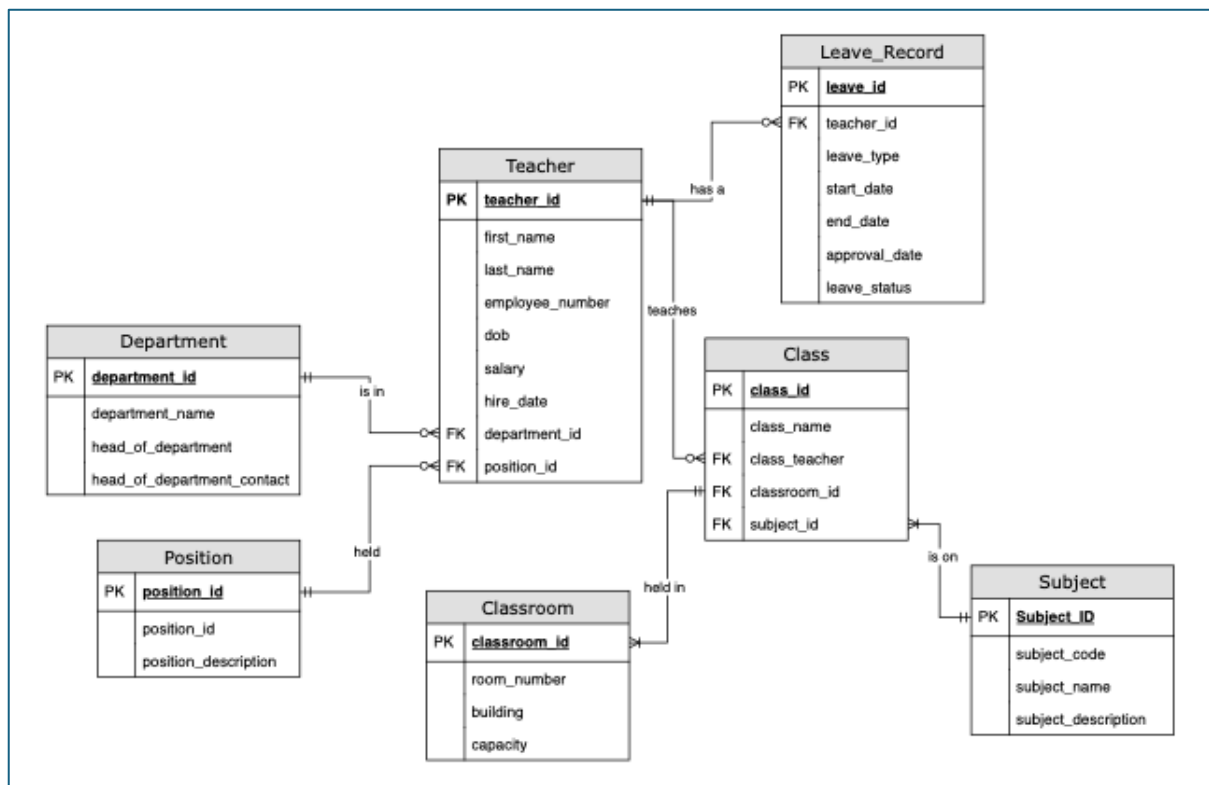
5.6 Extra curricular and student activities tracker system.



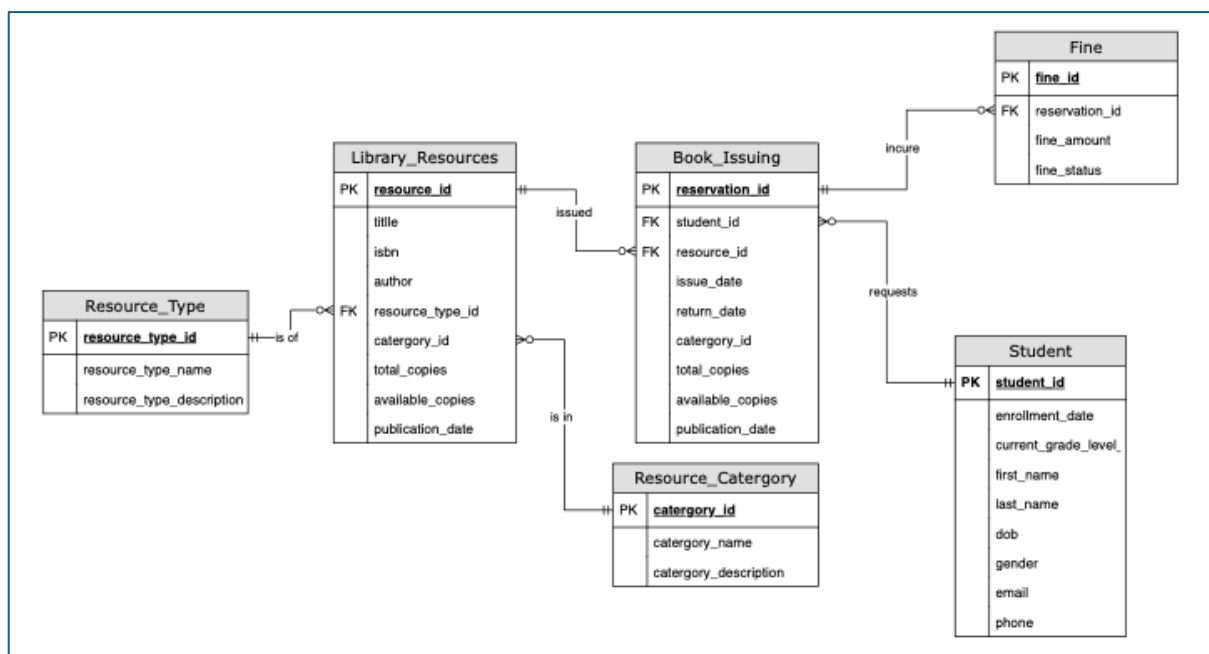
5.7 Communication and parental engagement platform.



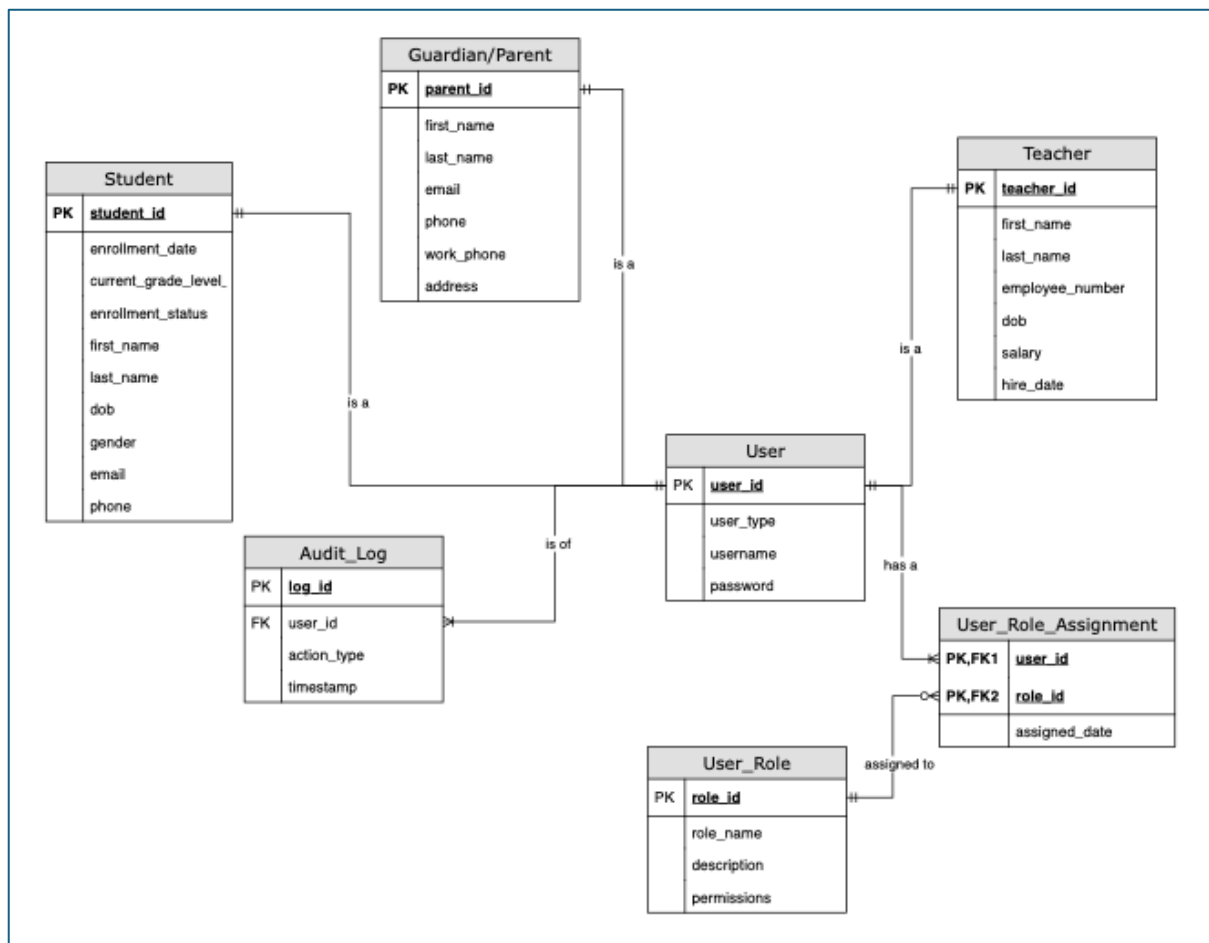
5.8 Staff management module.



5.9 Library and learning resources system.



5.10 Data privacy and security.

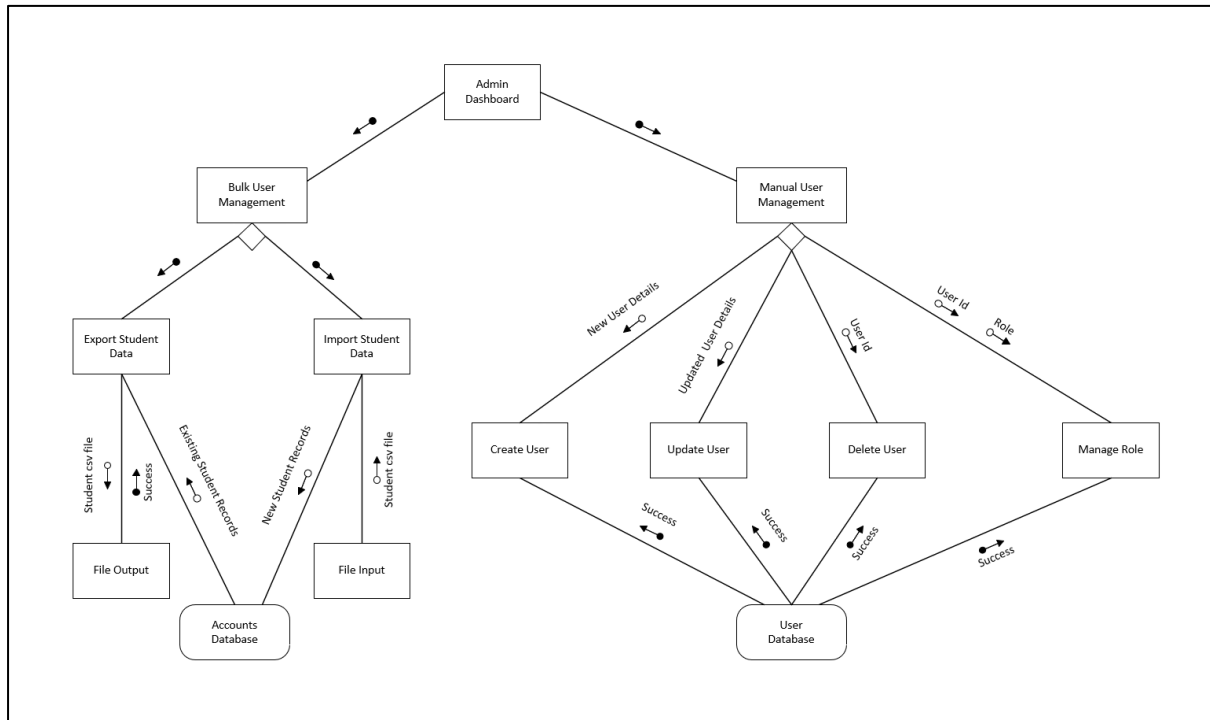


6.0 Application Architecture

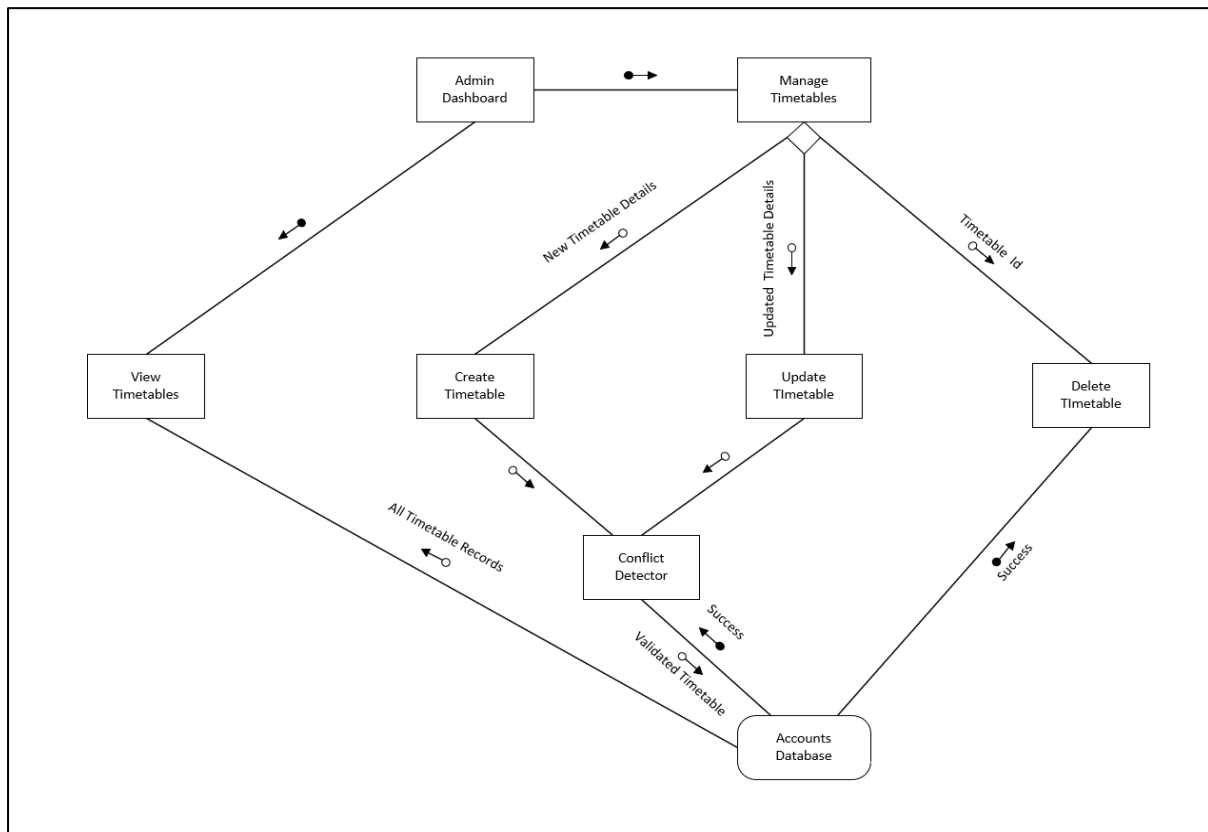
6.1 Structured design artifacts

6.1.1 Structured Chart

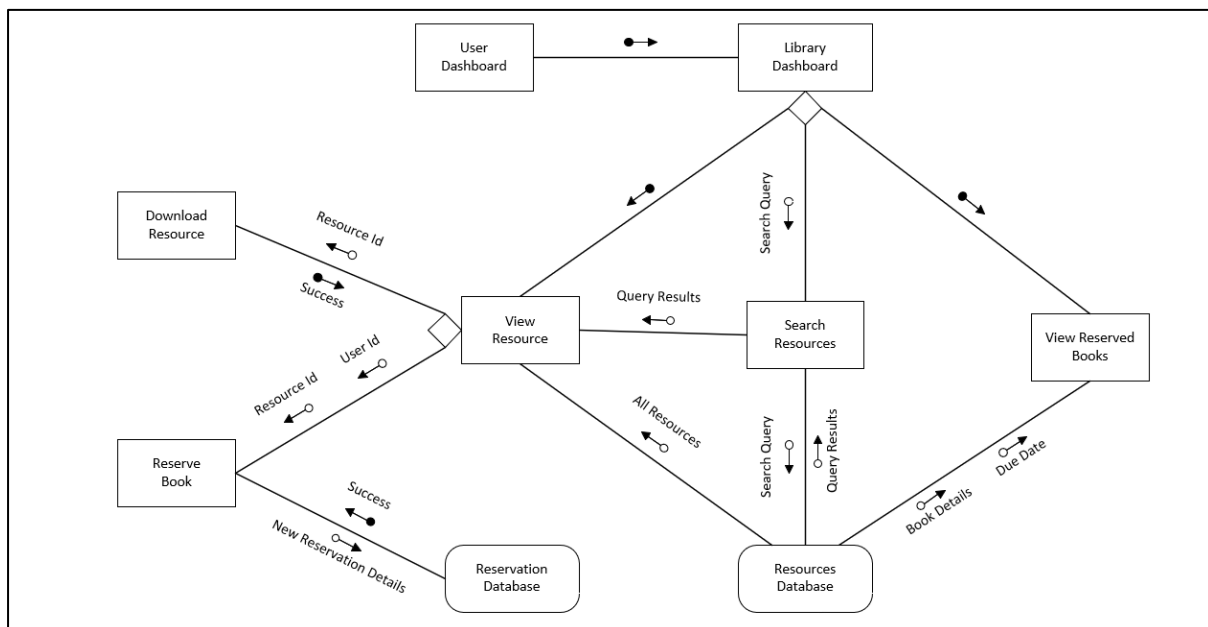
6.1.1.1 User Management



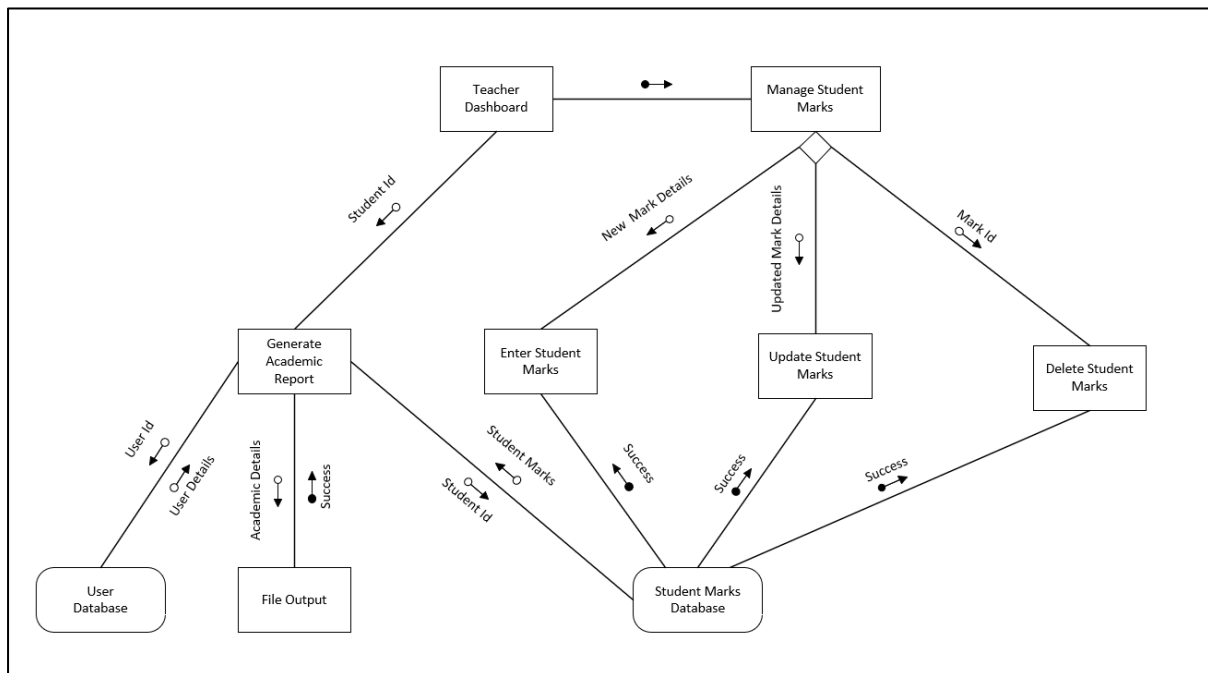
6.1.1.2 Timetable Management



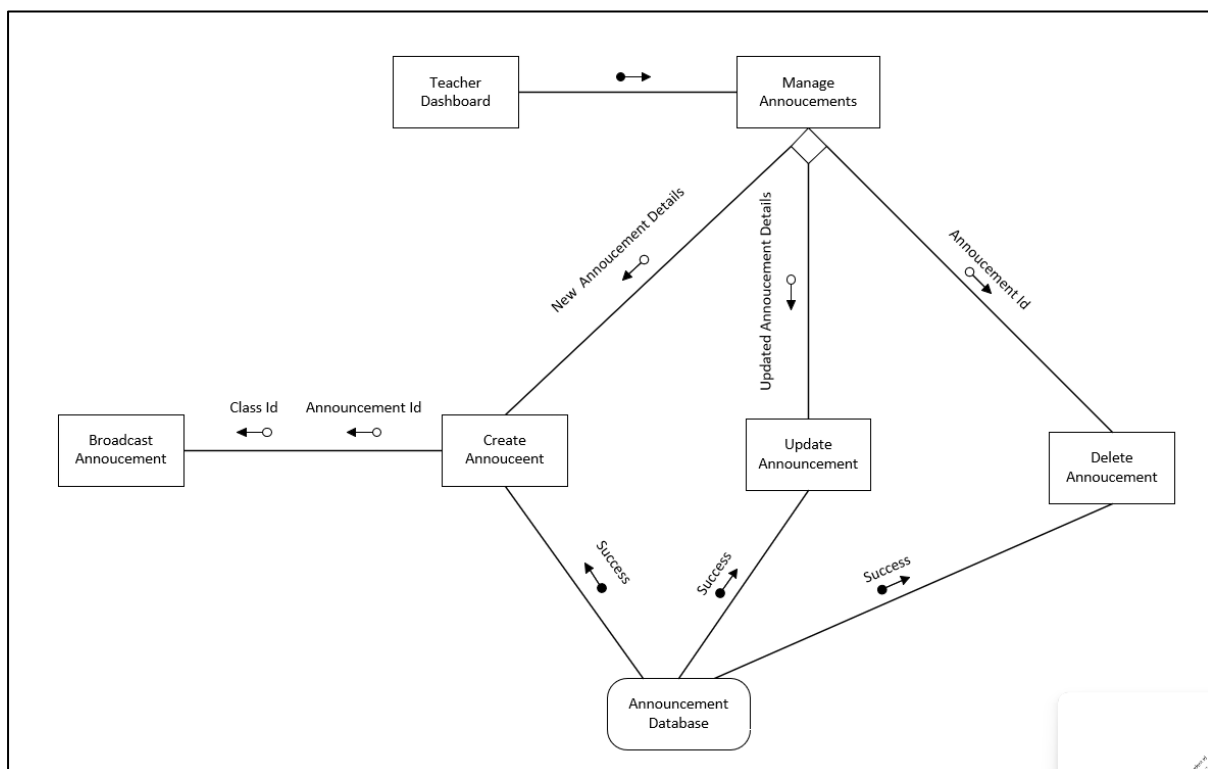
6.1.1.3 Library Features



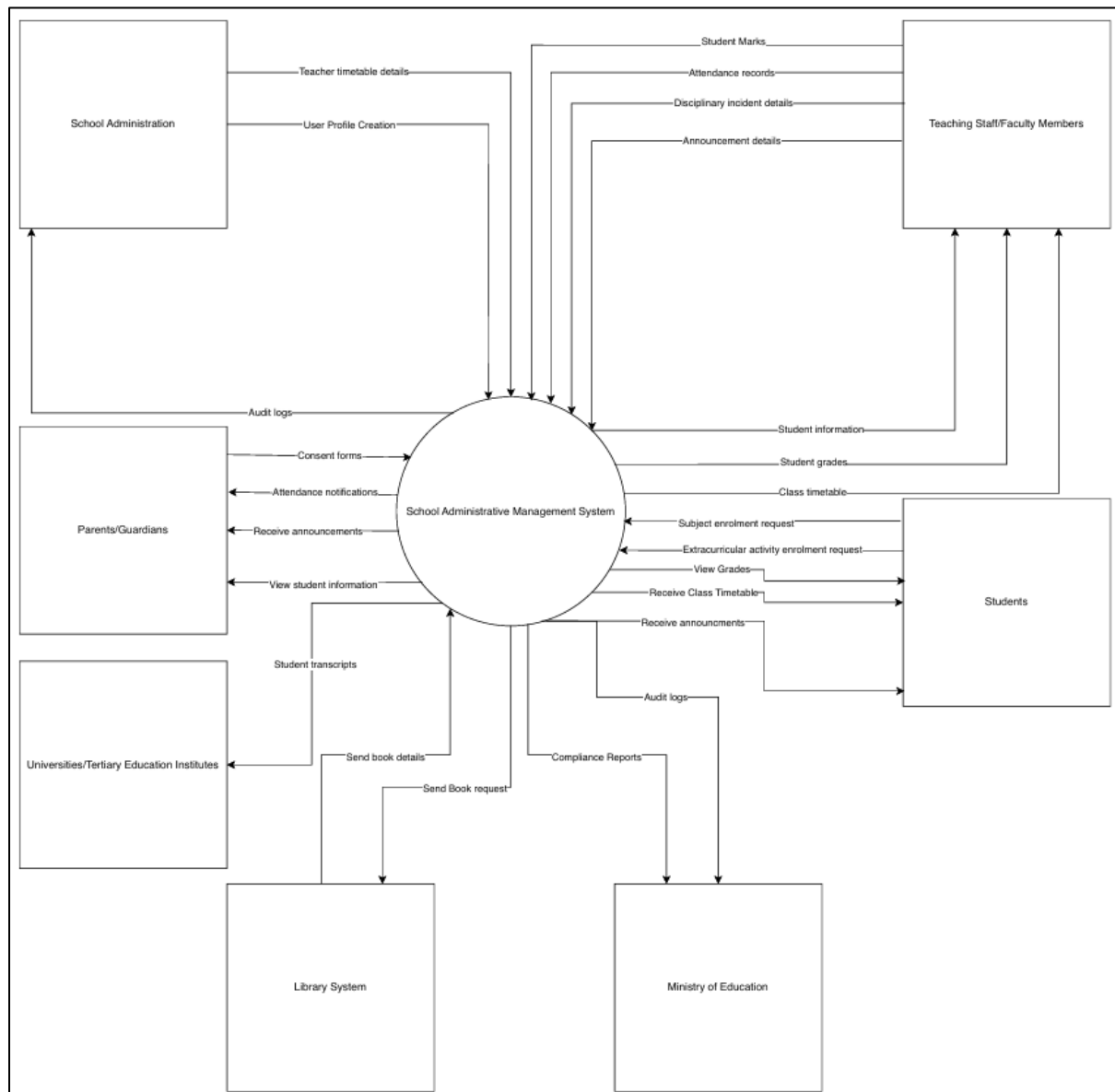
6.1.1.4 Student Mark Management



6.1.1.5 Announcement Management



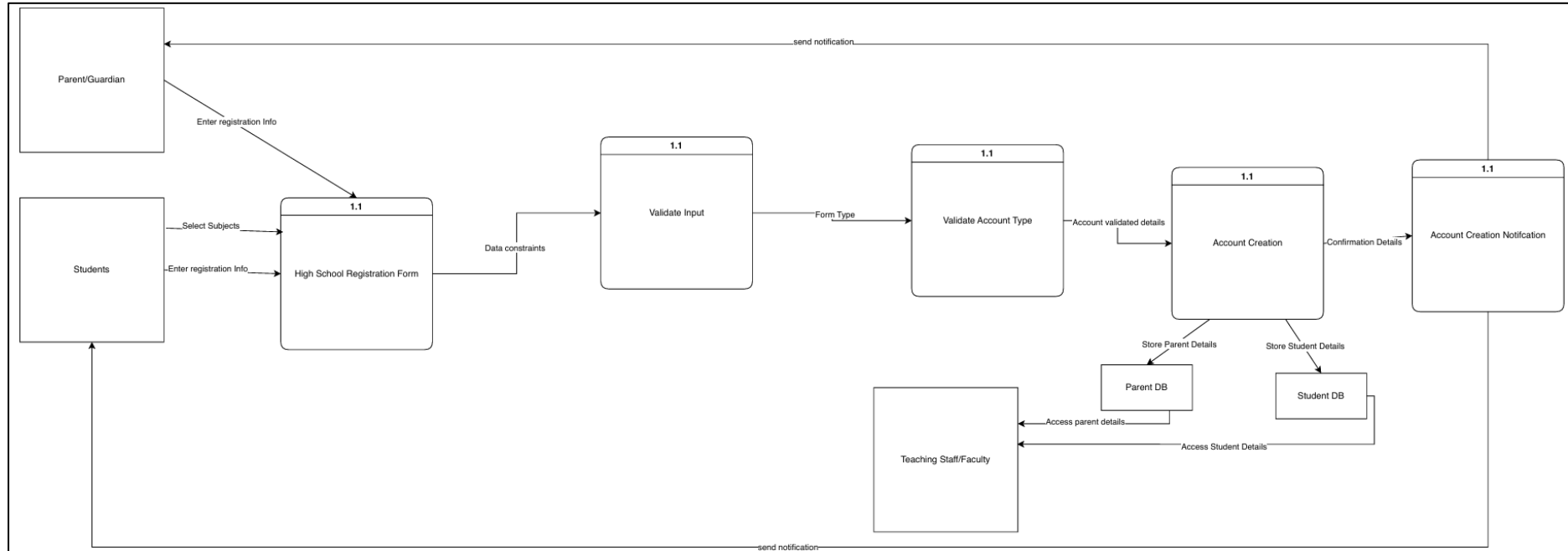
6.1.2 System Context Diagram/DFD Level 0 Diagram



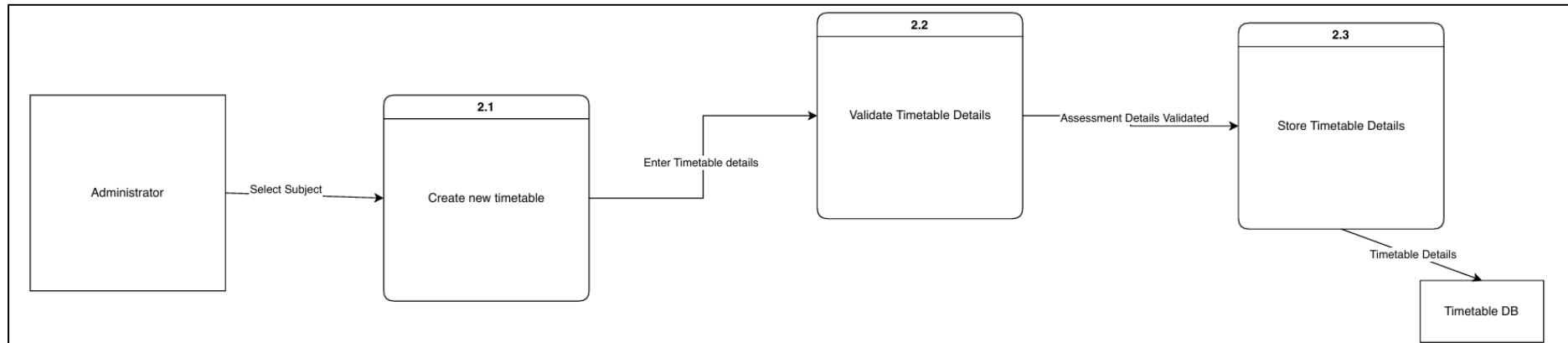


6.1.2.1 Primitive DFD diagram (Lowest level DFD)

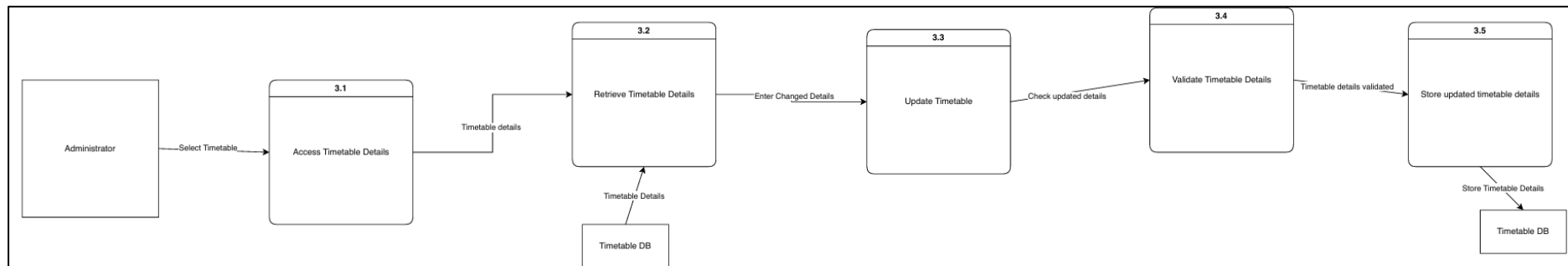
6.1.2.1.1 User Registration



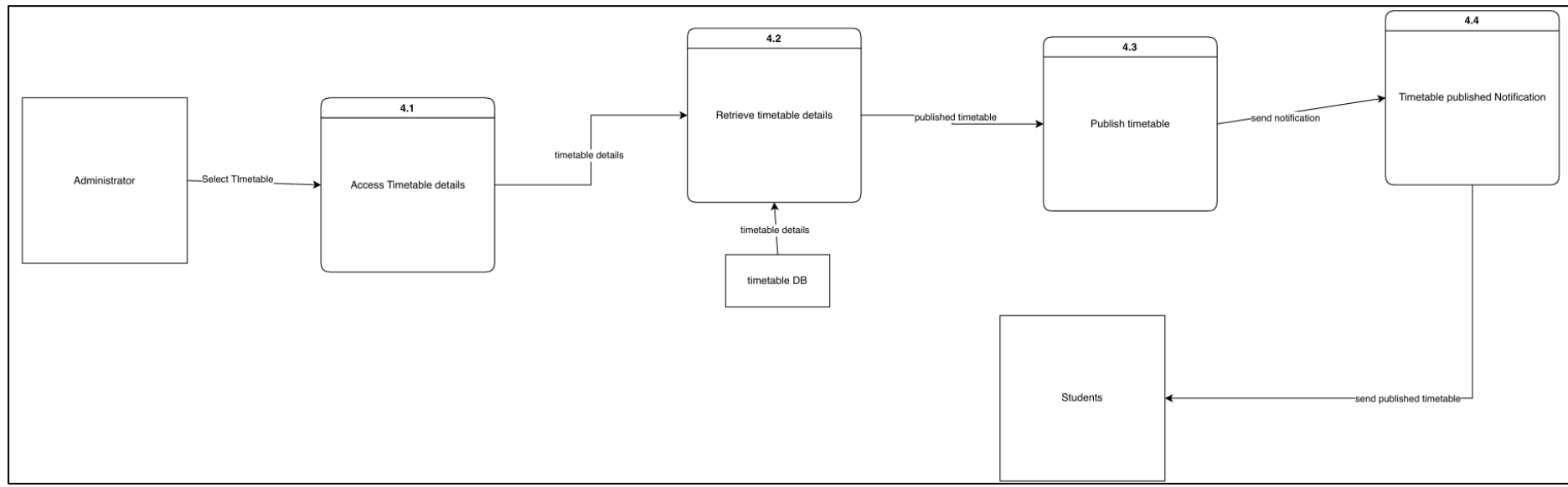
6.1.2.1.2 Create Timetable



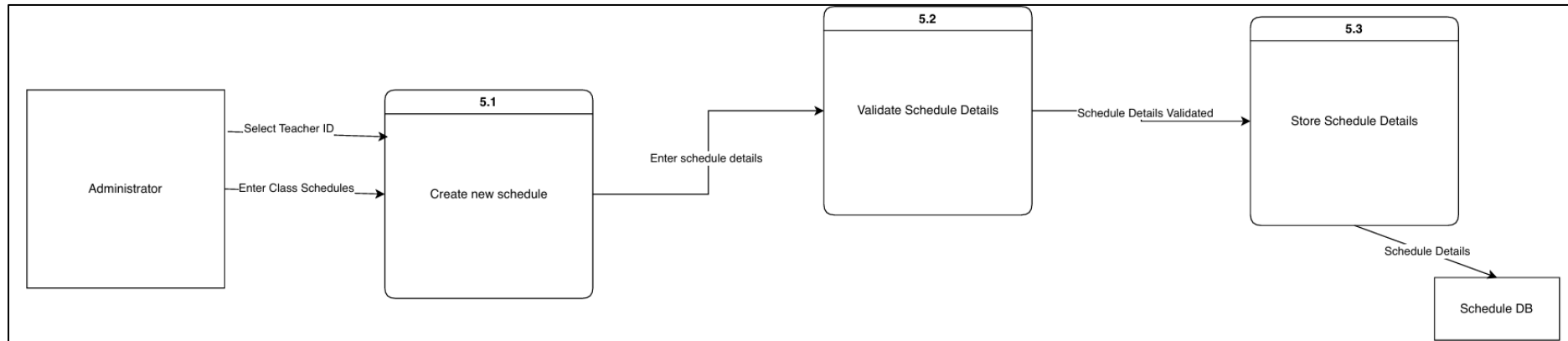
6.1.2.1.3 Update Timetable



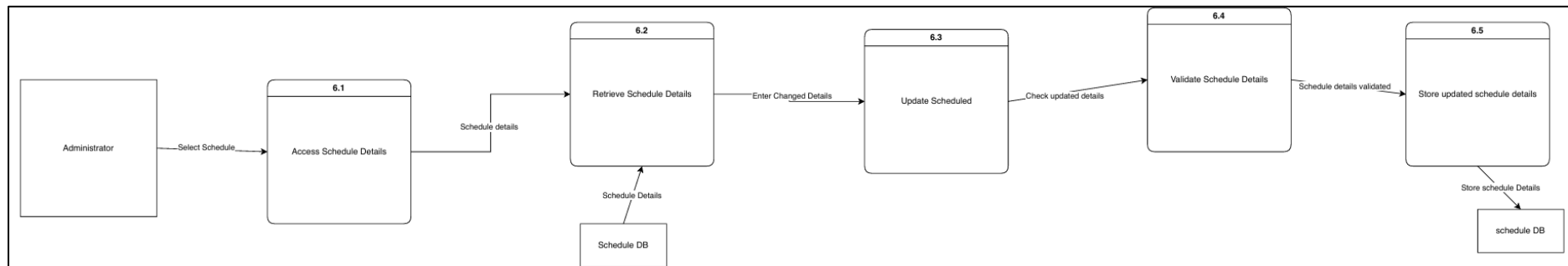
6.1.2.1.4 View Timetable



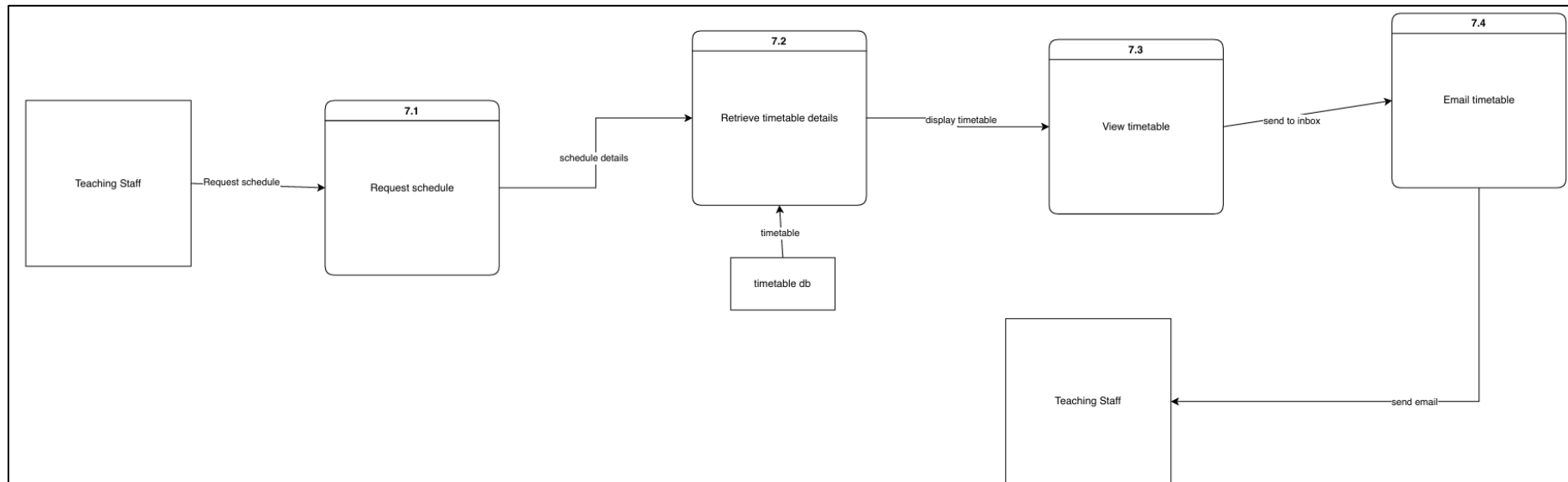
6.1.2.1.5 Create Teacher Schedule



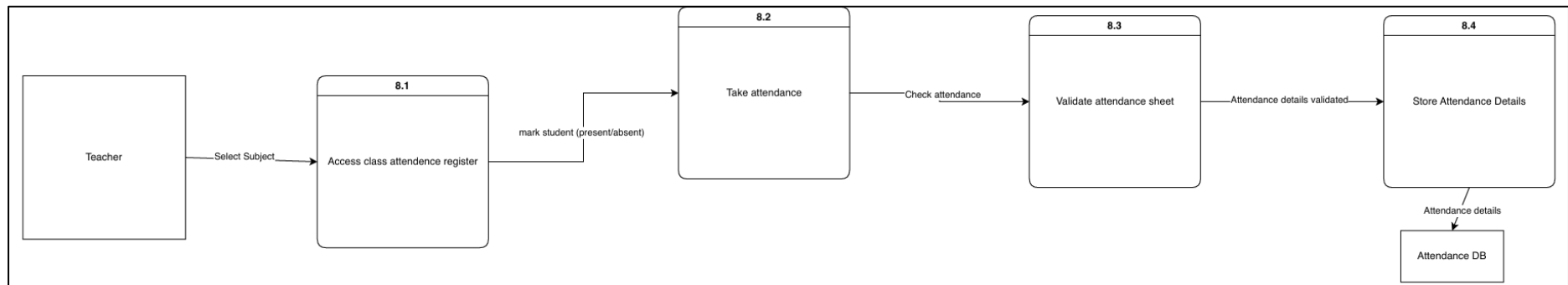
6.1.2.1.6 Update Teacher Schedule



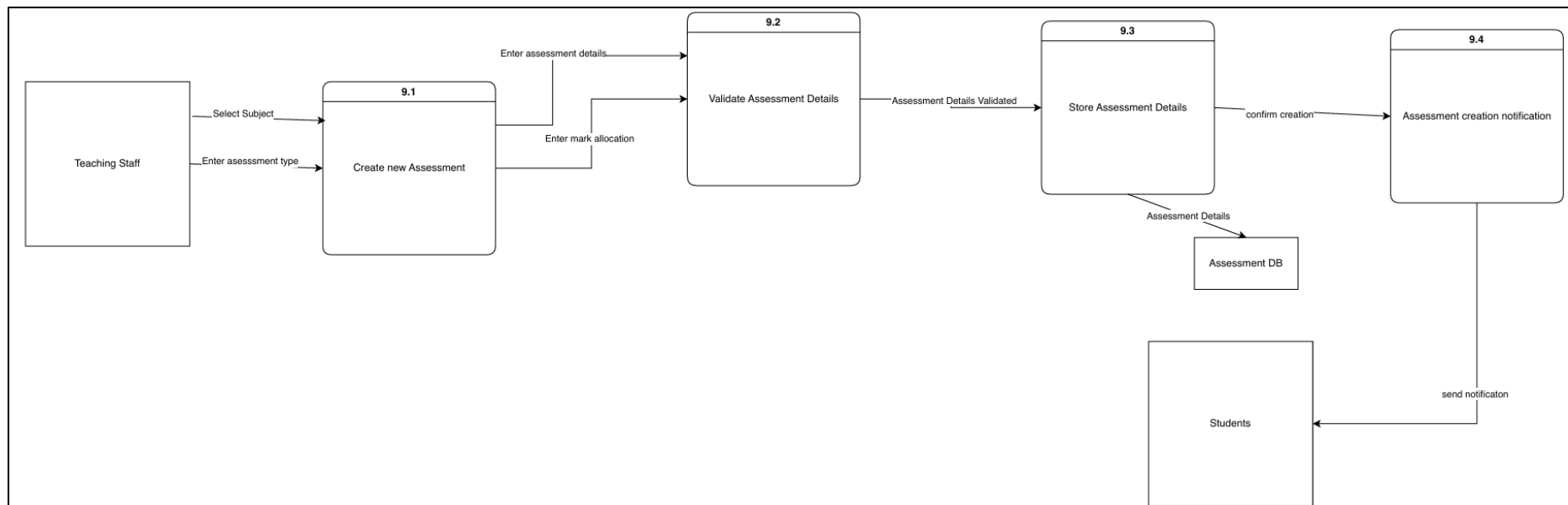
6.1.2.1.7 View Teacher Schedule



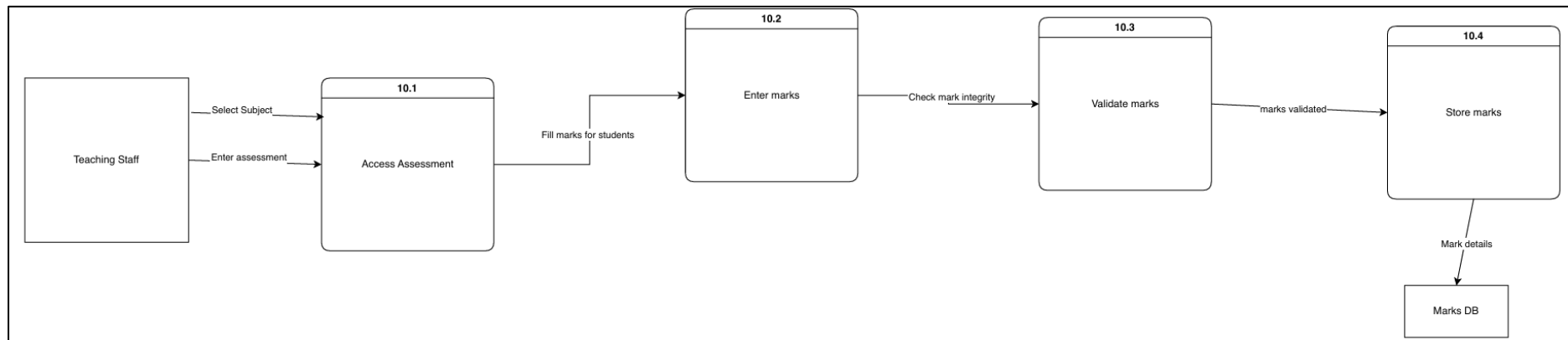
6.1.2.1.8 Take Class Attendance



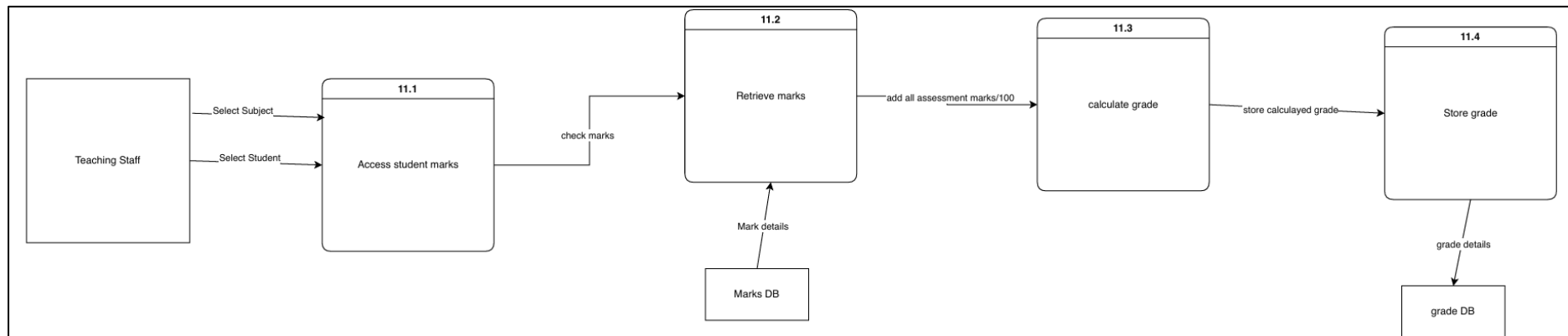
6.1.2.1.9 Create Assessment



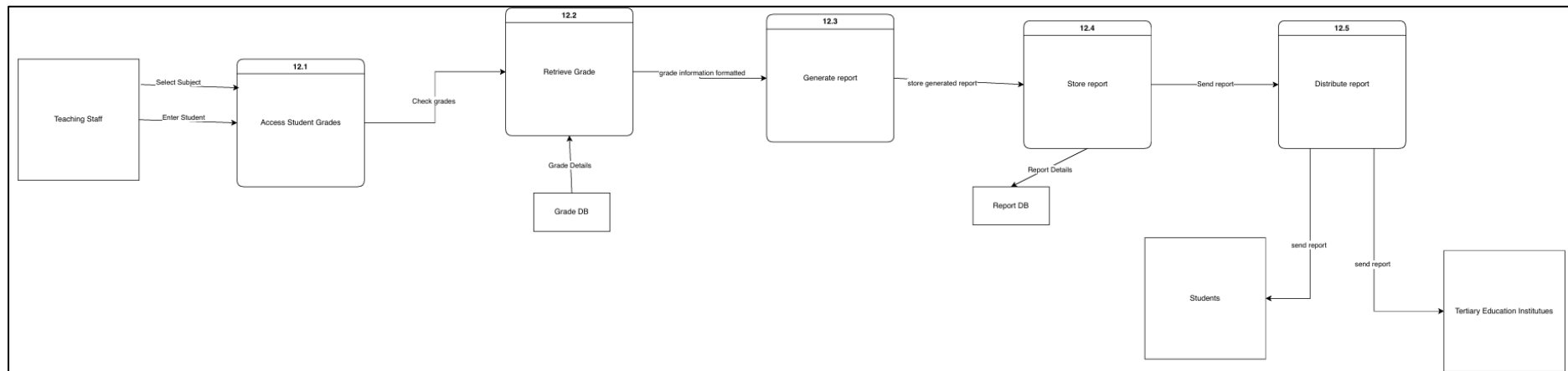
6.1.2.1.10 Enter Student Assessment Marks



6.1.2.1.11 Calculate Student Grade



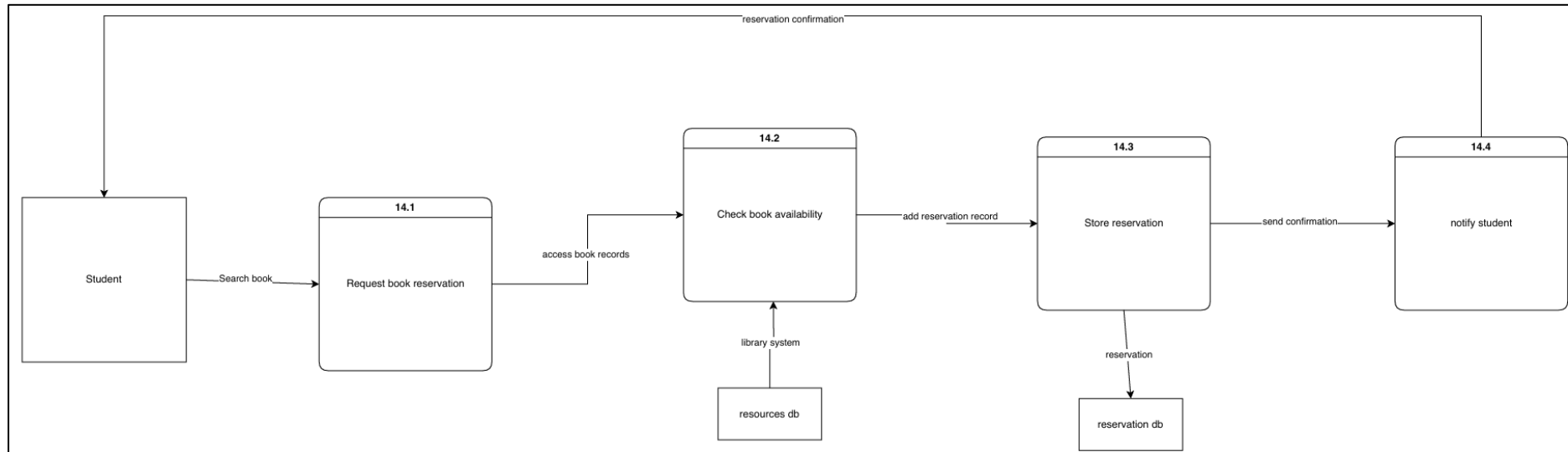
6.1.2.1.12 Generate Student Academic Report



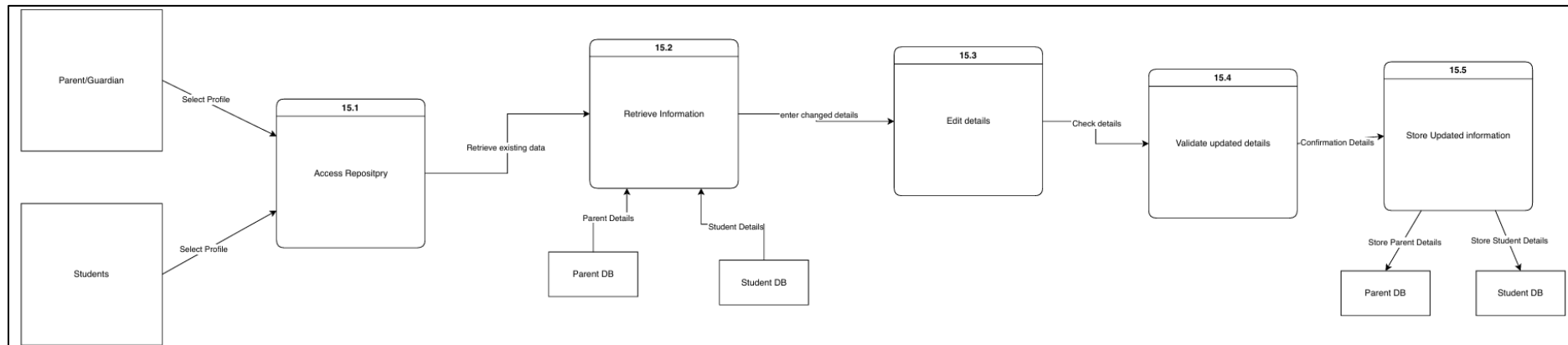
6.1.2.1.13 Browse Library Resources



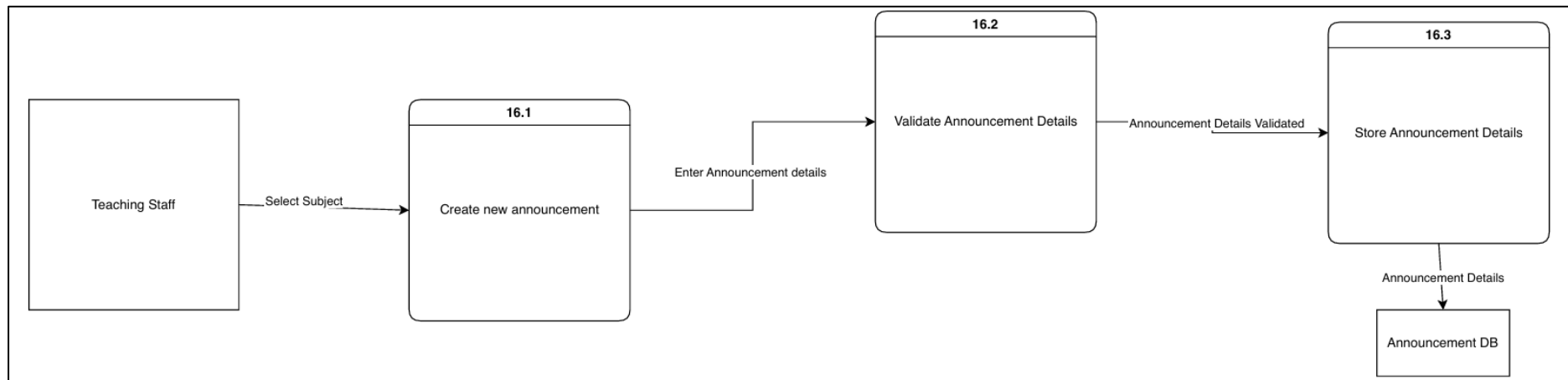
6.1.2.1.14 Library Book Reservation



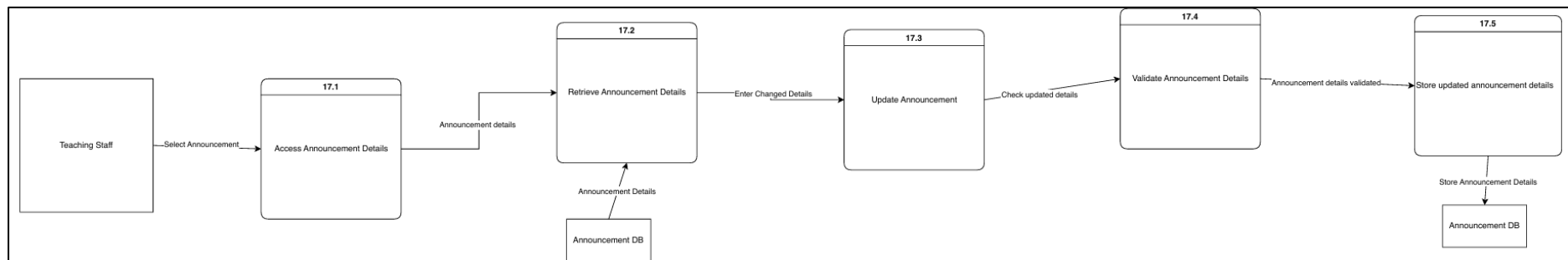
6.1.2.1.15 Change Repository Information



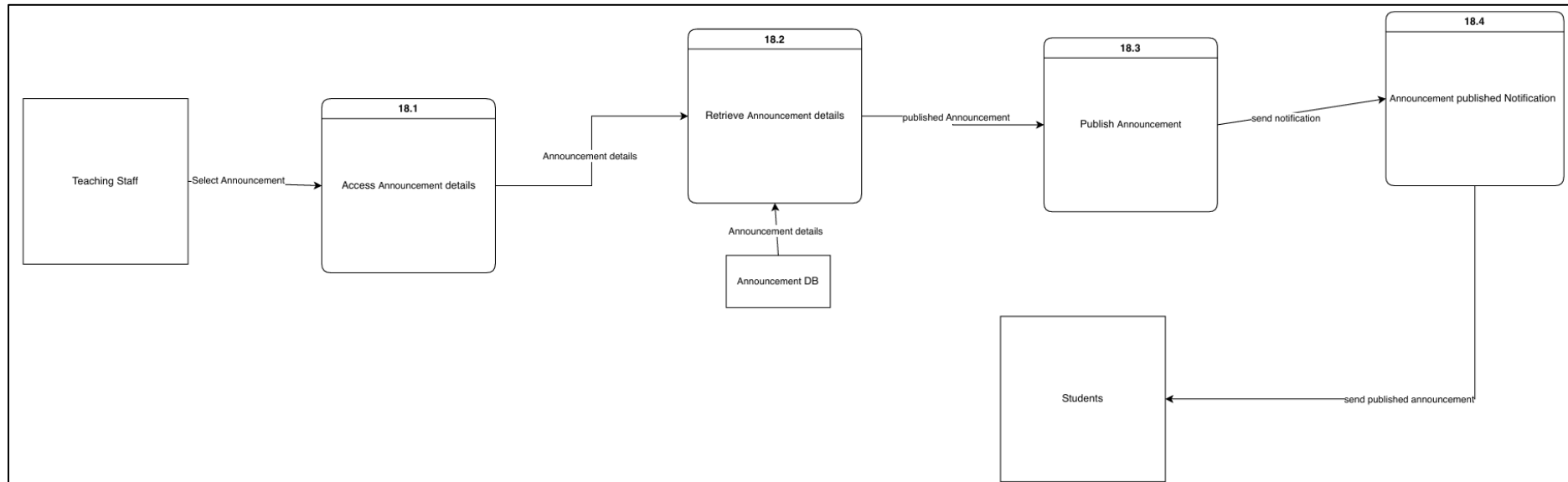
6.1.2.1.16 Create Announcement



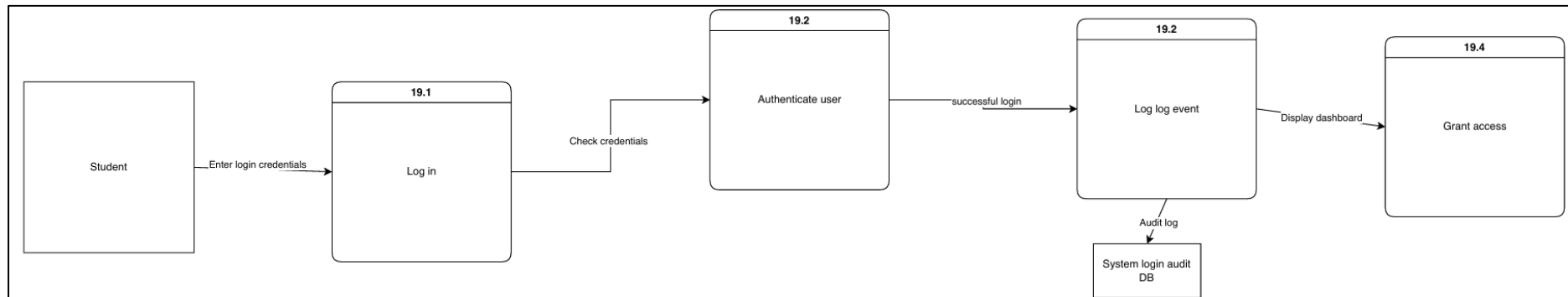
6.1.2.1.17 Update Announcement



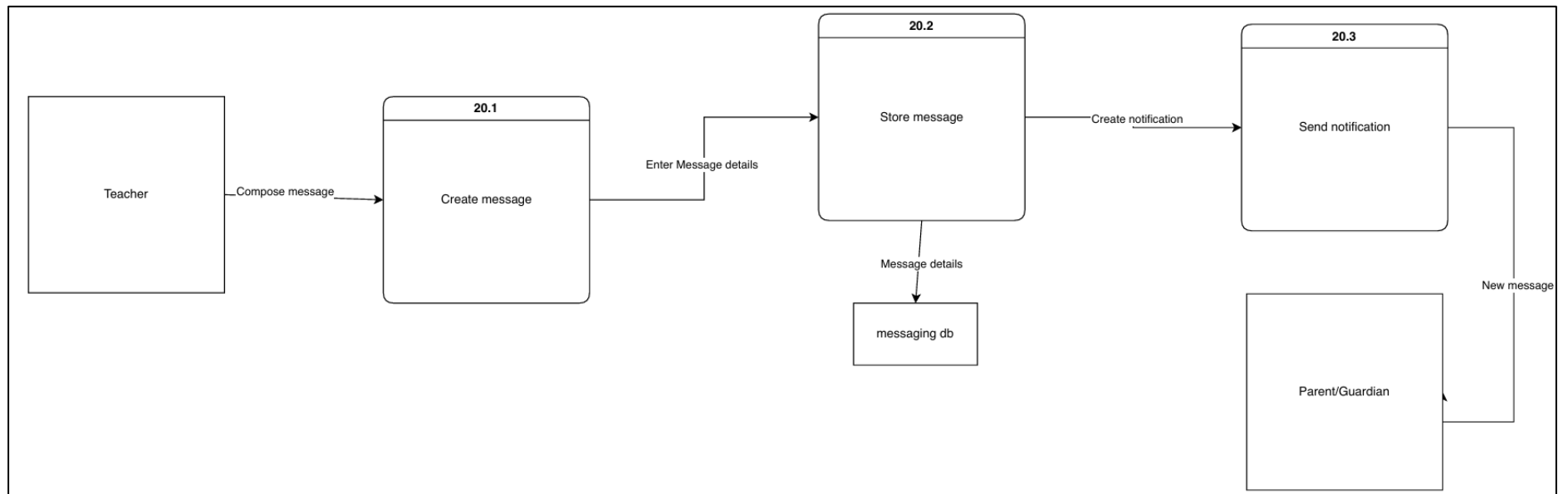
6.1.2.1.18 View Announcement



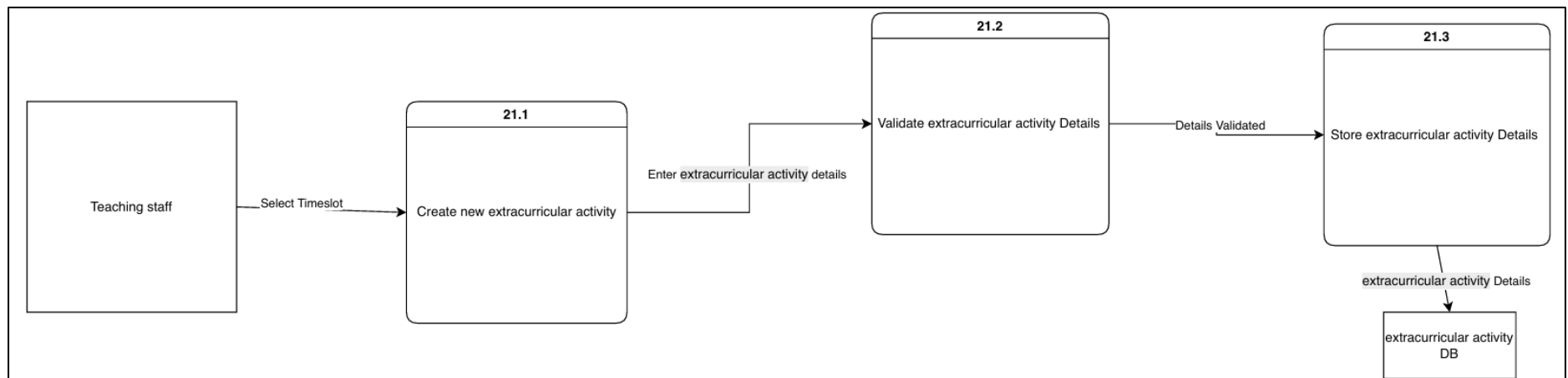
6.1.2.1.19 User Login Auditing



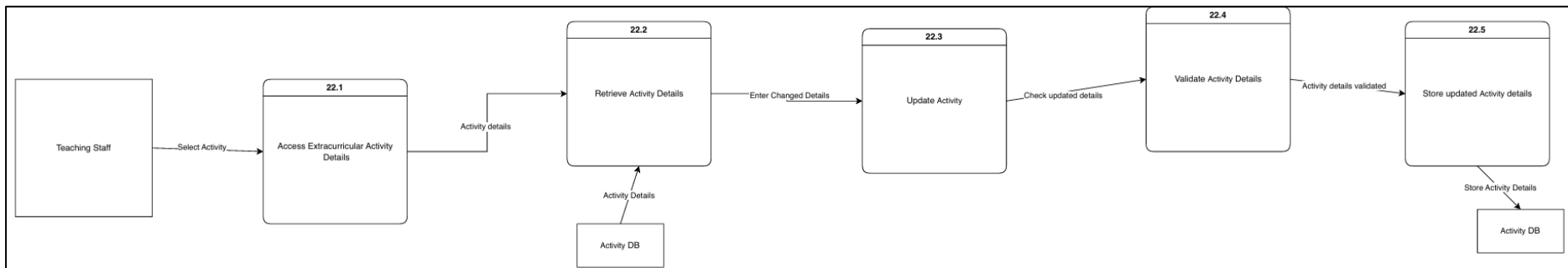
6.1.2.1.20 Create Parent-Teacher Message



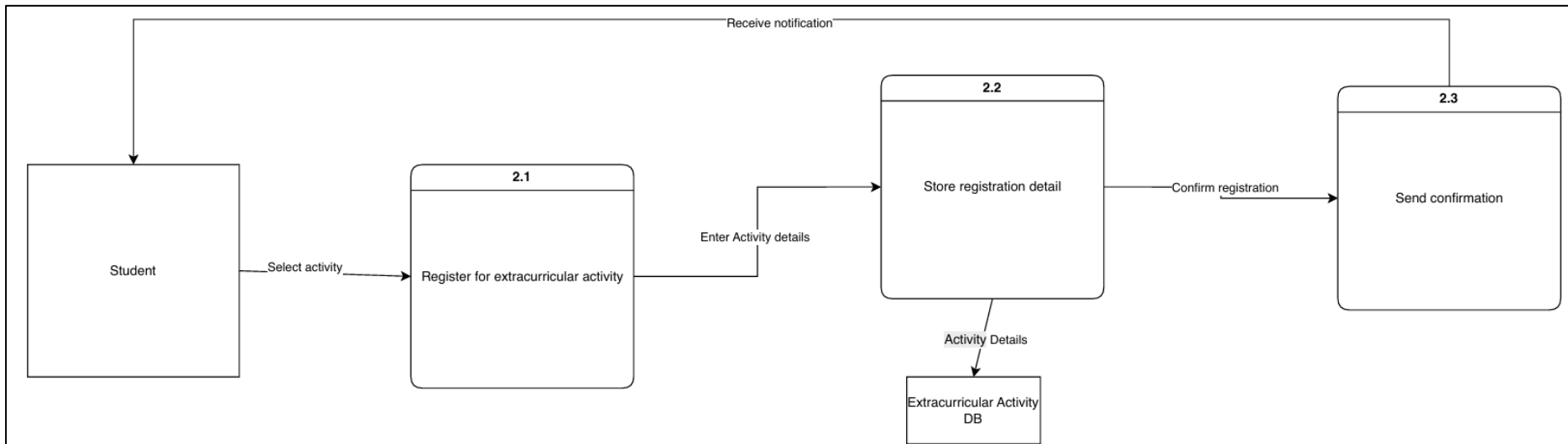
6.1.2.1.21 Create Extracurricular Activity



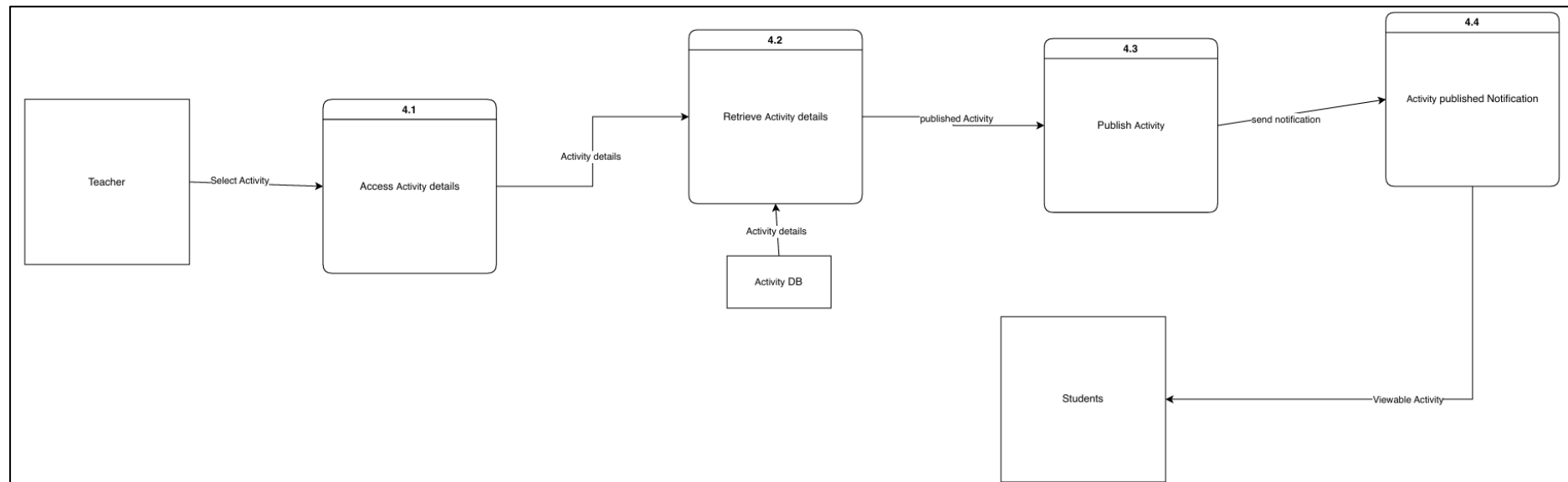
6.1.2.1.22 Update Extracurricular Activity Details



6.1.2.1.23 Register for Extracurricular Activity



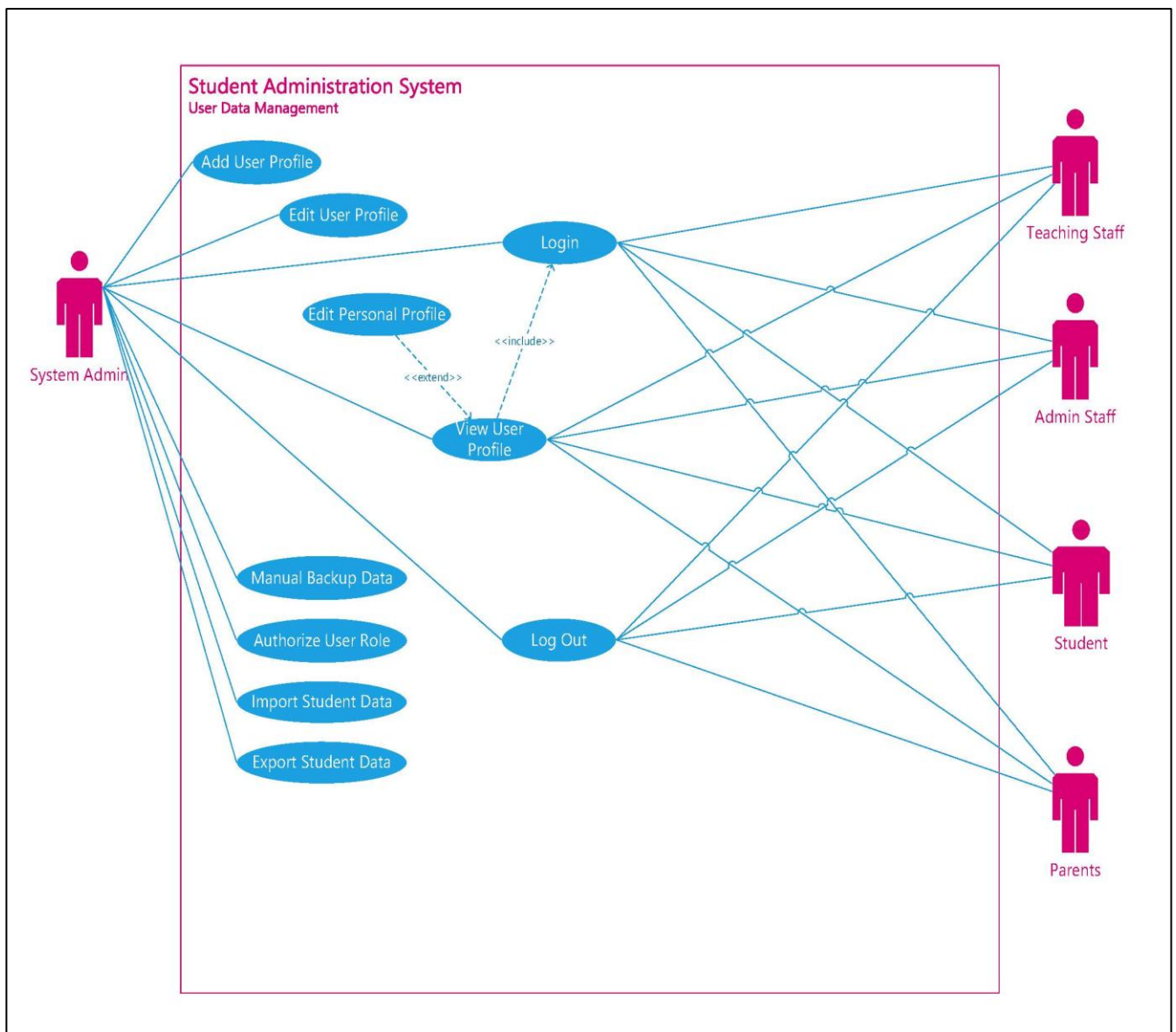
6.1.2.1.24 View Extracurricular Activity



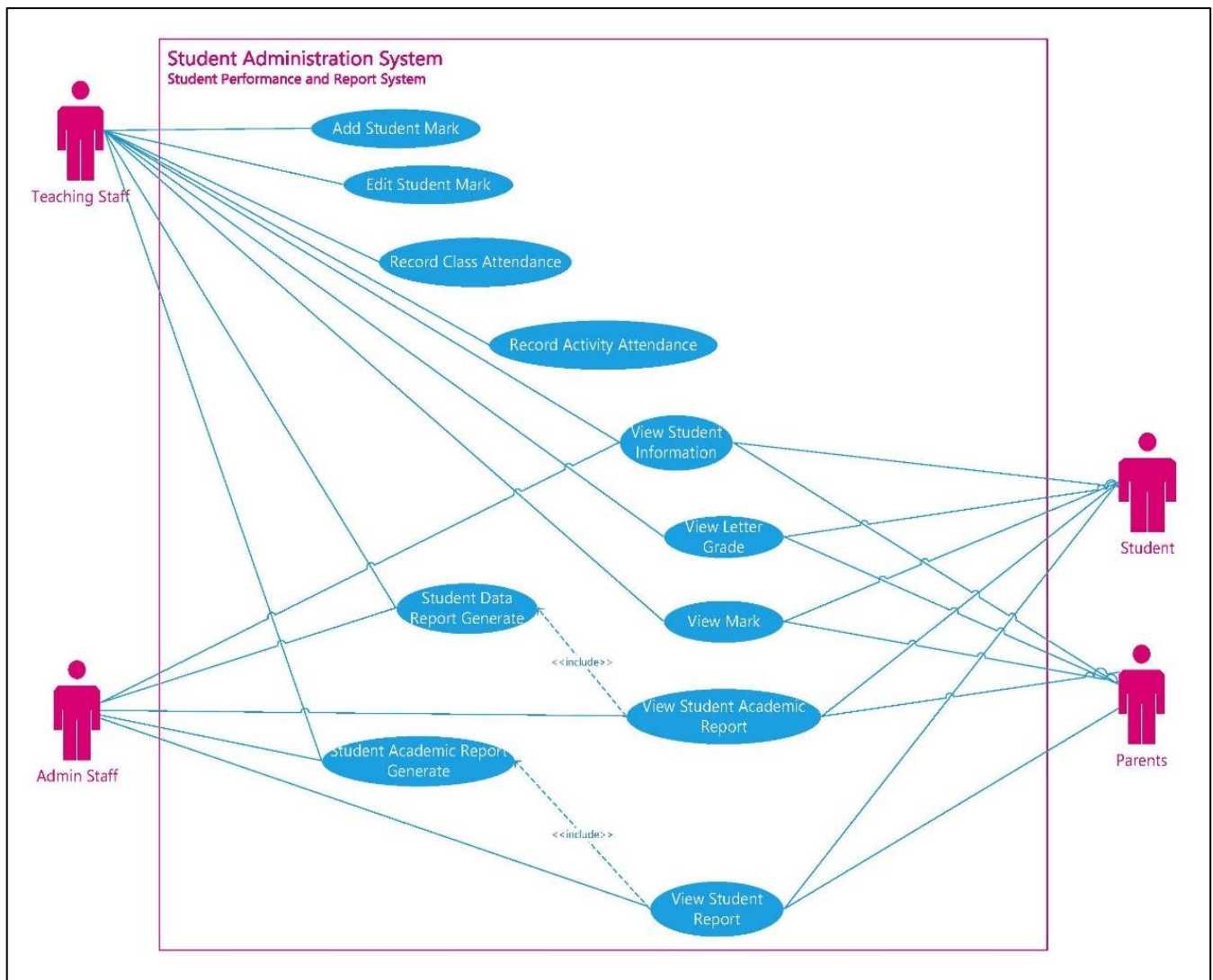
6.2.1.1 Major Use Case Diagram



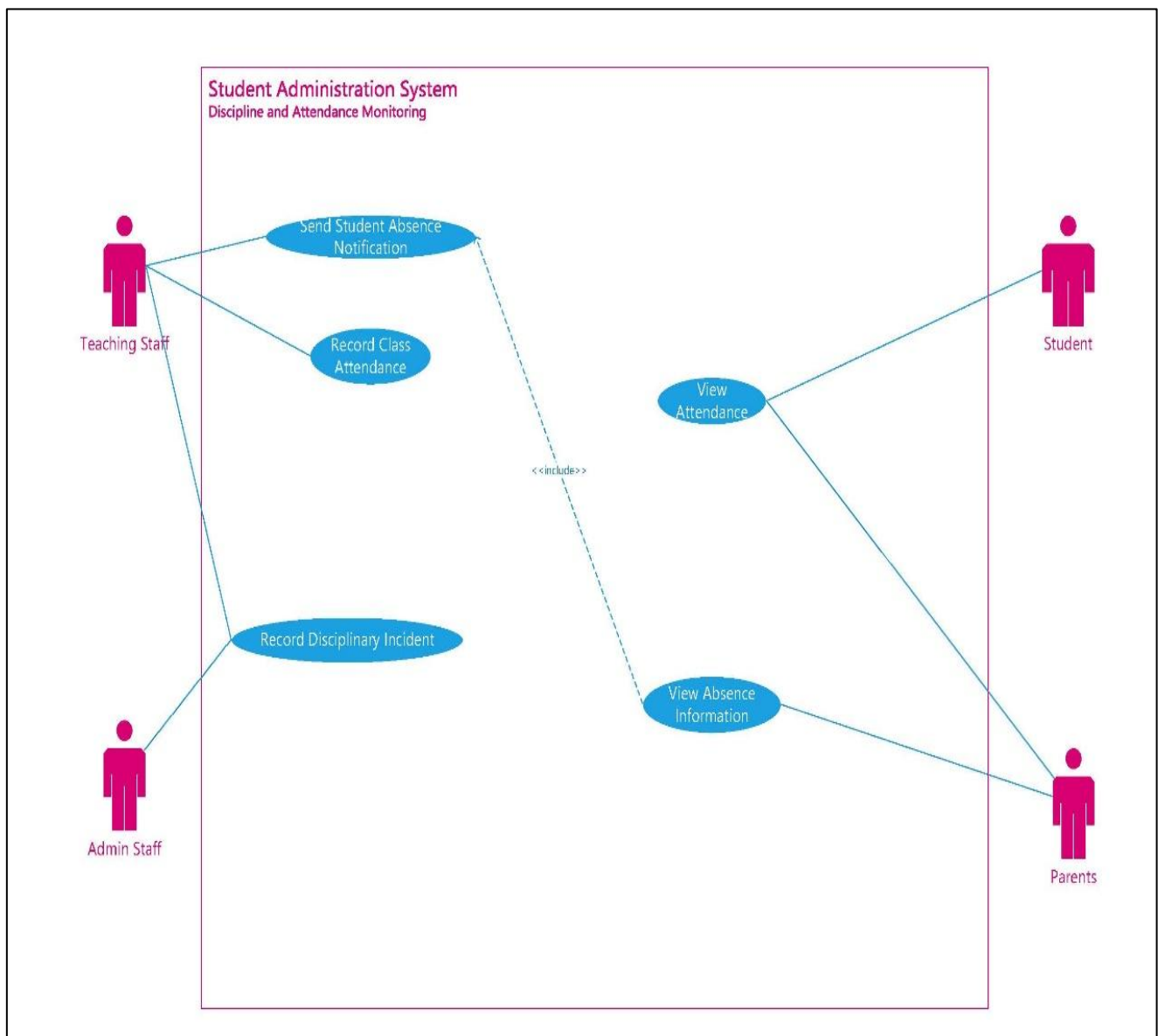
6.2.1.2 User Data Management



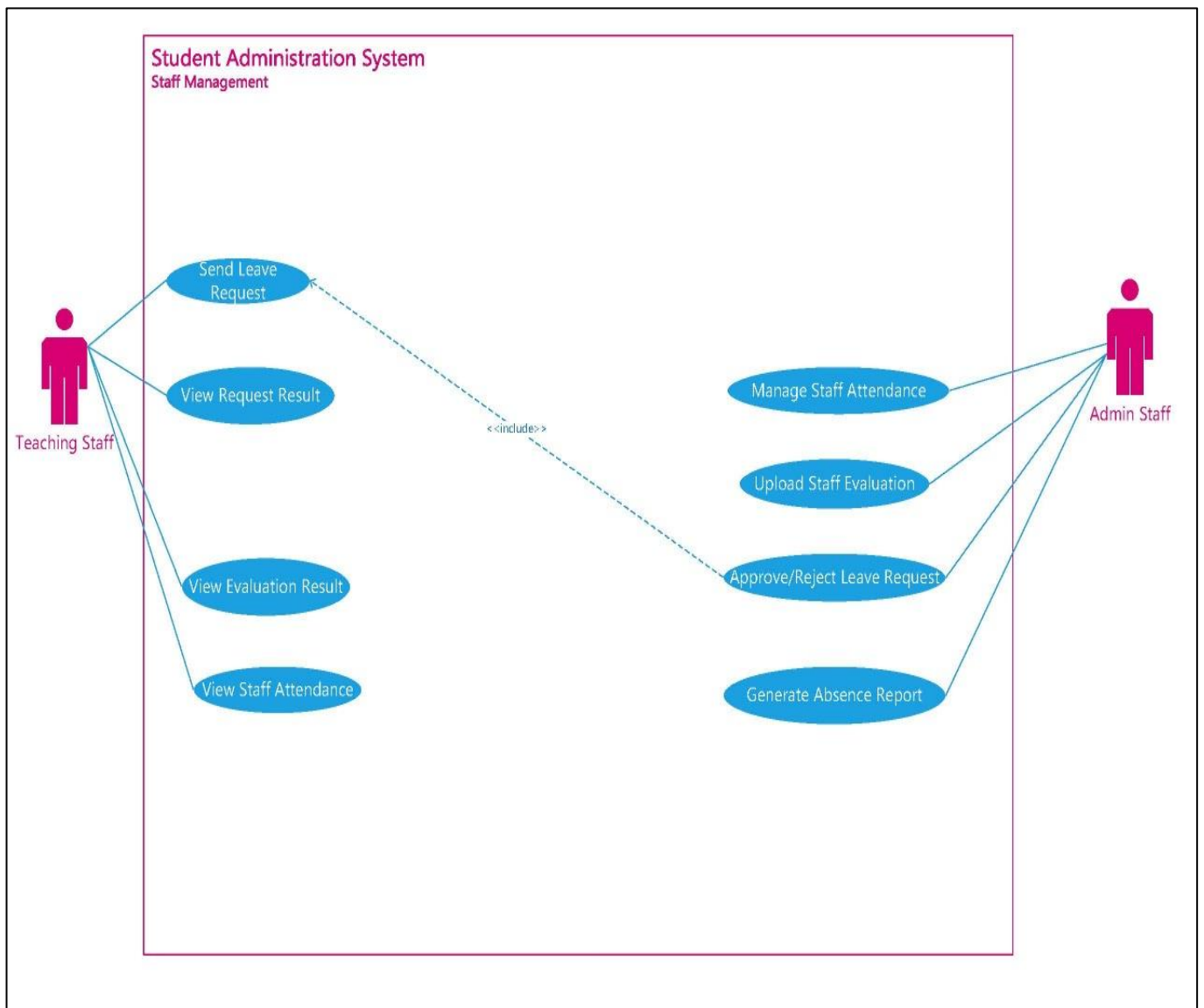
6.2.1.3 Student Performance and Report System



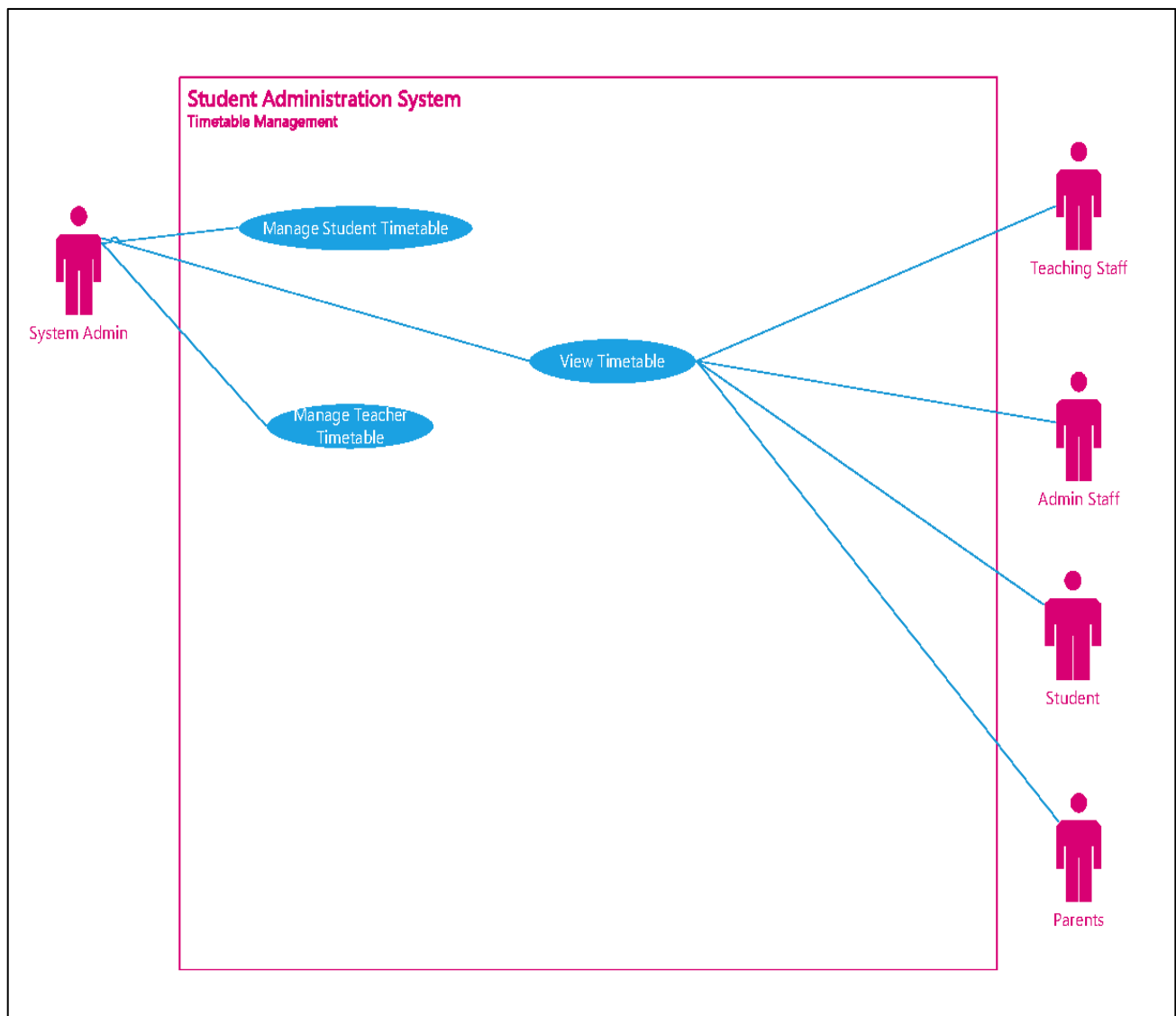
6.2.1.4 Discipline and Attendance Monitoring



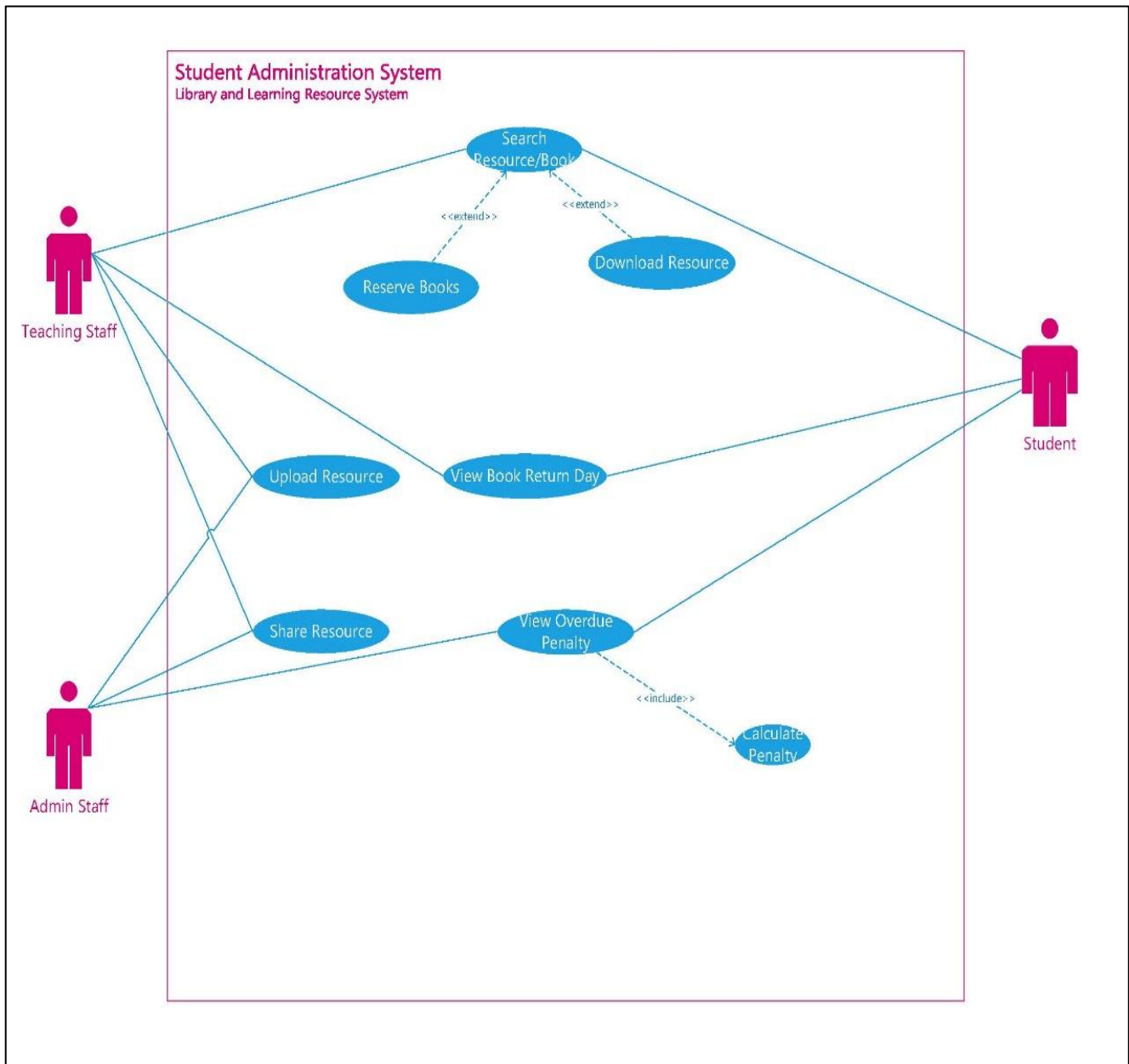
6.2.1.5 Staff Management



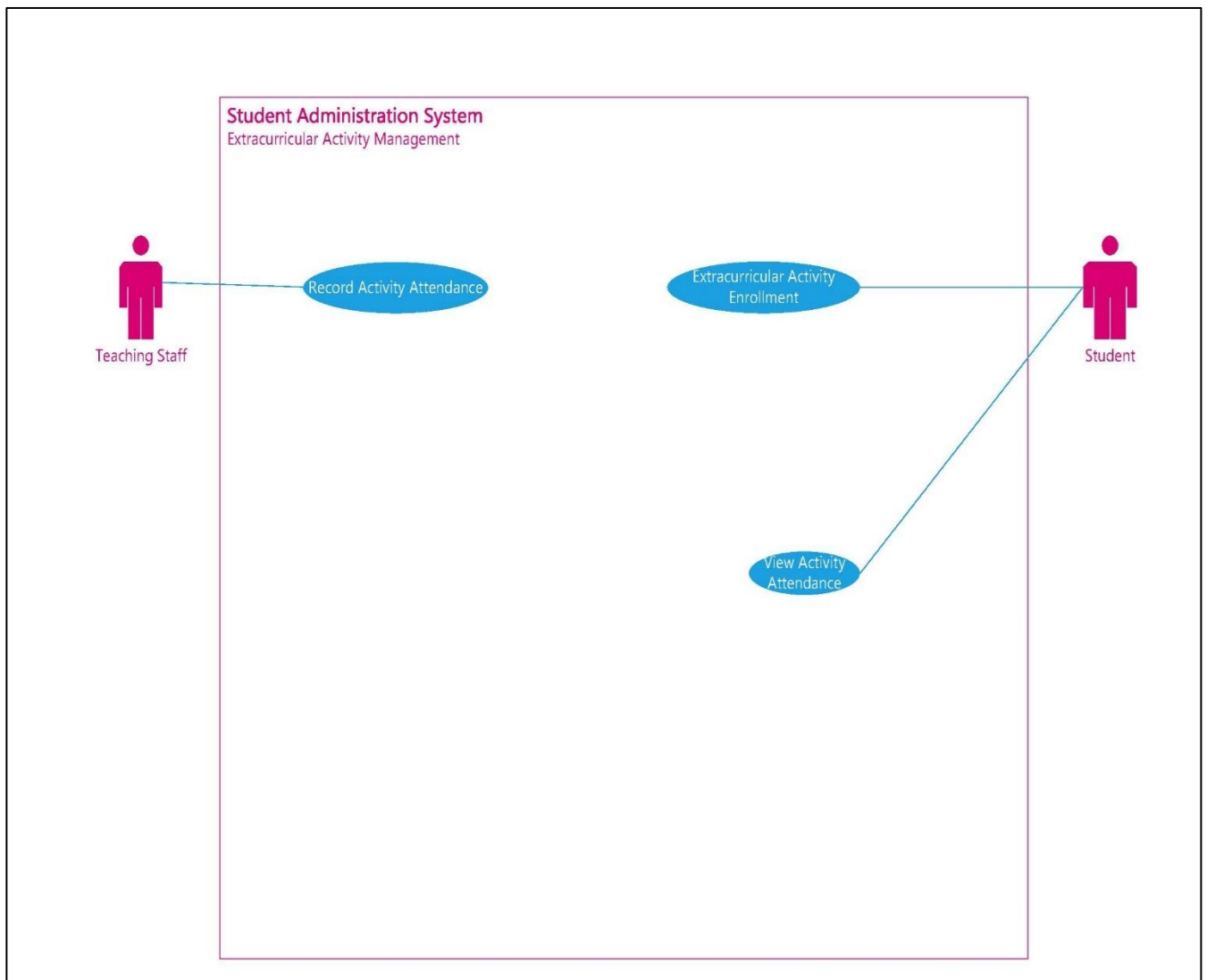
6.2.1.6 Class Timetable Management



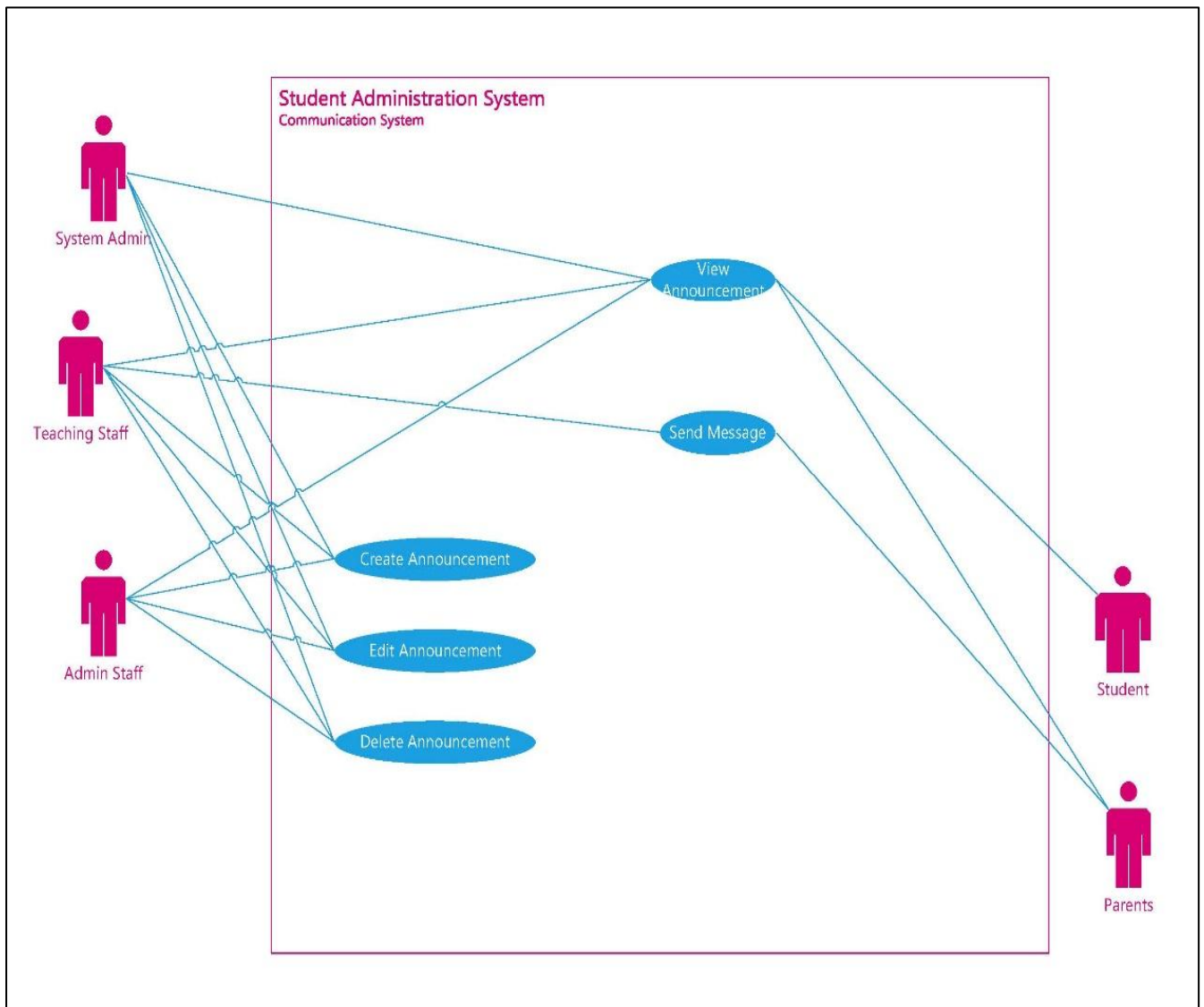
6.2.1.7 Library and Learning Resource System



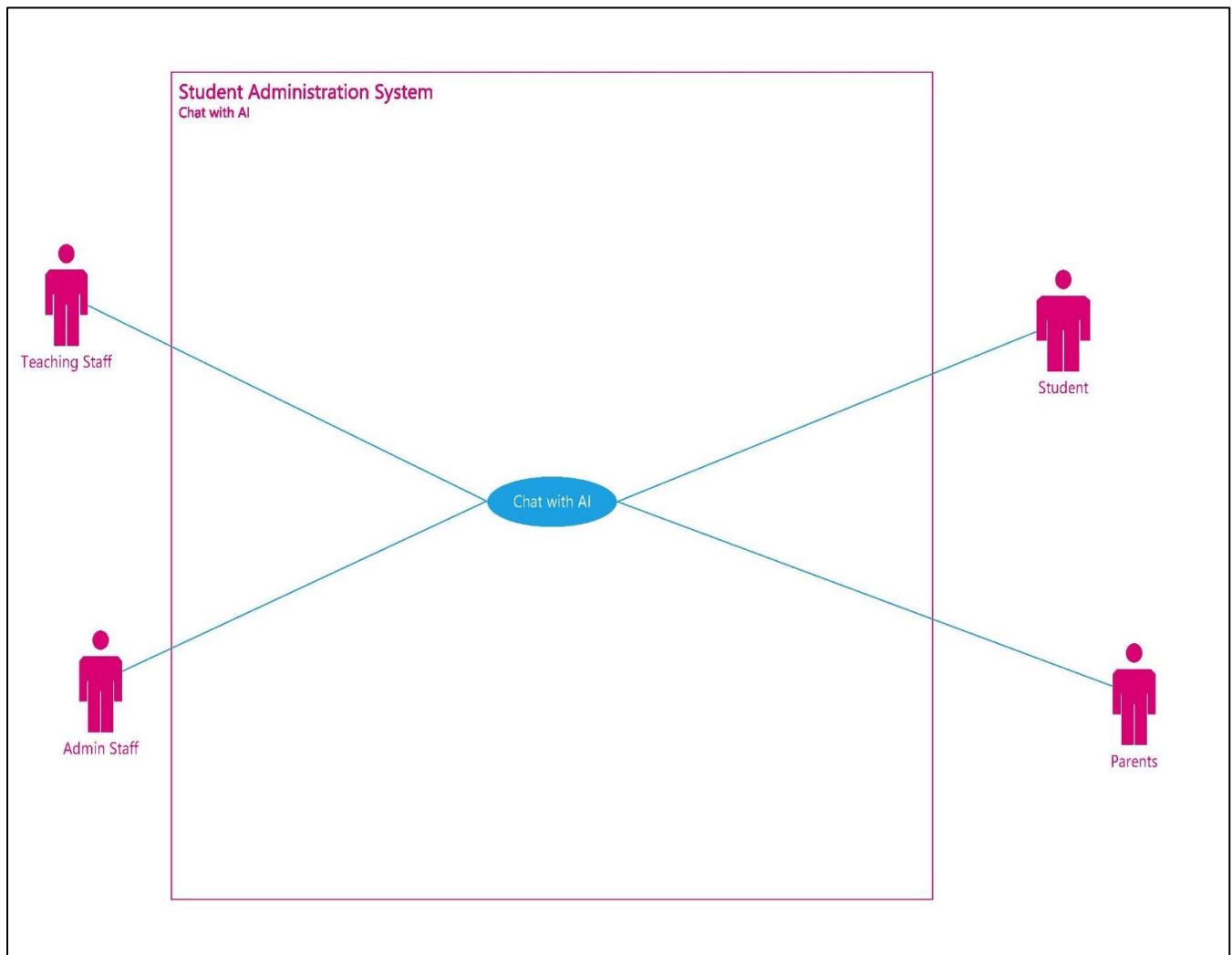
6.2.1.8 Extracurricular Activity Management



6.2.1.9 Communication System



6.2.1.10 Chat with AI



6.2.2 Use Case Descriptions

6.2.2.1 Create new user profile

Use Case Name:	Create new user profile	
Scenario:	Creating a new user profile	
Triggering Event:	The system administrator clicks on the “Create New User” button that is present in the Account Management interface.	
Brief Description:	The system administrator creates the profile/account of new users with the appropriate role. Creation of new user profile is done for the new users of the system	
Actors:	System Administrator	
Stakeholders:	- System Administrator - Administrative Staff - Teaching Staff - Students	
Preconditions:	- The system administrator must already be logged into the system. - The system administrator must already have the authorization to create new profiles.	
Postconditions:	- A new user profile will be created in the database of the system. - The new user profile will have the intended roles and permissions.	
Flow of Activities:	<u>Actor</u> 1. Navigates to the Account Management section. 2. Clicks on the “Create New User” button. 3. Enters the required new user details. 4. Clicks on the “submit” button.	<u>System</u> 1.1 Displays the Account Management section. 2.1 Displays the account creation form. 3.1 Validates the entered details. 4.1 Creates a new user profile in the database and displays account creation message.

Exception Conditions:	• Information to be filled in the required field(s) is missing. • The entered login credentials (email and password) are already associated with an existing account. • Invalid data input on the registration form.
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6.2.2.2 Login

Use Case Name:	Login	
Scenario:	User logs in to the system	
Triggering Event:	• User clicks on the “Login” button on the login page and enters login credentials.	
Brief Description:	The system verifies and authenticates user logins, providing the users with access to the system functionalities based on the user role.	
Actors:	System administrator, administrative staff, teaching staff and students.	
Stakeholders:	• System administrator • Administrative Staff • Teaching Staff • Students	
Preconditions:	• The user profile/account must already exist in the system.	
Postconditions:	• The user is verified and granted access to use the system. • The system creates a secure session for the logged in user(s). • The system creates a successful login record.	
Flow of Activities:	<u>Actor</u> 1. Clicks on the “Login” button. 2. Enters login credentials. 3. Clicks on the “submit” button.	<u>System</u> 1.1 Displays the login page. 2.1 Validates login credentials. 3.1 Verifies account status and redirects user to the home dashboard.

Exception Conditions:	• Trying to log in without filling the required fields. • Incorrect login credentials being entered.
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6.2.2.3 Record Student Medical Information

Use Case Name:	Record Student Medical Information			
Scenario:	Creating a new student medical information			
Triggering Event:	• A new student enrolment and registration. • Teaching staff accesses the stored medical records and clicks the “Add new record” button.			
Brief Description:	The teaching staff creates medical record for a new student. The record contains information such as health conditions, allergies and medications which is securely kept as this is confidential data.			
Actors:	Teaching staff			
Stakeholders:	• Teaching staff • Students			
Preconditions:	• Student profile must already exist in the system. • Teaching staff must have authorization to access student medical records and add a new record.			
Postconditions:	• New medical record is created and saved in the database. • System creates a log of the successful creation of a new medical record.			
Flow of Activities:	<table><tr><td><u>Actor</u> 1. Clicks on “Student profiles” button. 2. Clicks on “Medical Information” button. 3. Enters medical information. 4.Clicks on “submit” button.</td><td><u>System</u> 1.1 Displays the student profiles page. 2.1 Checks for permissions. 3.1 Validates entered information. 4.1 Creates and stores the new medical record.</td></tr></table>		<u>Actor</u> 1. Clicks on “Student profiles” button. 2. Clicks on “Medical Information” button. 3. Enters medical information. 4.Clicks on “submit” button.	<u>System</u> 1.1 Displays the student profiles page. 2.1 Checks for permissions. 3.1 Validates entered information. 4.1 Creates and stores the new medical record.
<u>Actor</u> 1. Clicks on “Student profiles” button. 2. Clicks on “Medical Information” button. 3. Enters medical information. 4.Clicks on “submit” button.	<u>System</u> 1.1 Displays the student profiles page. 2.1 Checks for permissions. 3.1 Validates entered information. 4.1 Creates and stores the new medical record.			

Exception Conditions:	• Restriction provided by the system on unauthorized users trying to create new medical record. • Student profile does not exist in the system database.
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6.2.2.4 Import Student Data

Use Case Name:	Import Student Data	
Scenario:	Importing student data from another school	
Triggering Event:	• System administrator clicks on the “Import Data” option, present on the Data Management page.	
Brief Description:	A student data import module is present in the system that enables importing of student information from external sources, for example, from other schools. This is mainly used for student(s) who have joined another school in the middle of a school term.	
Actors:	System administrator	
Stakeholders:	• System administrator • Student	
Preconditions:	• The system administrator must already be logged in to the system. • The data to be imported must be compatible. • The system must have enough resources to import the data.	
Postconditions:	• The data of a newly joined student is imported successfully. • The system saves the new record in the database.	
Flow of Activities:	<u>Actor</u> 1. Navigates to the “Data Management” page. 2. Clicks on “Import Data” button. 3. Selects the file to be imported.	<u>System</u> 1.1 Displays the “Data Management” page 2.1 Displays the import section. 3.1 Checks file size.

	4. Clicks on “Done” button	4.1 Validates the file content, performs importation and displays either “Import successful” or “Import unsuccessful” message.
Exception Conditions:	- Importing data which contains invalid format will prompt an “Import error” message. - Importing a duplicate data will prompt either to replace or skip.	

6.2.2.5 Enter Student Marks

Use Case Name:	Enter Student Marks			
Scenario:	Entering subject marks for students			
Triggering Event:	The teaching staff selects the “Enter Marks” option to initiate mark entry of respective students.			
Brief Description:	Teaching staff recording marks for the assessments done by students in their respective subjects.			
Actors:	Teaching staff			
Stakeholders:	- Teaching staff - Students			
Preconditions:	- The teaching staff must already be logged in to the system. - Student whose mark is being entered must already have a profile in the system.			
Postconditions:	- System records the marks in the database. - System automatically calculates class average and rankings of students after all the marks are entered.			
Flow of Activities:	<table><tr><td><u>Actor</u></td><td><u>System</u></td></tr></table>		<u>Actor</u>	<u>System</u>
<u>Actor</u>	<u>System</u>			

	1. Navigates to assessment marks interface. 2. Selects the year level, subject name and name of the assessment. 3. Enters the marks. 4. Clicks on the "Save" button.	1.1 Displays the assessment marks interface. 2.1 Displays all the year levels in the school, student and subject list. 3.1 Validates the marks being entered. 4.1 Saves the record in the database and displays confirmation message.
Exception Conditions:	- Entered data is in invalid format for example, alphabets entered instead of numbers. - Mark entered is greater than 100 or less than 0.	

6.2.2.6 View Student Grades

Use Case Name:	View Student Grades
Scenario:	Viewing grades of enrolled students
Triggering Event:	The system administrator, administrative staff, teaching staff, students or parents clicks on the view grades option.
Brief Description:	Viewing of grades of all the students will be allowed for the system administrator, administrative staff and the teaching staff. Students will only be allowed to view their own grades, and the parents will only be allowed to see the grades of their own child(ren).
Actors:	System administrator, administrative staff, teaching staff, students and parents
Stakeholders:	- System administrator - Administrative staff - Teaching staff - Students - Parents

Preconditions:	- The system administrator, administrative staff, teaching staff, students or the parent(s) must already be logged in to the system. - Student must have already completed the assessment in order for the grade to be available.	
Postconditions:	Grade is displayed under the “View grades” option.	
Flow of Activities:	<u>Actor</u> 1. Navigates to the View grades option.	<u>System</u> 1.1 Checks for user role and displays grade if available. 1.2 Provides an option to also print grades.
Exception Conditions:	Student account is disabled or deleted from the system.	

6.2.2.7 Generate academic report

Use Case Name:	Generate academic report
Scenario:	Generating academic report for individual students.
Triggering Event:	- Automated generation of academic reports at the end of every term. - Request from student(s) moving to another school.
Brief Description:	Generation of report which contains marks of the assessments done by the particular student, also containing the feedback from the respective subject teachers and overall behaviour in class.
Actors:	System administrator and teaching staff
Stakeholders:	- System administrator - Teaching staff - Student
Preconditions:	Student profile and other relevant details must already be present in the system.

	Authorization to generate academic reports must be given to the user.								
Postconditions:	- Academic report being requested for is successfully generated. - Academic report is available to print or download.								
Flow of Activities:	<table border="1"> <thead> <tr> <th><u>Actor</u></th><th><u>System</u></th></tr> </thead> <tbody> <tr> <td>1. Navigates to “Generate reports” interface.</td><td>1.1 Displays the “Generate reports” page.</td></tr> <tr> <td>2. Selects report type.</td><td>2.1 Validates report type.</td></tr> <tr> <td>3. Clicks “Generate report” button.</td><td>3.1 Generates and provides access to the academic report.</td></tr> </tbody> </table>	<u>Actor</u>	<u>System</u>	1. Navigates to “Generate reports” interface.	1.1 Displays the “Generate reports” page.	2. Selects report type.	2.1 Validates report type.	3. Clicks “Generate report” button.	3.1 Generates and provides access to the academic report.
<u>Actor</u>	<u>System</u>								
1. Navigates to “Generate reports” interface.	1.1 Displays the “Generate reports” page.								
2. Selects report type.	2.1 Validates report type.								
3. Clicks “Generate report” button.	3.1 Generates and provides access to the academic report.								
Exception Conditions:	Some marks for assessments are not present.								

6.2.2.8 Record class attendance

Use Case Name:	Record class attendance
Scenario:	Recording daily class attendance
Triggering Event:	Teaching staff clicks on “record class attendance” option.
Brief Description:	Teaching staff are able to create daily class attendance records.
Actors:	Teaching staff
Stakeholders:	- Teaching staff - Students
Preconditions:	- Teaching staff must already be logged into the system - Class list must be present in the system
Postconditions:	- Attendance of students will be updated - System saves the record into the database

Flow of Activities:	<u>Actor</u> 1. Logs in to the system. 2. Navigates to the attendance section. 3. Chooses the year level 4. Marks students as absent or present. 5. Clicks on the “Save attendance” button.	<u>System</u> 1.1 Displays home page 2.1 Displays the attendance interface. 3.1 Displays the list of all the student for that particular year level. 4.1 Validates and saves attendance marks. 5.1 Displays “Attendance saved” message.
Exception Conditions:	- Loss of internet connection will prompt the system to save the attendance data locally and sync when the connection reestablishes. - Details of certain students not updated in the class list, for instance a newly enrolled student.	

6.2.2.9 Notify attendance issues

Use Case Name:	Notify attendance issues
Scenario:	Sending notification/message regarding issues of students' attendance.
Triggering Event:	- Students being absent for more than four consecutive days without updating the teaching staff. - Teaching staff sends attendance message to parents.
Brief Description:	Teachers sending attendance issue notification/message to the students' parent(s).
Actors:	Teaching staff and parent(s)
Stakeholders:	- Teaching staff - Students - Parents
Preconditions:	The teaching staff must already be logged in to the system to send the message.

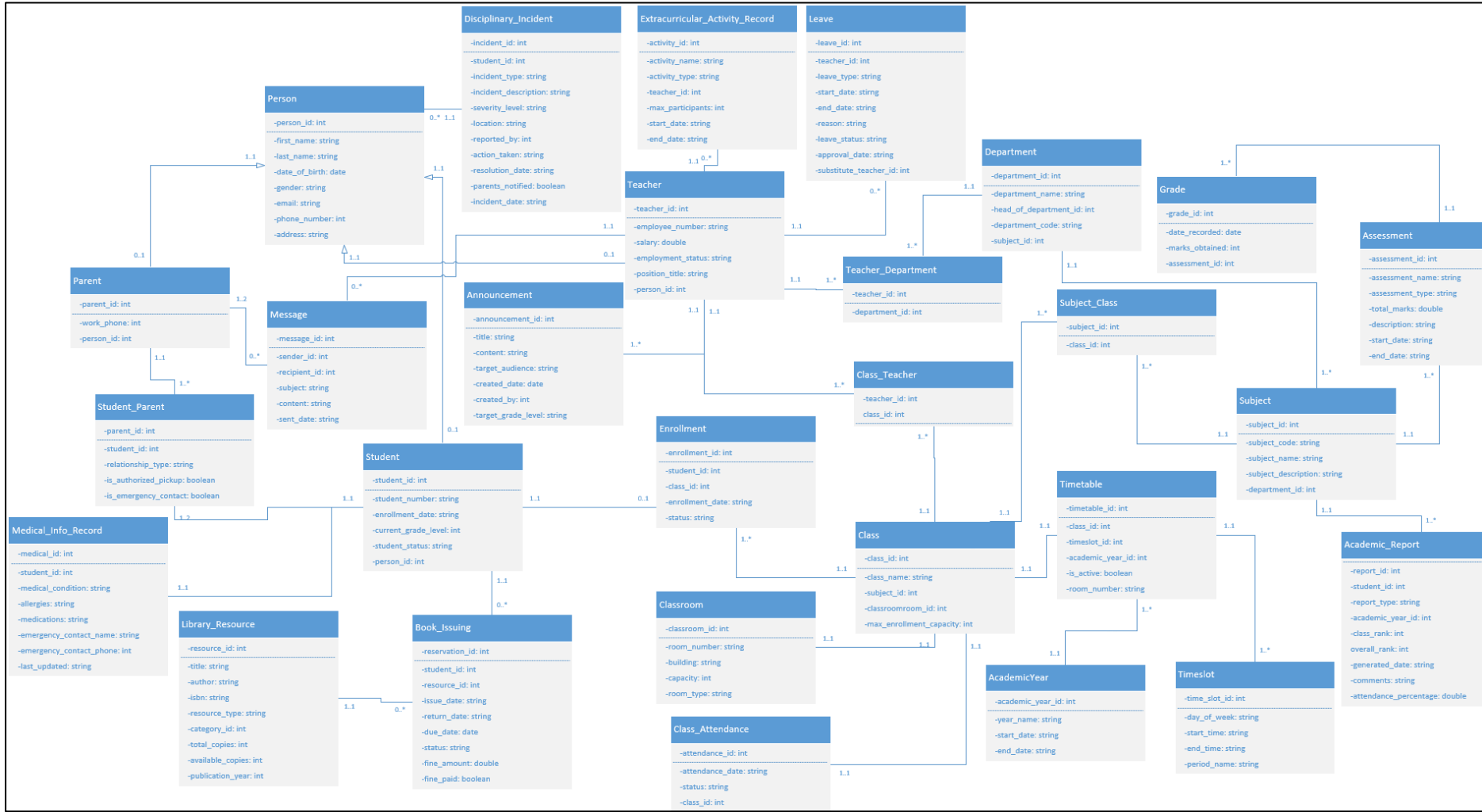
	Parent(s) must already be logged in to the system to receive the message.	
Postconditions:	Message is received by the parent(s) about their child(ren)'s attendance issues.	
Flow of Activities:	<u>Actor</u> 1. Logs in to the system. 2. Navigates to the attendance interface. 3. Selects highlighted students to get parent contact details. 4. Selects notify parents' option. 5. Types the message and chooses the notification delivery options (mobile app notification/email/SMS). 6. Clicks on the "Send message" button.	<u>6.System</u> 1.1 Redirects and displays the home page. 2.1 Displays the attendance page, highlighting the students who have been absent consecutively for 4 days. 3.1 Displays all the details of the highlighted student(s), including parent details. 4.1 Displays the form to fill in the details including message delivery options. 5.1 Validates message and ensuring the required fields are filled. 6.1 Delivers the notification to parent(s) and creates a log.
Exception Conditions:	Loss of internet connection while sending the notification will prompt the system to automatically retry when connection reestablishes.	

6.2.2.10 Create class timetable

Use Case Name:	Create class timetable	
Scenario:	Creating a new class timetable	
Triggering Event:	• The starting of a new academic term. • Addition of new subjects into the curriculum. • System administrator selects “create new class timetable” option.	
Brief Description:	The system administrator is able to create a new timetable for the intended class.	
Actors:	System administrator	
Stakeholders:	• System administrator	
Preconditions:	• The system administrator must already be logged in to the system. • System administrator must have the authorization to create new class timetables.	
Postconditions:	• New class timetable is created. • The users of the system are able to view the created timetable. • The system saves the timetable into the database.	
Flow of Activities:	<u>Actor</u> 1. Logs in to the system. 2. Navigates to the timetable management section. 3. Clicks on the “Create new class timetable” button. 4. Enters the required details.	<u>System</u> 1.1 Redirects to the home page. 2.1 Displays the timetable management page. 3.1 Displays the timetable template to enter required details. 4.1 Validates the entered information, verifies for conflicts.

	5. Checks the drafted timetable. 6. Clicks on the “Save timetable” button.	5.1 Displays the drafted timetable. 6.1 Creates the new class timetable and saves it into the database, together with make it available to the intended users.
Exception Conditions:	• Unavailability of teaching staff or classrooms. • Required fields not filled while creating a new class timetable.	

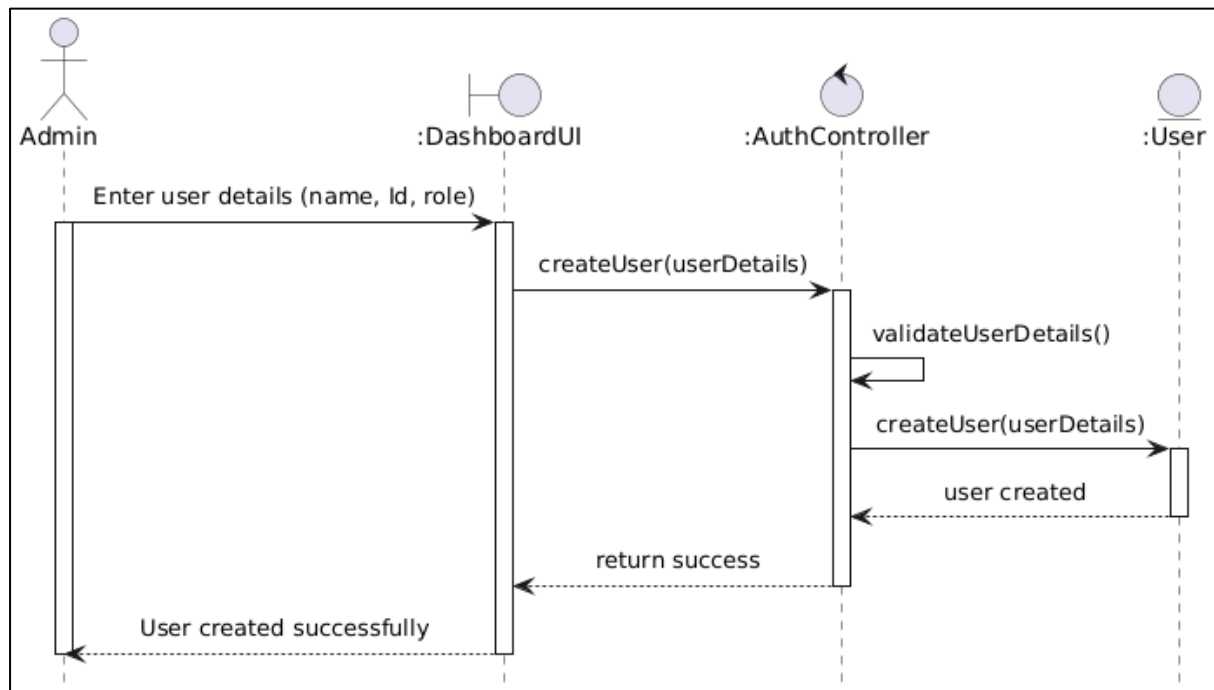
6.2.3 Domain Model Class diagram



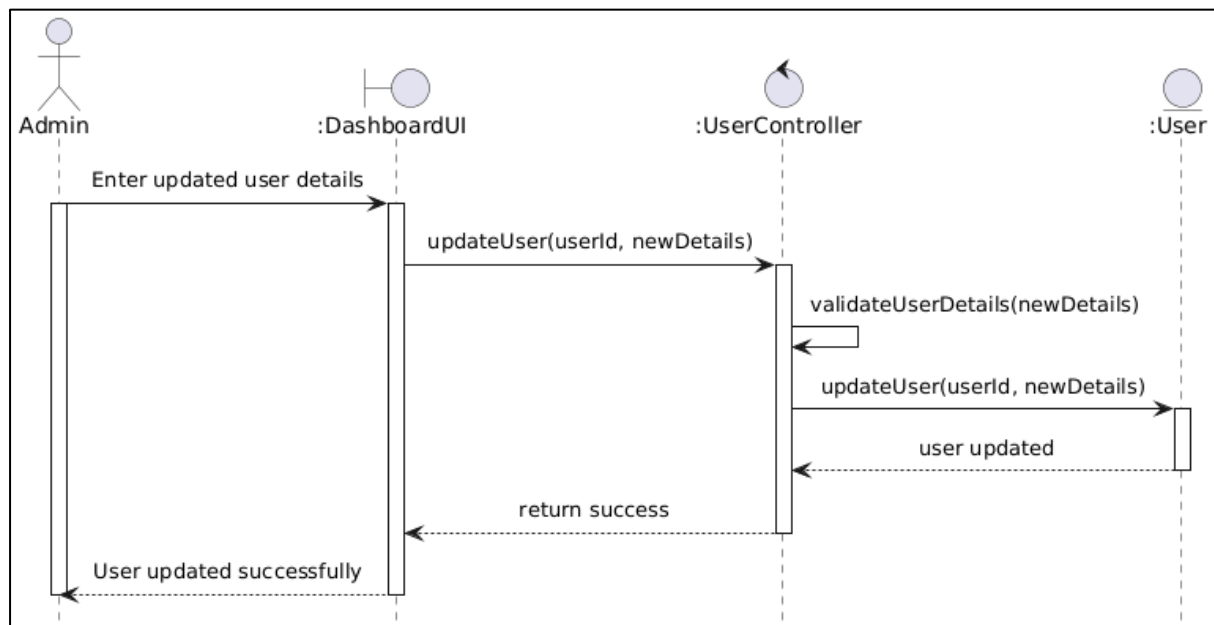
6.2.4 Sequence Diagrams

6.2.4.1 User Management Module

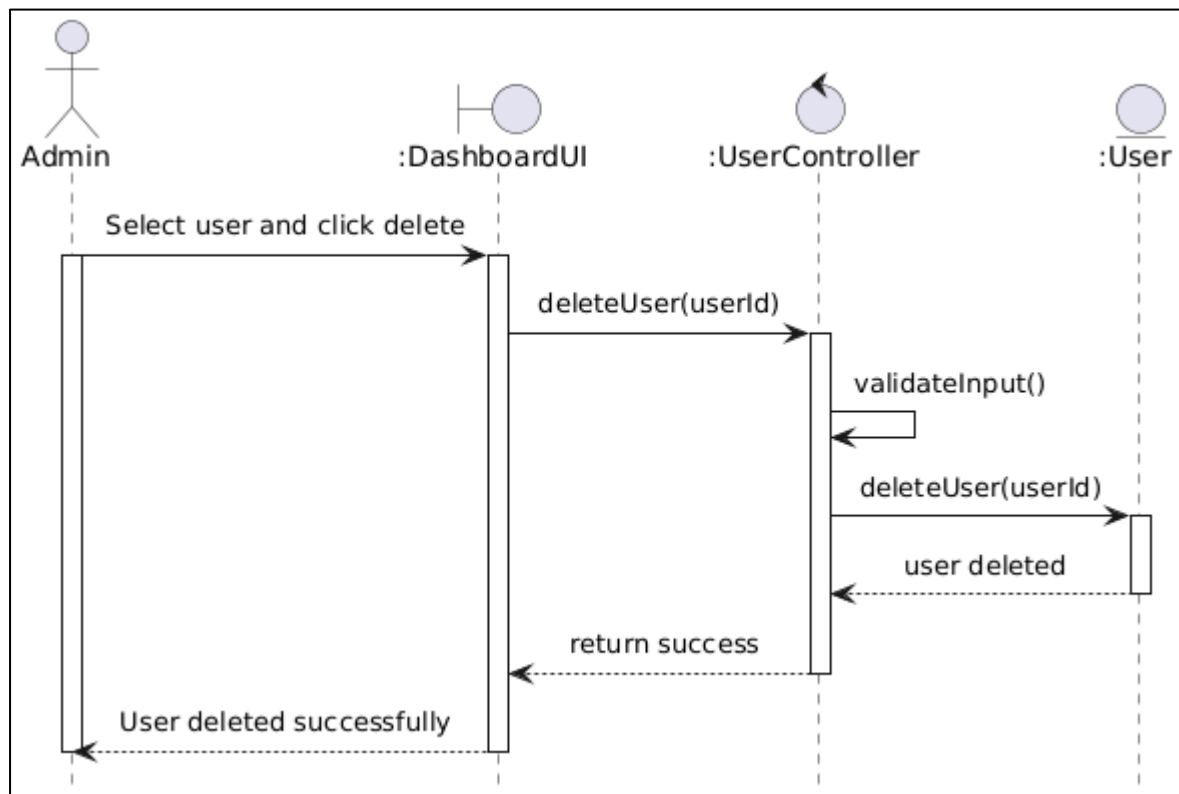
6.2.4.1.1 Create User



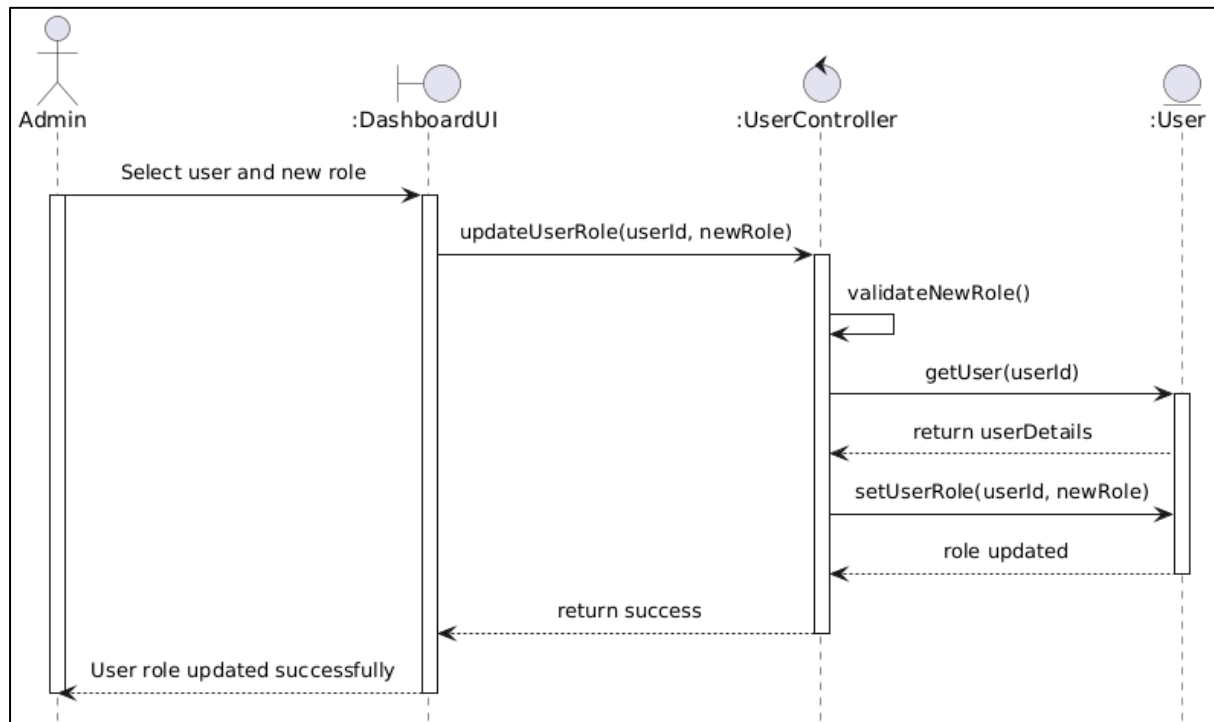
6.2.4.1.2 Update User



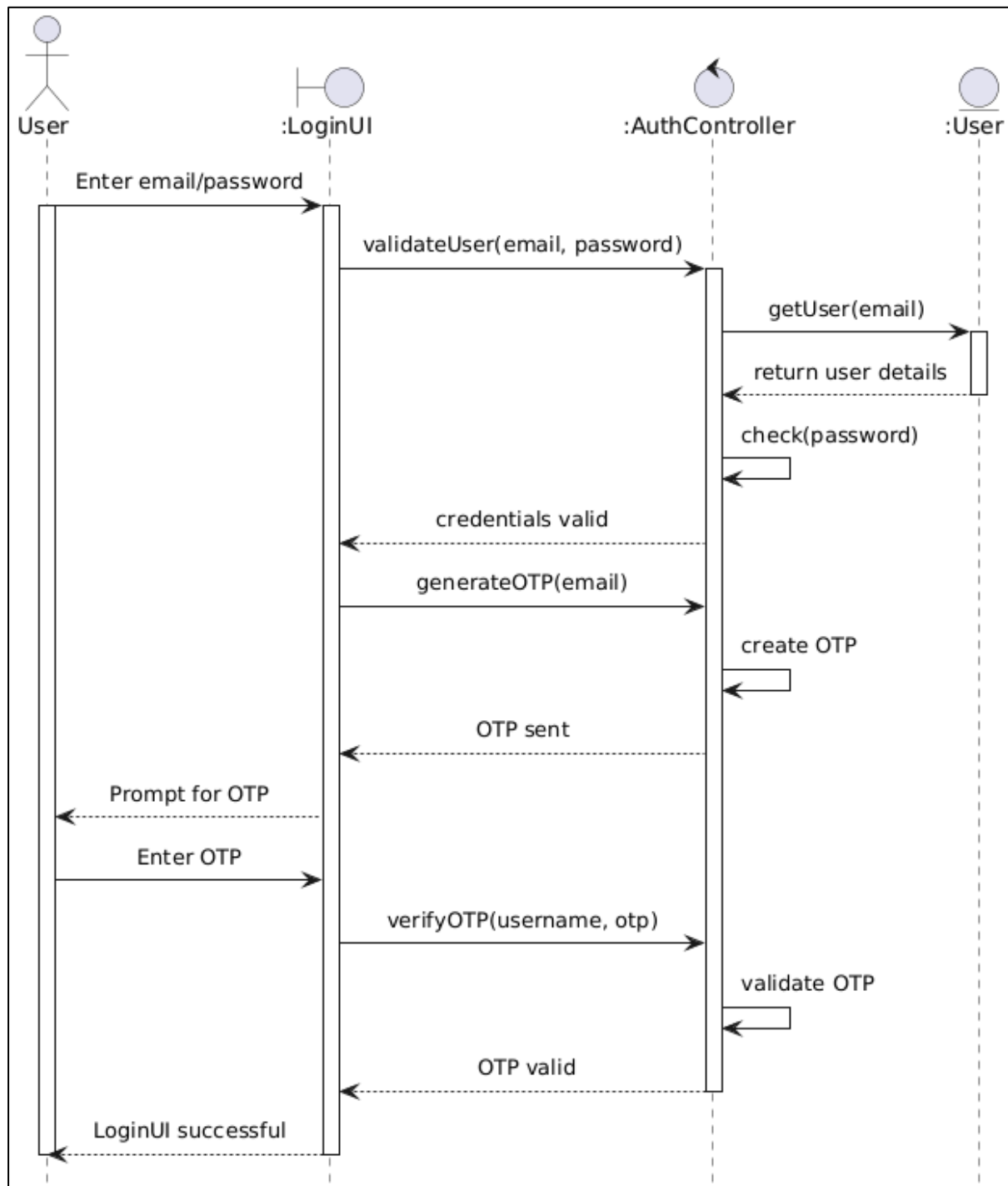
6.2.4.1.3 Delete User



6.2.4.1.4 Manage User Roles

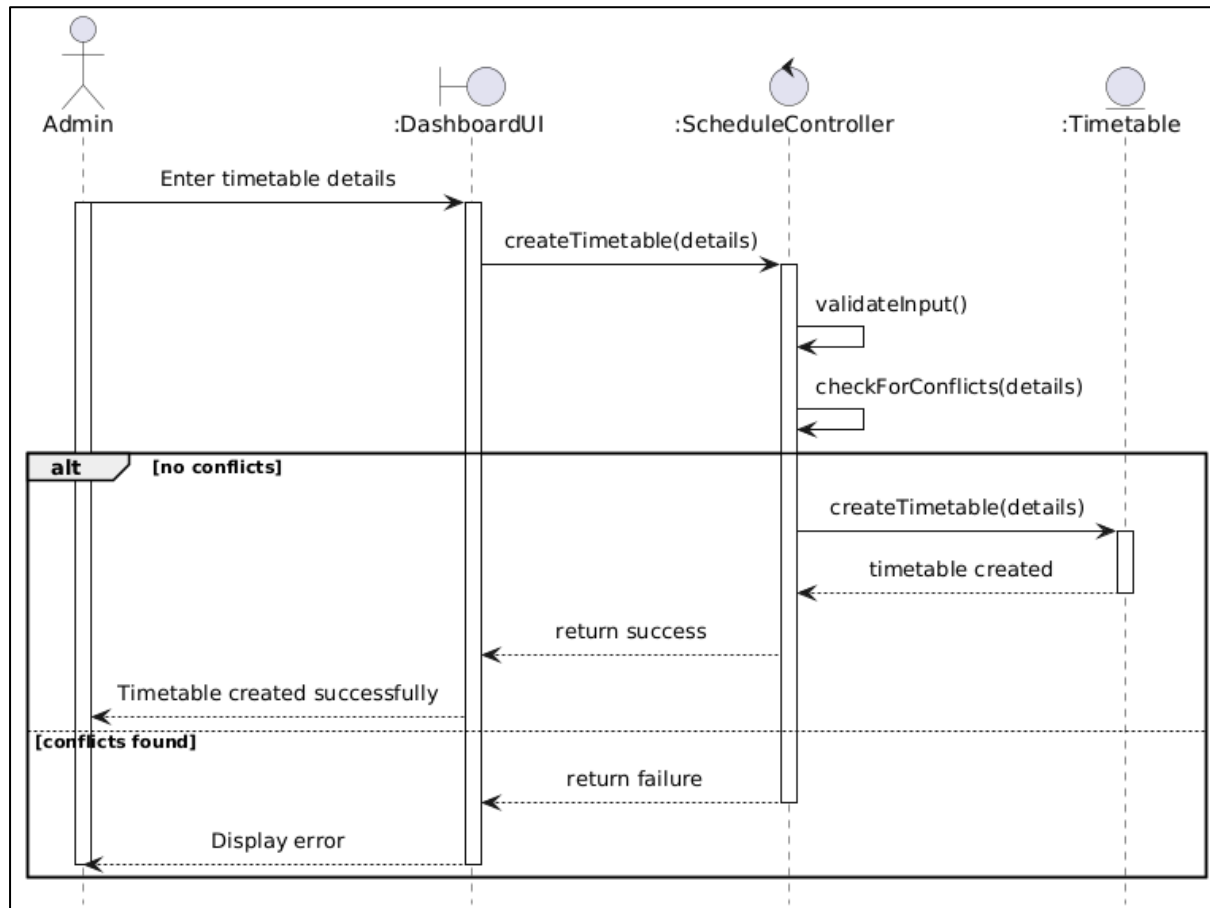


6.2.4.1.5 User Login

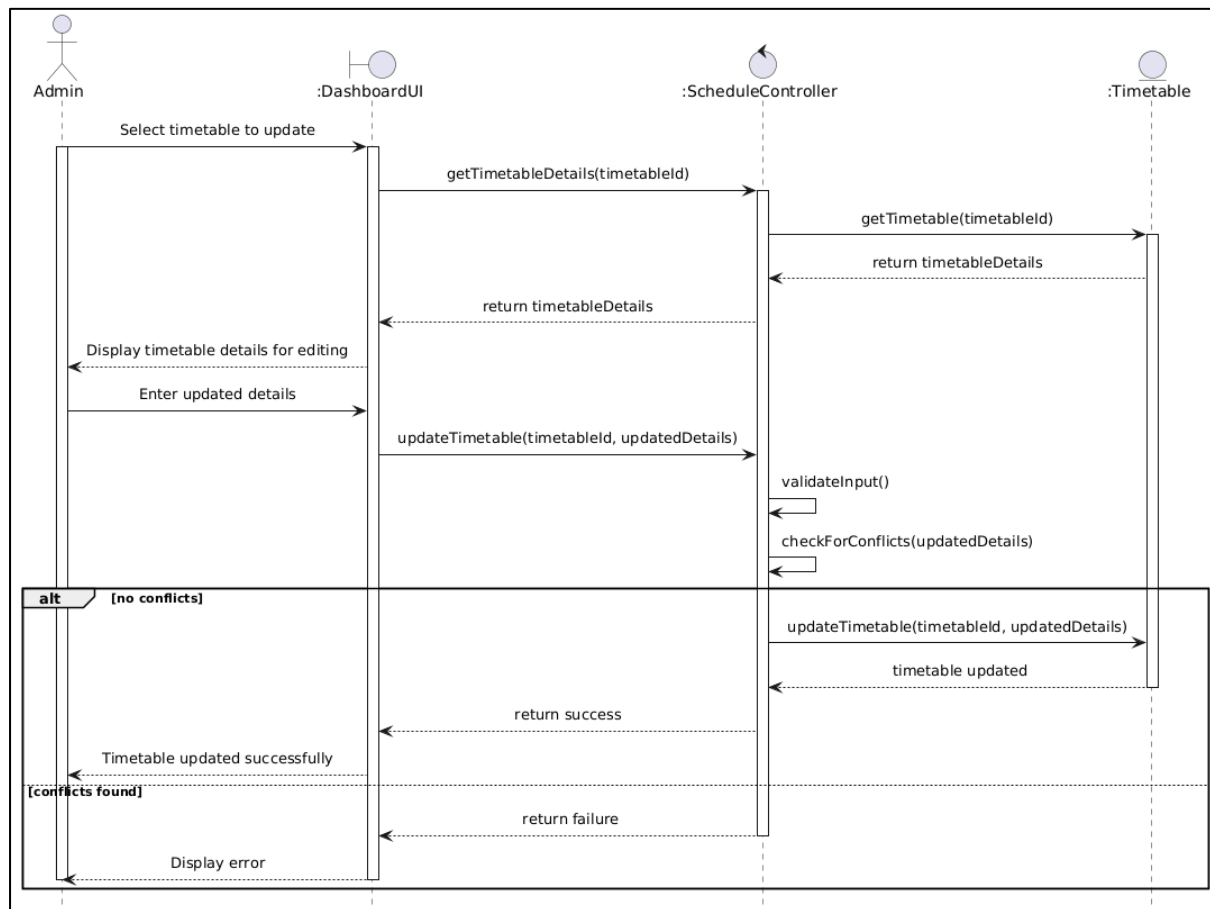


6.2.4.2 Timetable Management Module

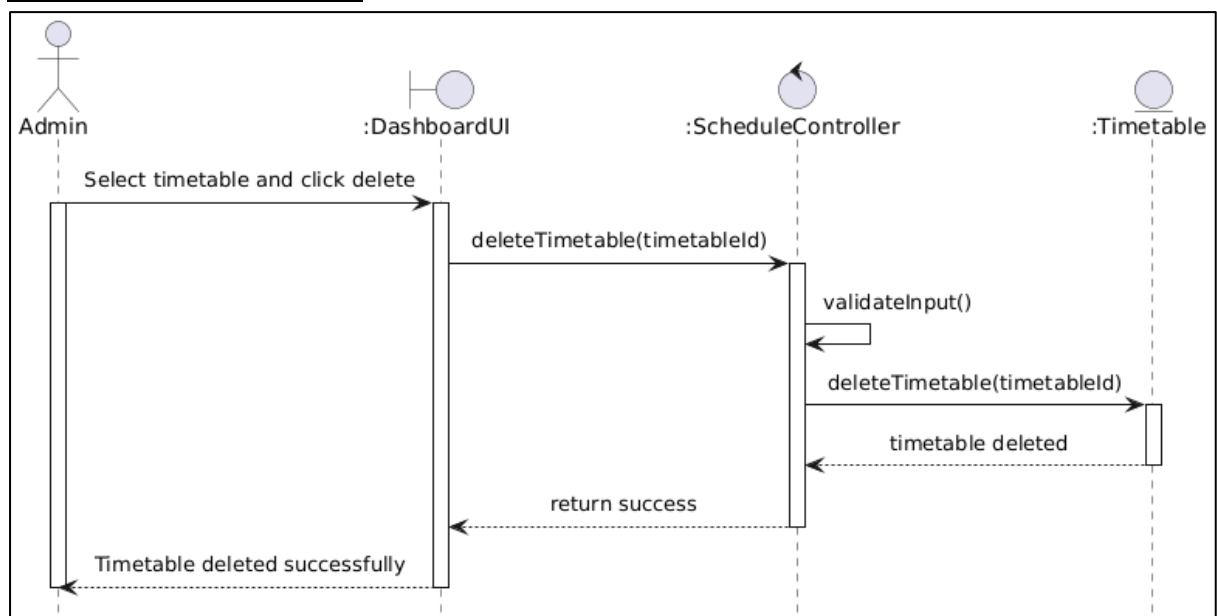
6.2.4.2.1 Create Timetable



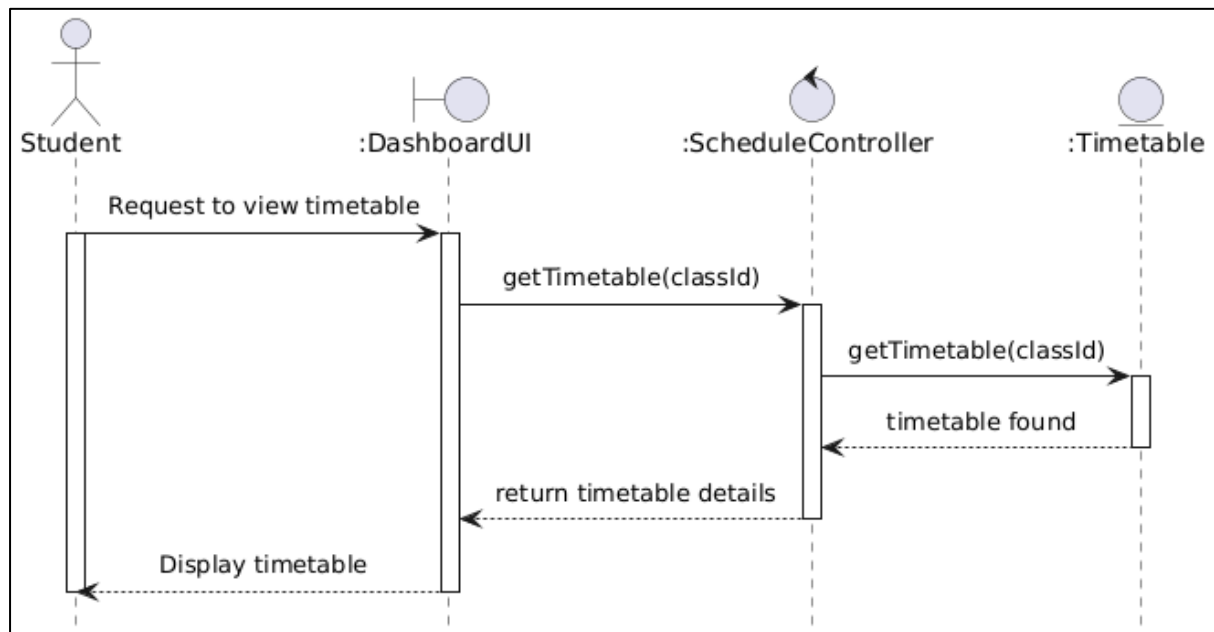
6.2.4.2.2 Update Timetable



6.2.4.2.3 Delete Timetable

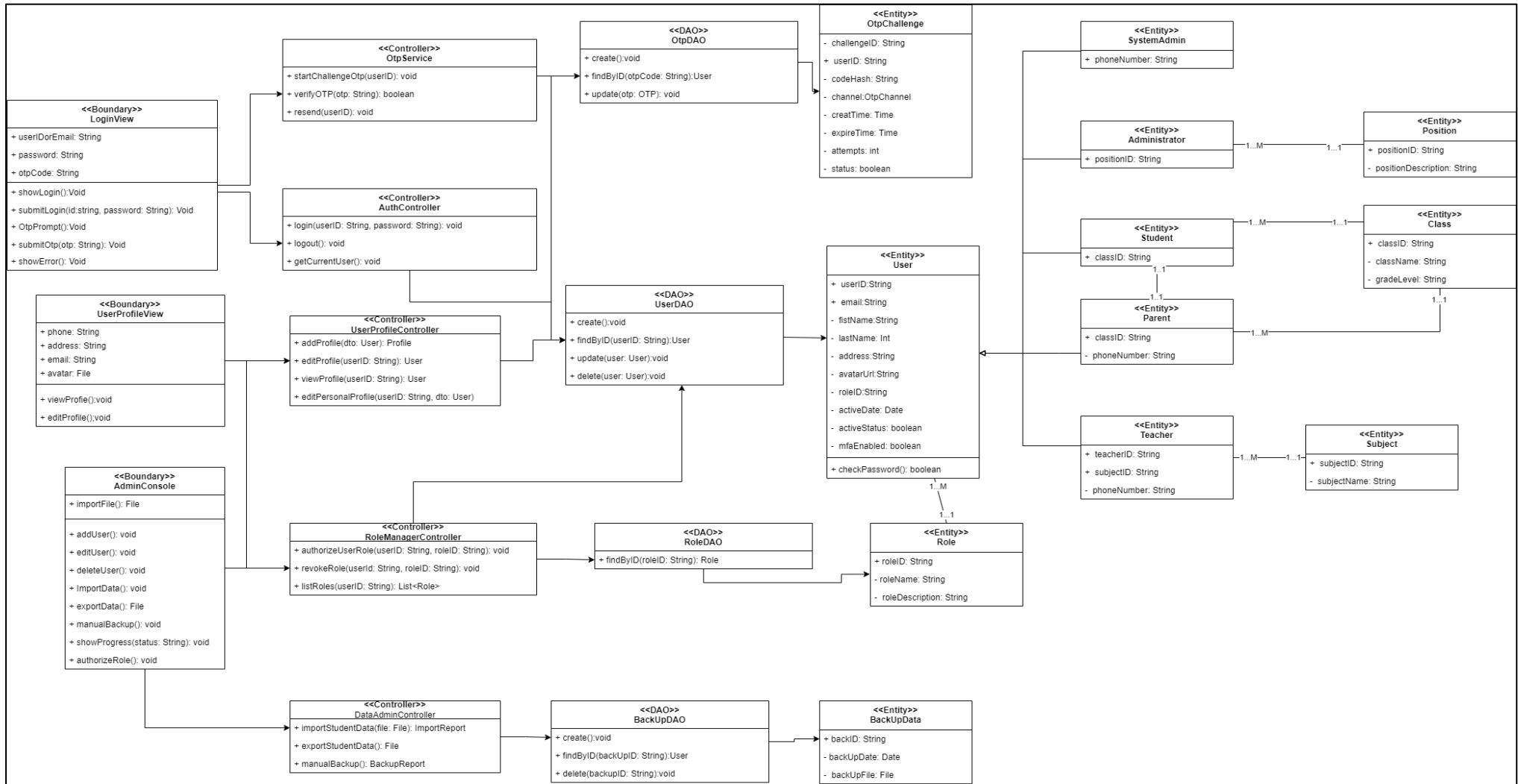


6.2.4.2.4 View Timetable

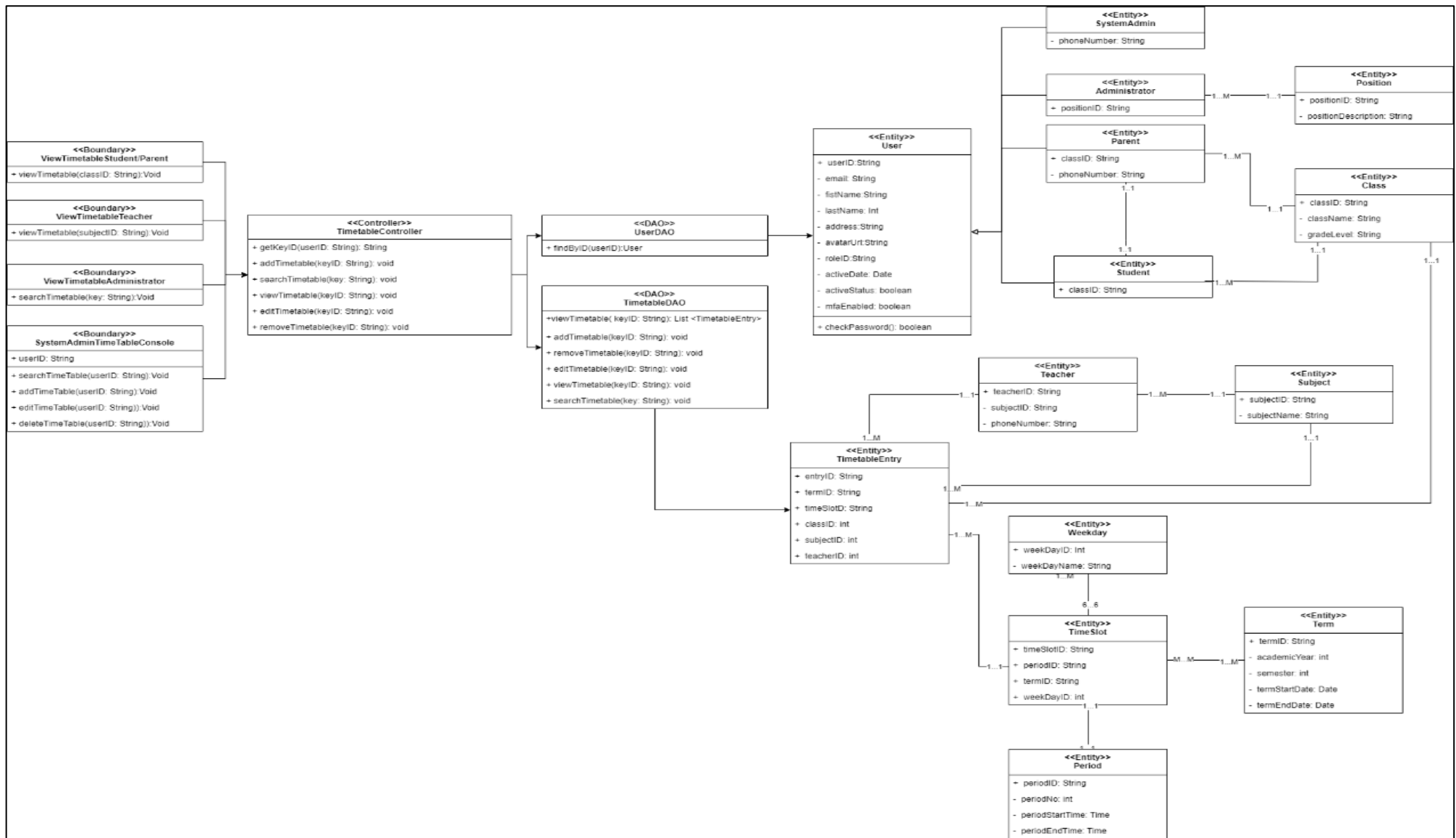


6.2.5 Detailed/Design Class Diagram

6.2.5.1 User Data Management

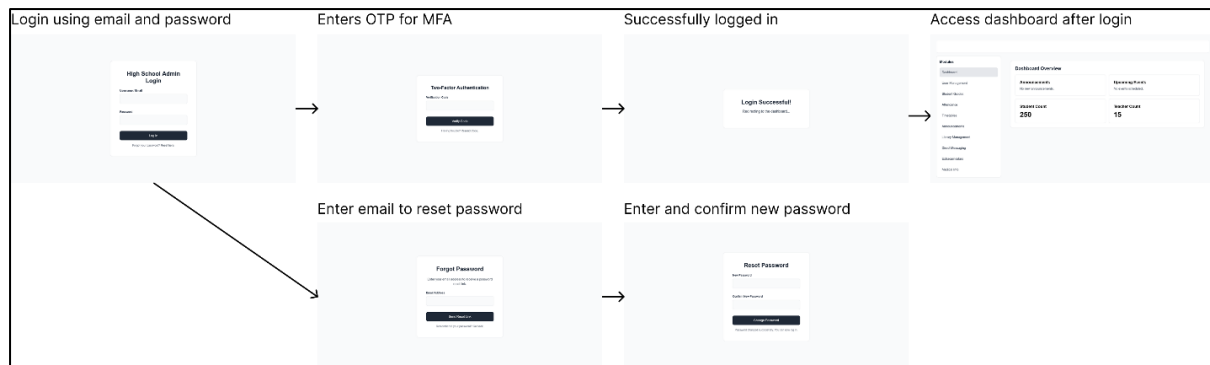


6.2.5.2 Timetable Management



7.0 User Interfaces

7.1 Authentication Module



7.1.1 Login

The login form is titled "High School Admin Login" and contains the following fields and buttons:

- Username / Email:** A text input field.
- Password:** A password input field.
- Log In:** A dark blue button.
- Forgot your password? Reset here.** A link below the password field.

7.1.2 Forgot Password

Forgot Password

Enter your email address to receive a password reset link.

Email Address

Send Reset Link

Remembered your password? [Go back.](#)

7.1.3 Reset Password

Reset Password

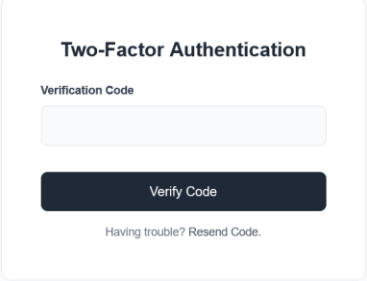
New Password

Confirm New Password

Change Password

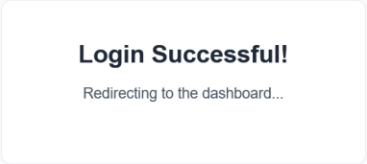
Password changed successfully. You can now log in.

7.1.4 MFA Code Verification



The image shows a 'Two-Factor Authentication' verification screen. It features a title 'Two-Factor Authentication' at the top. Below the title is a label 'Verification Code' followed by a text input field. Underneath the input field is a dark blue button labeled 'Verify Code'. At the bottom of the form, there is a link that says 'Having trouble? Resend Code.'

7.1.5 Successful Login



The image shows a 'Login Successful!' screen. It features a title 'Login Successful!' at the top. Below the title is a message that says 'Redirecting to the dashboard...'

7.2 Student Management Module

View list of all students

The 'User List' interface features a sidebar with modules: Dashboard, User Management (selected), Student Grades, Attendance, Timetables, Announcements, Library Management, Direct Messaging, Extracurriculars, and Medical Info. The main content area displays a table with the following data:

NAME	ROLE	EMAIL	ACTIONS
John Doe	Student	john.doe@gmail.com	Edit Delete

A 'Create New User' button is located at the top right of the table.

Create a new student

The 'Create New User Profile' form includes a sidebar with the same modules as the previous screen. The main content area contains input fields for 'Name' and 'Email', a 'Select Role' dropdown menu, and a 'Save Profile' button at the bottom.

Delete an existing student

The 'Delete User' dialog box shows a confirmation message: 'Please confirm you want to delete the user "John Doe"'. It features a red 'Delete User' button and a grey 'Cancel' button. The sidebar on the left remains consistent with the other screens.

7.2.1 Admin Dashboard

The Admin Dashboard features a sidebar with modules: Dashboard (selected), User Management, Student Grades, Attendance, Timetables, Announcements, Library Management, Direct Messaging, Extracurriculars, and Medical Info. The main content area is titled 'Dashboard Overview' and contains four summary cards:

- Announcements**: No new announcements.
- Upcoming Events**: No events scheduled.
- Student Count**: 250
- Teacher Count**: 15

7.2.2 View All Students

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

User List

Create New User

NAME	ROLE	EMAIL	ACTIONS
John Doe	Student	john.doe@email.com	Edit Delete

7.2.3 Create New Student

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

Create New User Profile

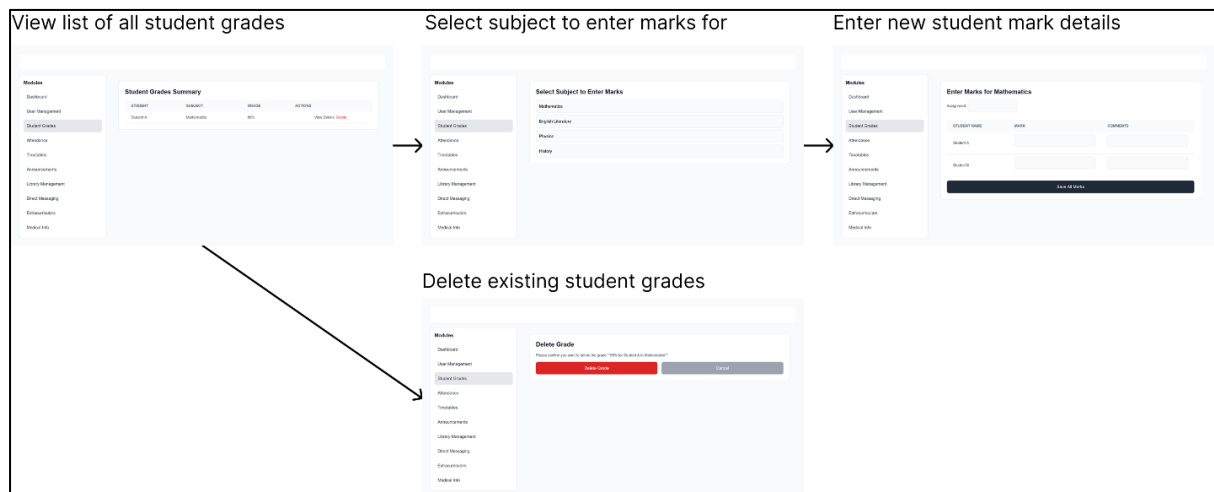
Select Role

Save Profile

7.2.4 Delete Existing Student

The screenshot shows the 'User Management' module selected in the sidebar. The main content area displays a 'Delete User' dialog box. The dialog contains the text 'Please confirm you want to delete the user **John Doe**.' and two buttons: a red 'Delete User' button and a grey 'Cancel' button.

7.3 Student Grades Module



7.3.1 View All Grades

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

Student Grades Summary

STUDENT	SUBJECT	GRADE	ACTIONS
Student A	Mathematics	85%	View Details Delete

7.3.2 Select Subject for Mark Entry

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

Select Subject to Enter Marks

Mathematics

English Literature

Physics

History

7.3.3 Enter New Student Mark Details

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

Enter Marks for Mathematics

Assignment:

STUDENT NAME	MARK	COMMENTS
Student A		
Student B		

Save All Marks

1.3.4 Delete Existing Grades

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

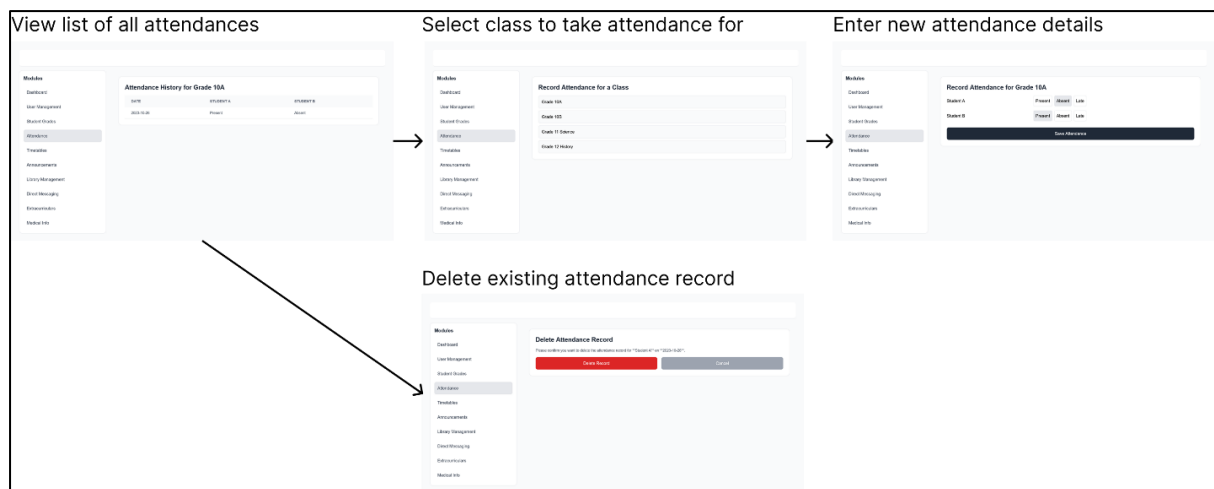
Delete Grade

Please confirm you want to delete the grade **85% for Student A in Mathematics**.

Delete Grade

Cancel

7.4 Student Attendance Module



7.4.1 View All Attendances

The screenshot shows the 'Attendance' module selected in the sidebar. The main content area displays the 'Attendance History for Grade 10A' table.

DATE	STUDENT A	STUDENT B
2023-10-26	Present	Absent

7.4.2 Select Class for Attendance Entry

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

Record Attendance for a Class

Grade 10A

Grade 10B

Grade 11 Science

Grade 12 History

7.4.3 Enter New Student Attendance Details

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

Record Attendance for Grade 10A

Student A

Present

Absent

Late

Student B

Present

Absent

Late

Save Attendance

7.4.4 Delete Existing Attendance Record

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

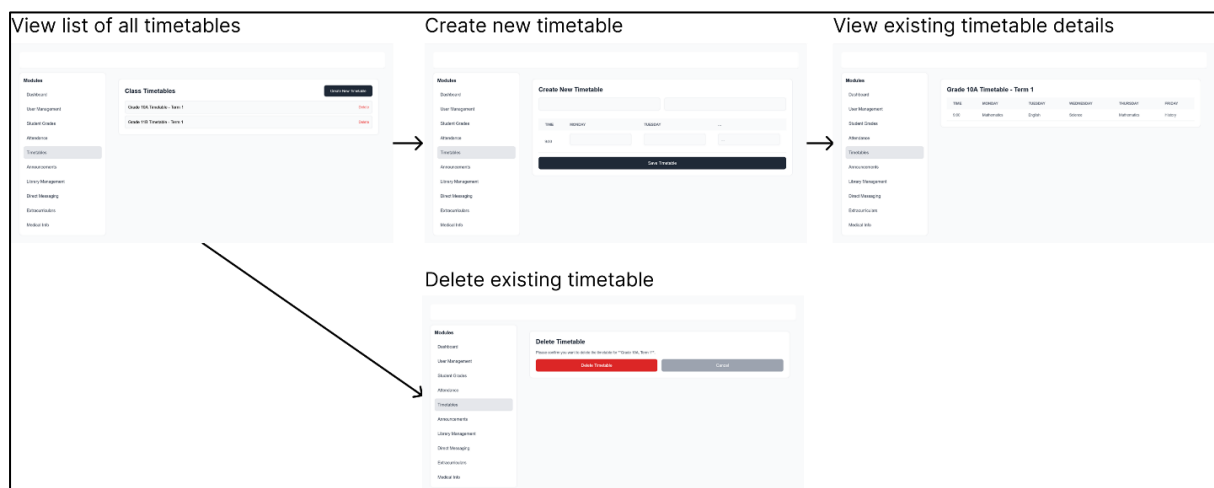
Delete Attendance Record

Please confirm you want to delete the attendance record for **Student A** on **2023-10-26**.

Delete Record

Cancel

7.5 Class Timetable Module



7.5.1 View All Timetables

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

Class Timetables

Create New Timetable

Grade 10A Timetable - Term 1

Delete

Grade 11B Timetable - Term 1

Delete

7.5.2 Create New Timetable

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

Create New Timetable

TIME	MONDAY	TUESDAY	...
9:00			...

Save Timetable

7.5.3 View Timetable Details

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

Grade 10A Timetable - Term 1

TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
9:00	Mathematics	English	Science	Mathematics	History

7.5.4 Delete Existing Timetable

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

Delete Timetable

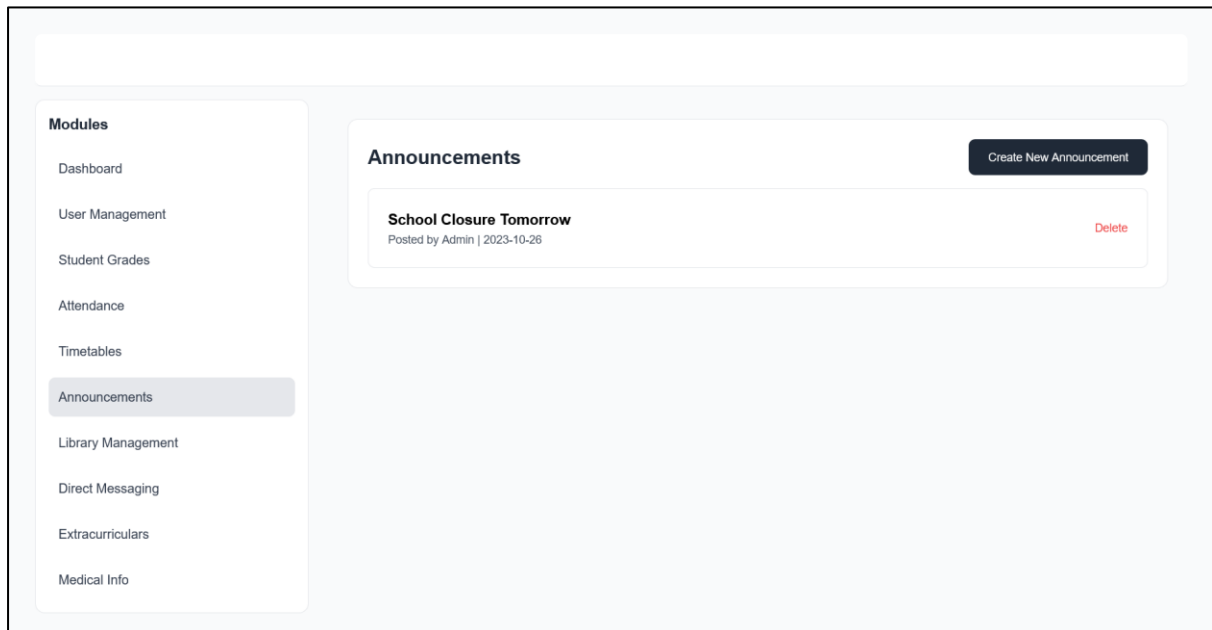
Please confirm you want to delete the timetable for **Grade 10A, Term 1**.

Delete Timetable

Cancel

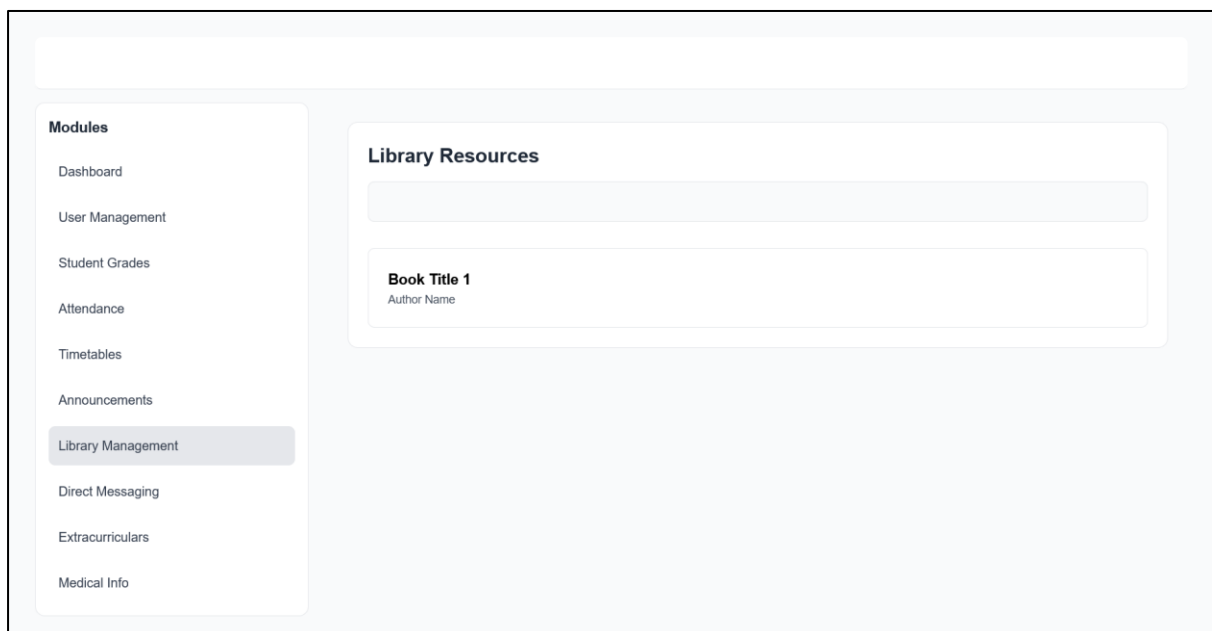
7.6 Announcements Module

7.6.1 View All Announcements



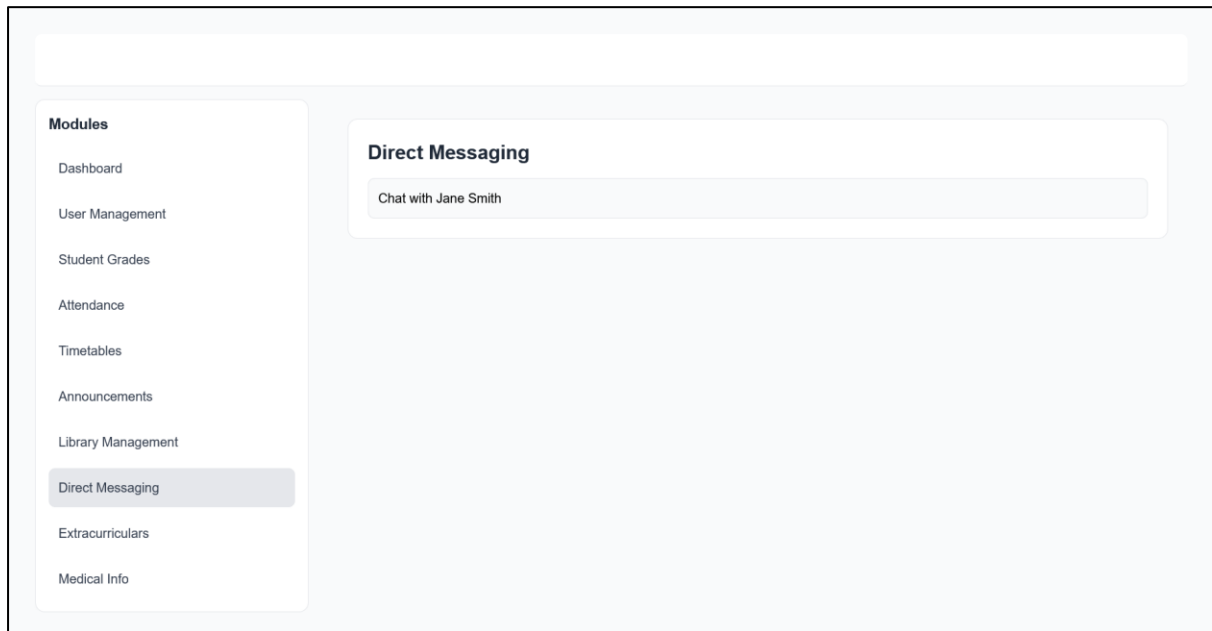
7.7 Library Module

7.7.1 View All Library Resources



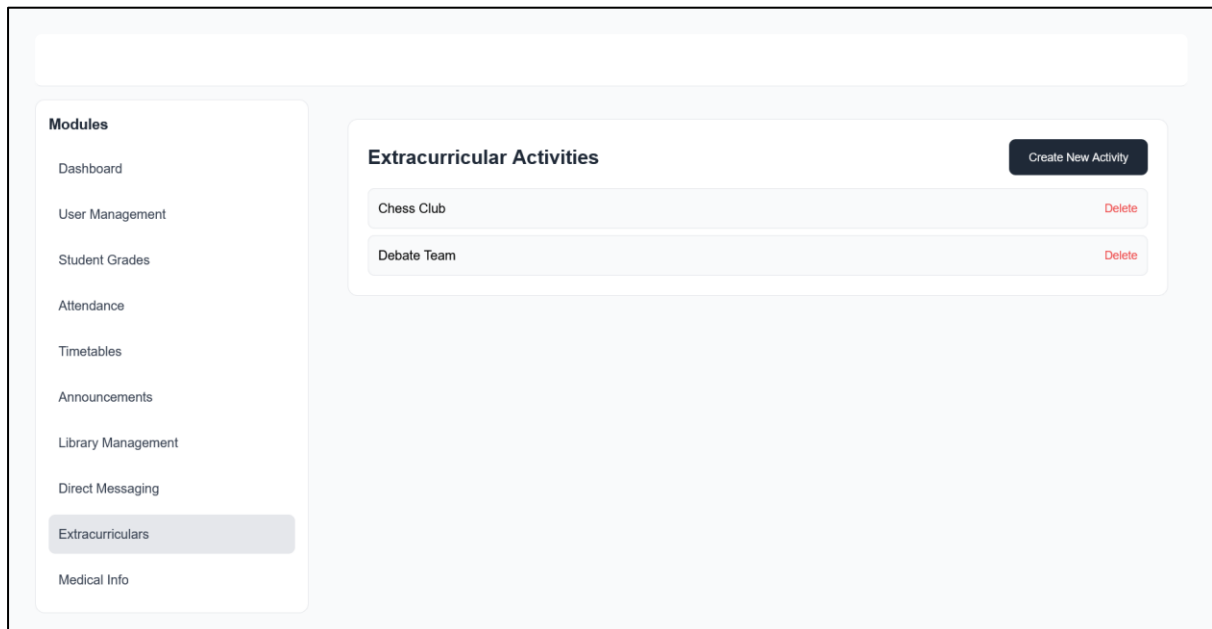
7.8 Direct Messaging Module

7.8.1 View Conversations



7.9 Extracurricular Module

7.9.1 View All Extracurricular Activities



7.10 Student Medical Module

7.10.1 View All Student Medical Details

Modules

Dashboard

User Management

Student Grades

Attendance

Timetables

Announcements

Library Management

Direct Messaging

Extracurriculars

Medical Info

Student Medical Information

STUDENT NAME	CONDITION	NOTES
Student A	Asthma	Inhaler with school nurse.

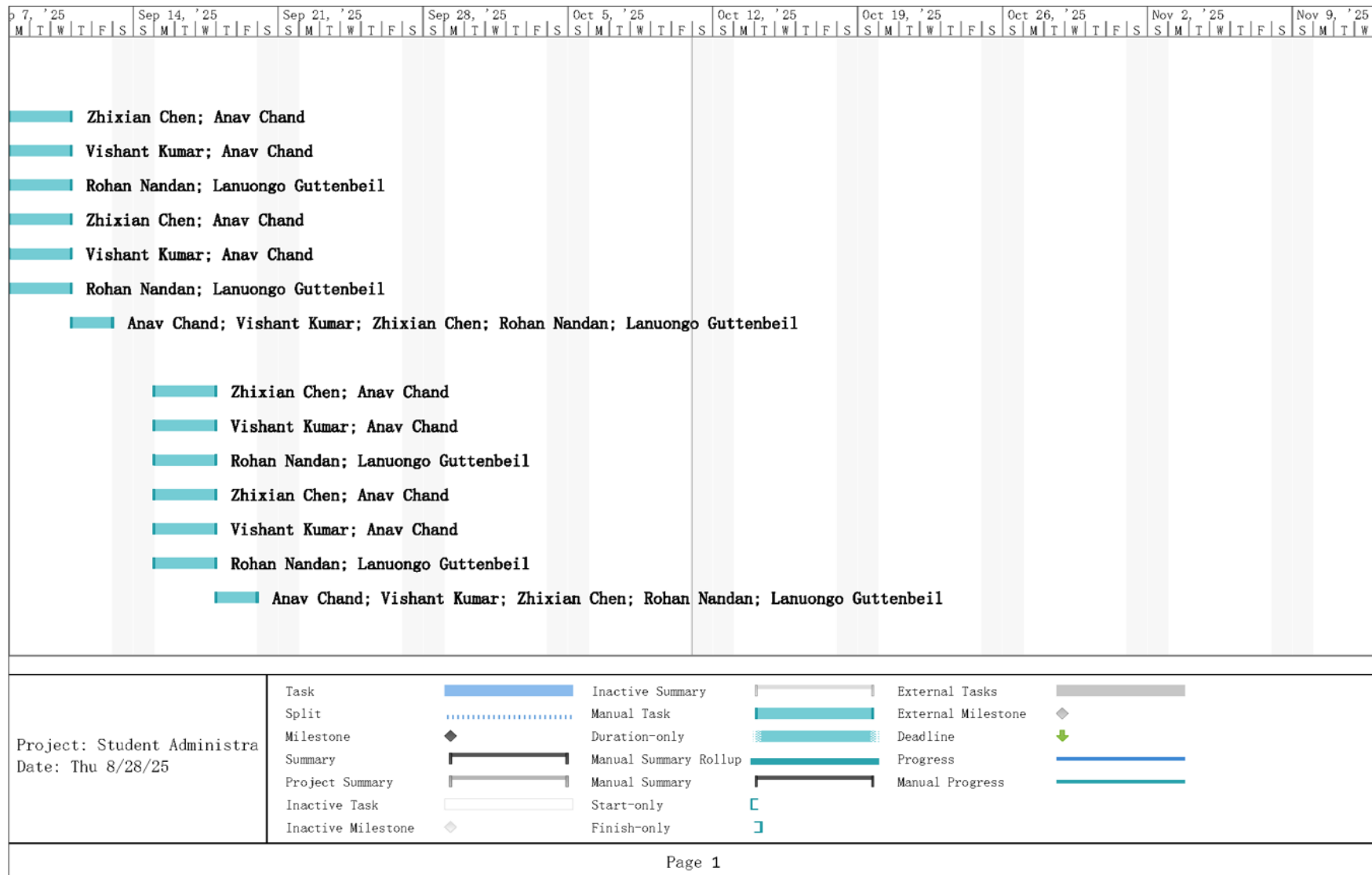
8.0 Implementation Plan

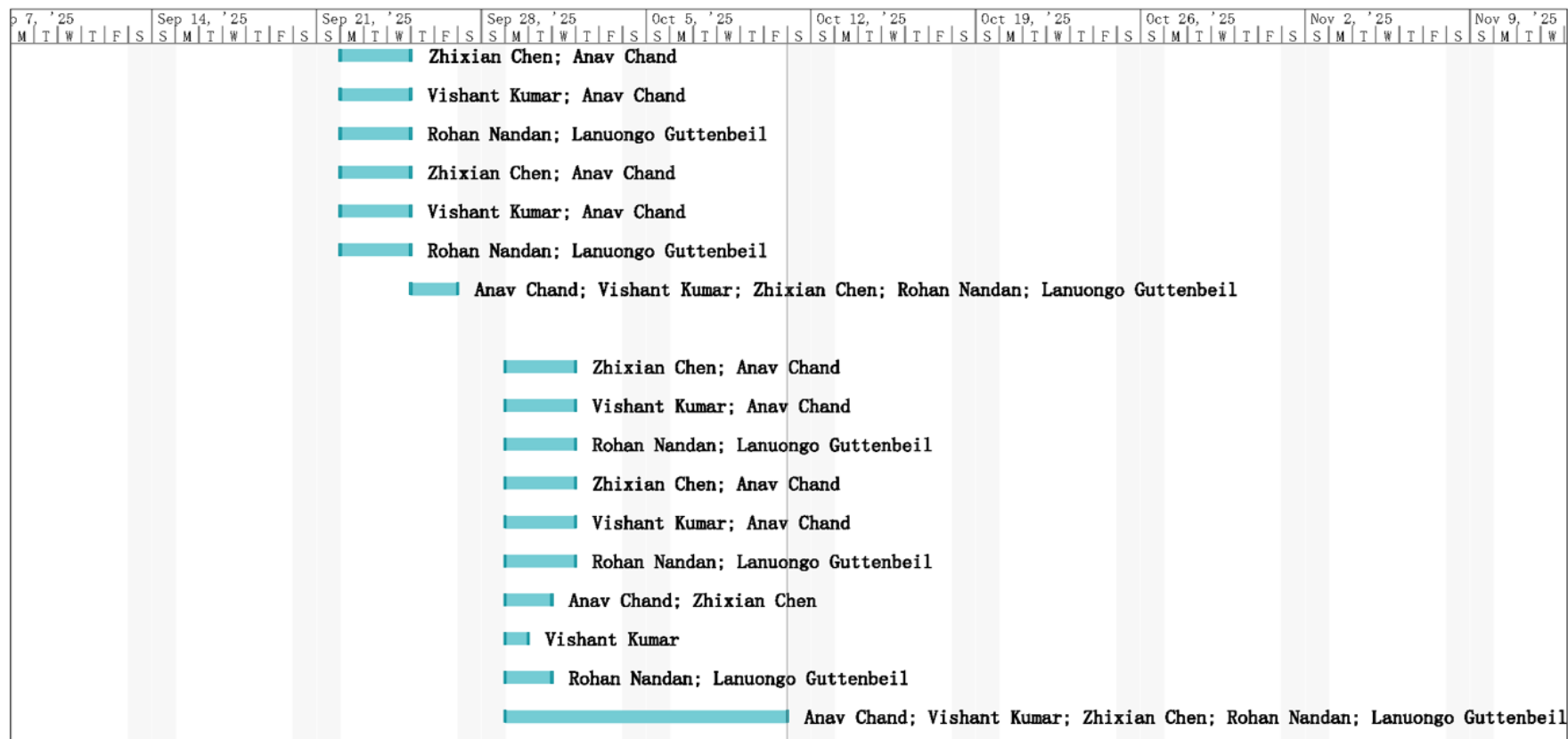
8.1 Gantt Chart

A1					
ID	Task Name	Duration	Start	Finish	Resource Names
1	SAS – Modules Implementation & Testing Plan (4 Weeks)				
2	W1 – User Data Management & Student Performance				
3	User Data Management – Backend (auth/roles/profile/import/export)	3 days	Mon 9/8/25	Wed 9/10/25	Zhixian Chen; Anav Chand
4	User Data Management – Frontend (login/profile/role views)	3 days	Mon 9/8/25	Wed 9/10/25	Vishant Kumar; Anav Chand
5	User Data Management – Unit Tests & Test Records	3 days	Mon 9/8/25	Wed 9/10/25	Rohan Nandan; Lanuongo Guttenbeil
6	Student Performance – Backend (marks CRUD, letter grade)	3 days	Mon 9/8/25	Wed 9/10/25	Zhixian Chen; Anav Chand
7	Student Performance – Frontend (enter/edit/view marks)	3 days	Mon 9/8/25	Wed 9/10/25	Vishant Kumar; Anav Chand
8	Student Performance – Unit Tests & Test Records	3 days	Mon 9/8/25	Wed 9/10/25	Rohan Nandan; Lanuongo Guttenbeil
9	Weekly Integration Test & Fix (W1 modules)	2 days	Thu 9/11/25	Fri 9/12/25	Anav Chand; Vishant Kumar; Zhixian Chen; Rohan Nandan; Lanuongo Guttenbeil
10	W2 – Discipline/Attendance & Timetable				
11	Discipline & Attendance – Backend (attendance record, incidents, absence info)	3 days	Mon 9/15/25	Wed 9/17/25	Zhixian Chen; Anav Chand
12	Discipline & Attendance – Frontend (attendance entry, views)	3 days	Mon 9/15/25	Wed 9/17/25	Vishant Kumar; Anav Chand
13	Discipline & Attendance – Unit Tests & Test Records	3 days	Mon 9/15/25	Wed 9/17/25	Rohan Nandan; Lanuongo Guttenbeil
14	Class Timetable – Backend (teacher/student, conflict check)	3 days	Mon 9/15/25	Wed 9/17/25	Zhixian Chen; Anav Chand
15	Class Timetable – Frontend (weekly grid, role views)	3 days	Mon 9/15/25	Wed 9/17/25	Vishant Kumar; Anav Chand
16	Class Timetable – Unit Tests & Test Records	3 days	Mon 9/15/25	Wed 9/17/25	Rohan Nandan; Lanuongo Guttenbeil
17	Weekly Integration Test & Fix (W2 modules)	2 days	Thu 9/18/25	Fri 9/19/25	Anav Chand; Vishant Kumar; Zhixian Chen; Rohan Nandan; Lanuongo Guttenbeil
18	W3 – Library/Resources & Communication				
19	Library/Resources – Backend (search/upload/download/reserve, penalty)	3 days	Mon 9/22/25	Wed 9/24/25	Zhixian Chen; Anav Chand
20	Library/Resources – Frontend (search/list/detail/reserve)	3 days	Mon 9/22/25	Wed 9/24/25	Vishant Kumar; Anav Chand
21	Library/Resources – Unit Tests & Test Records	3 days	Mon 9/22/25	Wed 9/24/25	Rohan Nandan; Lanuongo Guttenbeil
22	Communication – Backend (announcements CRUD, messaging)	3 days	Mon 9/22/25	Wed 9/24/25	Zhixian Chen; Anav Chand
Page 1					

A1					
ID	Task Name	Duration	Start	Finish	Resource Names
23	Communication – Frontend (annnc pages, send/view messages)	3 days	Mon 9/22/25	Wed 9/24/25	Vishant Kumar; Anav Chand
24	Communication – Unit Tests & Test Records	3 days	Mon 9/22/25	Wed 9/24/25	Rohan Nandan; Lanuongo Guttenbeil
25	Weekly Integration Test & Fix (W3 modules)	2 days	Thu 9/25/25	Fri 9/26/25	Anav Chand; Vishant Kumar; Zhixian Chen; Rohan Nandan; Lanuongo Guttenbeil
26	W4 – Activities, Staff Management, Chat with AI & Final Testing				
27	Extracurricular Activities – Backend (enroll, attendance view)	3 days	Mon 9/29/25	Wed 10/1/25	Zhixian Chen; Anav Chand
28	Extracurricular Activities – Frontend (enroll, attendance view)	3 days	Mon 9/29/25	Wed 10/1/25	Vishant Kumar; Anav Chand
29	Extracurricular Activities – Unit Tests & Test Records	3 days	Mon 9/29/25	Wed 10/1/25	Rohan Nandan; Lanuongo Guttenbeil
30	Staff Management – Backend (leave flow, attendance, evaluation)	3 days	Mon 9/29/25	Wed 10/1/25	Zhixian Chen; Anav Chand
31	Staff Management – Frontend (approval, evaluations, reports)	3 days	Mon 9/29/25	Wed 10/1/25	Vishant Kumar; Anav Chand
32	Staff Management – Unit Tests & Test Records	3 days	Mon 9/29/25	Wed 10/1/25	Rohan Nandan; Lanuongo Guttenbeil
33	Chat with AI – Backend (FAQ/KB retrieval-based Q&A)	2 days	Mon 9/29/25	Tue 9/30/25	Anav Chand; Zhixian Chen
34	Chat with AI – Frontend (simple chat UI)	1 day	Mon 9/29/25	Mon 9/29/25	Vishant Kumar
35	Chat with AI – Unit Tests & Test Records	2 days	Mon 9/29/25	Tue 9/30/25	Rohan Nandan; Lanuongo Guttenbeil
36	System-wide Integration & Regression Tests (all modules)	10 days	Mon 9/29/25	Fri 10/10/25	Anav Chand; Vishant Kumar; Zhixian Chen; Rohan Nandan; Lanuongo Guttenbeil
Page 2					

8.2 Timeline Chart





Project: Student Administra Date: Thu 8/28/25	Task		Inactive Summary		External Tasks	
	Split		Manual Task		External Milestone	
	Milestone		Duration-only		Deadline	
	Summary		Manual Summary Rollup		Progress	
	Project Summary		Manual Summary		Manual Progress	
	Inactive Task		Start-only			
	Inactive Milestone		Finish-only			

9.0 Definitions and Acronyms

Acronyms	Long Form	Definitions
SDD	Software Design Document	A document that provides a detailed overview of the process of developing a software system.
ERD	Entity Relationship Diagram	A diagram that outlines the structure of a database, their entities, attributes and relationship between them.
DFD	Data Flow Diagram	A diagram that shows the flow of data as it enters and exits a system.
UI/UX	User Interface/User Experience	Refers to the design of user-friendly interfaces for a system.
SAS	Student Administration System	The name of the software system being designed for Bula High School to manage student administration.
KISS	Keep it Simple, Stupid	A design principle that focuses on creating simple and easy to understand code and architectural design.
MVC	Model View Controller	Software framework composed of three components Model, View, Controller that separates business logic, user interface and input logic.
S3	Simple Storage Services	An Amazon web services object storage service used for storing files like images and documents.
RDS	Rational Database Service	An Amazon web services relation database service used to store structured data.
	DynamoDB	An Amazon web services NoSQL database service used for storing unstructured data.
ORM	Object Relational Mapping	A tool used to handle database connections and queries, allowing developers to interact with the database using object-oriented code.
SDK	Software Development Kit	A set of development tools that allows for communication with databases and storage buckets on Amazon Web Services.
RBAC	Role Based Access Control	A security mechanism that allows users to access only the parts of the system that their roles permit.
MFA	Multifactor Authentication	A security measure that allows users to verify their identity.
TOTP	Time-Based One-Time Password	A time-based password used as an additional authentication.
AI	Artificial Intelligence	Chatbot that has knowledge of anything.
API	Application Programming Interface	Interface that connects the AI chatbot to the OpenAI model.
HTML	HyperText Markup Language	Used to render the streamed messages from the chatbot on the client.
ACID	Atomicity, Consistency, Isolation, Durability	Guarantees that database transactions are processed reliably.
ARIA	Accessible Rich Internet Application	Labels used for screen readers to attach labels to HTML elements.

AWS	Amazon Web Services	Cloud hosting provider for the system.
CI/CD	Continuous Integration/Deployment	Automated process for continuous integration and deployment of the system using Jenkins.
CSV	Comma-Separated Values	A file format for importing student data.
GDPR	General Data Protection Regulation	Set of data privacy regulations that the system will comply with.
HTTPS	HyperText Transfer Protocol Secure	Protocol that ensures encrypted transmission of data between the server and the users device.
NFR	Non-Functional Requirements	System requirement that does not relate to its functionality but its performance, reliability etc.
PDF	Portable Document Format	A file format for exporting student data.
RPO	Recovery Point Objective	Key metric in disaster recovery that defines maximum amount of data that can be lost in the event of a system failure.
RTO	Recovery Time Objective	Key metric in disaster recovery that defines maximum acceptable time to restore the system after a failure.
SSL	Secure Socket Layer	A cryptographic protocol used to establish an encrypted link between a server and a client.
UAT	User Acceptance Testing	A type of testing conducted before a full rollout to ensure the system meets the user needs.
UTC	Coordinated Universal Time	Primary time standard by which the world regulates clocks.
XML	eXtensible Markup Language	A file format used for importing legacy data from other school systems.

10.0 References

Atlassian. (n.d.). *Software Design Document [Tips & Best Practices]* | *The Workstream*.

<https://www.atlassian.com/work-management/knowledge-sharing/documentation/software-design-document#:~:text=Benefits%20of%20a%20software%20design%20document&text=Design%20documents%20improve%20stakeholder%20communication,and%20how%20to%20achieve%20them>.

Atlassian. (n.d.). *Software Design Document [Tips & Best Practices]* | *The Workstream*.

<https://www.atlassian.com/work-management/knowledge-sharing/documentation/software-design-document>

GeeksforGeeks. (n.d.). *GeeksforGeeks*. <https://www.geeksforgeeks.org/>

GeeksforGeeks. (2025, July 23). *MVC Framework Introduction*. GeeksforGeeks.

<https://www.geeksforgeeks.org/software-engineering/mvc-framework-introduction/>

Gupta, S. (2024, April 13). *Software Development & Design Principles, and Architectural Styles: Quick refresher*. Medium. <https://medium.com/@saurabh.engg.it/software-development-and-design-principles-and-architectural-styles-quick-refresher-a95497ef514f>

Understanding PDDs and SDDs: Essential documentation for automation projects. (n.d.).

<https://www.blueprintsys.com/blog/understanding-pdd-and-sdd>

10 Software development Best Practices (2025 Checklist). (2025, April 28). 2am.tech.

<https://www.2am.tech/blog/software-development-best-practices>

Assignment 1 Mark Allocation Sheet

After having discussed it as a group, we recommend the following mark allocation to each group member based on contribution or lack of it throughout the assignment.

Group number: 4 Project Manager: Vishant Kumar

Member Student ID	Percentage Contribution	Responsibilities
S11230430	20%	Parts 1, 2, 6.2.2 & 6.2.3
S11221203	20%	Parts 6.1.1, 6.2.4, 7 & ToR
S11234883	20%	Parts 3, 4, 5 & 6.1.2
S11230686	20%	Parts 6.2.1, 6.2.5 & 8
S11215322	20%	Parts 2, 3, 9 & ToR

Signed

Member Name	ID	Signature
<u>Vishant Kumar</u>	<u>S11230430</u>	<u>vishant</u>
<u>Anav Chand</u>	<u>S11221203</u>	<u>anav</u>
<u>Rohan Nandan</u>	<u>S11234883</u>	<u>rohan</u>
<u>Zhixian Chen</u>	<u>S11230686</u>	<u>zhixian</u>
<u>Lanuongo Guttenbeil</u>	<u>S11215322</u>	<u>lanuongo</u>